

ANALYTIC NUMBER THEORY

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1. ARITHMETIC FUNCTIONS

Definition 1.0.1 (Arithmetic function). Let R be a ring. An *arithmetic function on $\mathbb{Z}_{>0}$ with values in R* is a function $f : \mathbb{Z}_{>0} \rightarrow R$.

Definition 1.0.2 (Multiplicative function). Let R be a ring. A arithmetic function on $\mathbb{Z}_{>0}$ with values in R (Definition 1.0.1) $f : \mathbb{Z}_{>0} \rightarrow R$ is called *multiplicative* if $f(1) = 1$ and for

all $m, n \in \mathbb{Z}_{>0}$ with $\gcd(m, n) = 1$,

$$f(mn) = f(m)f(n).$$

Definition 1.0.3 (Completely multiplicative function). Let R be a ring. A arithmetic function (Definition 1.0.1) $f : \mathbb{Z}_{>0} \rightarrow R$ on $\mathbb{Z}_{>0}$ with values in R is called *completely multiplicative* if $f(1) = 1$ and for all $m, n \in \mathbb{Z}_{>0}$,

$$f(mn) = f(m)f(n).$$

Definition 1.0.4. Let $f, g : \mathbb{Z}_{>0} \rightarrow R$ be arithmetic functions (Definition 1.0.1). Their *Dirichlet convolution*, denoted by $(f * g)$, is the arithmetic function defined by

$$(f * g)(n) = \sum_{d|n} f(d) g\left(\frac{n}{d}\right)$$

for each $n \in \mathbb{Z}_{>0}$. As a binary operation on the set of arithmetic functions, Dirichlet convolution is commutative and associative (Proposition 1.0.5), with identity element ε , where $\varepsilon(1) = 1$ and $\varepsilon(n) = 0$ for $n > 1$.

Proposition 1.0.5 (Algebraic properties of Dirichlet convolution). (♠ TODO: AI generated, verify) Let R be a ring, and let $f, g, h : \mathbb{Z}_{>0} \rightarrow R$ be arithmetic functions (Definition 1.0.1). The Dirichlet convolution (Definition 1.0.4) satisfies the following properties:

1. **Commutativity:** $f * g = g * f$.
2. **Associativity:** $(f * g) * h = f * (g * h)$.
3. **Identity:** The arithmetic function ε defined by $\varepsilon(1) = 1$ and $\varepsilon(n) = 0$ for $n > 1$ is the identity element for Dirichlet convolution, i.e., $f * \varepsilon = \varepsilon * f = f$.

Proof. (♠ TODO: AI generated, verify) We prove each property directly from the definition of Dirichlet convolution (Definition 1.0.4). Let $n \in \mathbb{Z}_{>0}$ be arbitrary.

Commutativity. By Definition 1.0.4,

$$(f * g)(n) = \sum_{d|n} f(d) g\left(\frac{n}{d}\right).$$

As d ranges over the divisors of n , so does n/d . Substituting $d' = n/d$, we obtain

$$(f * g)(n) = \sum_{d'|n} f\left(\frac{n}{d'}\right) g(d') = (g * f)(n),$$

where the last equality uses commutativity of multiplication in the ring R .

Associativity. We compute

$$((f * g) * h)(n) = \sum_{d|n} (f * g)(d) h\left(\frac{n}{d}\right) = \sum_{d|n} \sum_{e|d} f(e) g\left(\frac{d}{e}\right) h\left(\frac{n}{d}\right).$$

The pair (e, d) satisfies $e \mid d$ and $d \mid n$, hence $e \mid n$. Writing $d = ef$ with $f \in \mathbb{Z}_{>0}$, the condition $d \mid n$ becomes $ef \mid n$, i.e., $f \mid (n/e)$. Setting $g' = d/e$ and $h' = n/d$, we have

$eg'h' = n$. Thus

$$((f * g) * h)(n) = \sum_{\substack{e, g', h' \in \mathbb{Z}_{>0} \\ eg'h' = n}} f(e) g(g') h(h').$$

By the same computation with g and h in place of f and g , we obtain

$$(f * (g * h))(n) = \sum_{\substack{e, g', h' \in \mathbb{Z}_{>0} \\ eg'h' = n}} f(e) g(g') h(h'),$$

so both expressions are equal.

Identity. We compute for $n \geq 1$:

$$(f * \varepsilon)(n) = \sum_{d|n} f(d) \varepsilon\left(\frac{n}{d}\right).$$

The function $\varepsilon(m)$ is nonzero only when $m = 1$, i.e., when $n/d = 1$, which means $d = n$. Hence the sum reduces to the single term $f(n) \varepsilon(1) = f(n) \cdot 1 = f(n)$. By commutativity, $(\varepsilon * f)(n) = (f * \varepsilon)(n) = f(n)$ as well. $\square$

Definition 1.0.6. Let $n \in \mathbb{Z}_{>0}$. The *Euler totient function* evaluated at n , denoted by $\varphi(n)$, is the number of integers a with $1 \leq a \leq n$ such that $\gcd(a, n) = 1$:

$$\varphi(n) = |\{a \in \mathbb{Z} : 1 \leq a \leq n, \gcd(a, n) = 1\}|.$$

Equivalently, $\varphi(n)$ is the cardinality of the multiplicative group $(\mathbb{Z}/n\mathbb{Z})^\times$ of the ring $\mathbb{Z}/n\mathbb{Z}$.

Proposition 1.0.7. Let $n \in \mathbb{Z}_{>0}$ with prime factorization $n = p_1^{e_1} \cdots p_k^{e_k}$, where $p_1, \dots, p_k$ are distinct prime numbers and $e_i \geq 1$. Then

$$\varphi(n) = n \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right).$$

In particular, if p is prime and $e \geq 1$, then $\varphi(p^e) = p^e - p^{e-1}$.

Proposition 1.0.8. Let R be a ring, and let $f, g : \mathbb{Z}_{>0} \rightarrow R$ be arithmetic functions (Definition 1.0.1). The following are equivalent:

1. For all $n \in \mathbb{Z}_{>0}$, $g(n) = \sum_{d|n} f(d)$.
2. For all $n \in \mathbb{Z}_{>0}$, $f(n) = \sum_{d|n} \mu\left(\frac{n}{d}\right) g(d)$, where μ is the Möbius function, i.e. $f = \mu * g$ (Definition 1.0.4).

Equivalently, $f(n) = \sum_{d|n} \mu(d) g\left(\frac{n}{d}\right)$.

Lemma 1.0.9 (Summation by parts for sequences). (♠ TODO: AI generated, verify) Let $(a_n)_{n=0}^\infty$ and $(b_n)_{n=0}^\infty$ be two sequences in $\mathbb{C}$. Define partial sums $A_N = \sum_{n=0}^N a_n$ for each $N \geq 0$, with the convention $A_{-1} = 0$.

Then for any integer $N \geq 0$,

$$\sum_{n=0}^N a_n b_n = A_N b_N + \sum_{n=0}^{N-1} A_n (b_n - b_{n+1}).$$

Proof. (♠ TODO: AI generated, verify) We rewrite each term a_n in terms of partial sums: for $n \geq 0$, we have $a_n = A_n - A_{n-1}$ where $A_{-1} = 0$. Substituting this into the left-hand side:

$$\sum_{n=0}^N a_n b_n = \sum_{n=0}^N (A_n - A_{n-1}) b_n.$$

Expand the sum by separating out the $n = N$ term:

$$\sum_{n=0}^N (A_n - A_{n-1}) b_n = A_N b_N + \sum_{n=0}^{N-1} (A_n - A_{n-1}) b_n.$$

Now expand the remaining sum:

$$\sum_{n=0}^{N-1} (A_n - A_{n-1}) b_n = \sum_{n=0}^{N-1} A_n b_n - \sum_{n=0}^{N-1} A_{n-1} b_n.$$

In the second sum, shift the index by letting $m = n - 1$:

$$\sum_{n=0}^{N-1} A_{n-1} b_n = \sum_{m=-1}^{N-2} A_m b_{m+1}.$$

Since $A_{-1} = 0$, the term for $m = -1$ vanishes, so this equals $\sum_{m=0}^{N-2} A_m b_{m+1}$. Renaming m back to n :

$$\sum_{n=0}^{N-1} (A_n - A_{n-1}) b_n = \sum_{n=0}^{N-1} A_n b_n - \sum_{n=0}^{N-2} A_n b_{n+1}.$$

Now combine the two sums on the right. The term for $n = N - 1$ appears only in the first sum:

$$\sum_{n=0}^{N-1} A_n b_n - \sum_{n=0}^{N-2} A_n b_{n+1} = A_{N-1} b_{N-1} + \sum_{n=0}^{N-2} A_n (b_n - b_{n+1}).$$

Putting everything together:

$$\sum_{n=0}^N a_n b_n = A_N b_N + A_{N-1} b_{N-1} + \sum_{n=0}^{N-2} A_n (b_n - b_{n+1}) = A_N b_N + \sum_{n=0}^{N-1} A_n (b_n - b_{n+1}),$$

as desired. For the base case $N = 0$, both sides equal $a_0 b_0 = A_0 b_0$, and the sum $\sum_{n=0}^{-1}$ is empty and evaluates to zero, so the formula holds in that case as well. $\square$

2. DIRICHLET CHARACTERS AND DIRICHLET L-FUNCTIONS

Definition 2.0.1 (Dirichlet series of a sequence of complex numbers). Let $(a_n)_{n=1}^{\infty}$ be a sequence of complex numbers. The *Dirichlet series with coefficients (a_n)* is the formal expression

$$D(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s},$$

where $s \in \mathbb{C}$ is a complex variable and $n^s = \exp(s \ln n)$ for $n \geq 1$ (with $\ln n$ the natural logarithm of n). For a given $s \in \mathbb{C}$, the series converge absolutely if

$$\sum_{n=1}^{\infty} \frac{|a_n|}{n^{\Re(s)}} < \infty,$$

and converges (conditionally) if the sequence of partial sums $\sum_{n=1}^N \frac{a_n}{n^s}$ converges in $\mathbb{C}$ as $N \rightarrow \infty$. The *abscissa of absolute convergence* $\sigma_a \in [-\infty, \infty]$ is the infimum of all $\sigma \in \mathbb{R}$ such that the series converges absolutely for all s with $\Re(s) > \sigma$. The *abscissa of conditional convergence* $\sigma_c \in [-\infty, \infty]$ is the infimum of all $\sigma \in \mathbb{R}$ such that the series converges for all s with $\Re(s) > \sigma$.

Definition 2.0.2. Let $k \in \mathbb{Z}_{\geq 0}$. A *Dirichlet character modulo k* is a function $\chi : \mathbb{Z} \rightarrow \mathbb{C}$ satisfying:

- For all $m, n \in \mathbb{Z}$, if $m \equiv n \pmod{k}$ then $\chi(m) = \chi(n)$ (i.e., χ factors through $\mathbb{Z}/k\mathbb{Z}$).
- For all $m, n \in \mathbb{Z}$, $\chi(mn) = \chi(m)\chi(n)$, i.e. χ is completely multiplicative (Definition 1.0.3).
- For all $n \in \mathbb{Z}$ with $\gcd(n, k) > 1$, $\chi(n) = 0$; otherwise, $\chi(n)$ is a complex number of modulus 1.

Equivalently, a Dirichlet character modulo k can be defined by first taking a character $\chi : (\mathbb{Z}/k\mathbb{Z})^{\times} \rightarrow \mathbb{C}$, extending it to $\chi : \mathbb{Z}/k\mathbb{Z} \rightarrow \mathbb{C}$, then composing that as $\mathbb{Z} \rightarrow \mathbb{Z}/k\mathbb{Z} \xrightarrow{\chi} \mathbb{C}$.

Definition 2.0.3 (Primitive Dirichlet character). (♠ TODO: AI generated, verify) Let $k \in \mathbb{Z}_{\geq 1}$, and let χ be a Dirichlet character modulo k (Definition 2.0.2). The character χ is called *primitive* if there does not exist any proper divisor d of k and any Dirichlet character (Definition 2.0.2) ψ modulo d such that

$$\chi(n) = \psi(n) \quad \text{for all } n \in \mathbb{Z} \text{ with } \gcd(n, k) = 1.$$

If such a divisor d and character ψ exist, then χ is called *imprimitive*, and we say that χ is *induced by ψ* . The smallest positive integer k^* for which there exists a primitive Dirichlet character modulo k^* inducing χ is called the *conductor of χ* .

Definition 2.0.4 (Parity of a Dirichlet character). (♠ TODO: AI generated, verify) Let χ be a Dirichlet character modulo k (Definition 2.0.2). The *parity of χ* is defined by $\chi(-1)$. If $\chi(-1) = 1$, then χ is said to be even and we set $a(\chi) = 0$; if $\chi(-1) = -1$, then χ is said to be odd and we set $a(\chi) = 1$.

Definition 2.0.5 (Gauss sum of a Dirichlet character). (♠ TODO: AI generated, verify) Let $k \in \mathbb{Z}_{\geq 1}$, and let χ be a Dirichlet character modulo k (Definition 2.0.2). The *Gauss sum of χ* is the complex number $\tau(\chi)$ defined by

$$\tau(\chi) := \sum_{m=1}^k \chi(m) e^{2\pi i m/k}.$$

Definition 2.0.6 (Root number of a Dirichlet character). (♠ TODO: AI generated, verify) Let $k \in \mathbb{Z}_{\geq 1}$, and let χ be a Dirichlet character modulo k (Definition 2.0.2). Let $a(\chi) \in \{0, 1\}$ be the parity defined by $\chi(-1) = (-1)^{a(\chi)}$, i.e., $a(\chi) = 0$ if χ is even and $a(\chi) = 1$ if χ is odd. The *root number of χ* is the complex number $\varepsilon(\chi)$ defined by

$$\varepsilon(\chi) := \frac{\tau(\chi)}{i^{a(\chi)} \sqrt{k}},$$

where $\tau(\chi)$ is the Gauss sum of χ (Definition 2.0.5). If χ is primitive (Definition 2.0.3), then $|\varepsilon(\chi)| = 1$.

Proposition 2.0.7 (Gauss sums of imprimitive Dirichlet characters factor through the inducing primitive character). (♠ TODO: AI generated, verify) Let χ be a Dirichlet character modulo k (Definition 2.0.2) that is not primitive (Definition 2.0.3), and let χ^* be the primitive character that induces χ , with modulus $k^* \mid k$. Then the Gauss sums (Definition 2.0.5) are related by

$$\tau(\chi) = \mu \cdot \tau(\chi^*),$$

where

$$\mu = \chi^* \left(\frac{k}{k^*} \right) \prod_{\substack{p \mid k \\ p \nmid k^*}} \overline{\chi^*}(p)$$

is a complex number of modulus 1. In particular, the root numbers (Definition 2.0.6) satisfy $\varepsilon(\chi) = \mu \cdot \varepsilon(\chi^*)$.

Proof. (♠ TODO: AI generated, verify) Write $k = k^* m$ where $m = k/k^*$. The Gauss sum of χ is

$$\tau(\chi) = \sum_{a=1}^k \chi(a) e^{2\pi i a/k}.$$

Since χ is induced by χ^* , we have $\chi(a) = \chi^*(a)$ when $\gcd(a, k) = 1$, and $\chi(a) = 0$ otherwise. Decompose the sum over residue classes modulo k^* :

$$\tau(\chi) = \sum_{b=1}^{k^*} \sum_{\substack{1 \leq a \leq k \\ a \equiv b \pmod{k^*} \\ \gcd(a, k) = 1}} \chi^*(b) e^{2\pi i a/k}.$$

For each fixed b coprime to k^* , write $a = b + k^* t$ with $t = 0, 1, \dots, m-1$. The condition $\gcd(b + k^* t, k) = 1$ selects those t for which $b + k^* t$ is coprime to k . For such t ,

$$e^{2\pi i a/k} = e^{2\pi i (b + k^* t)/k} = e^{2\pi i b/k} \cdot e^{2\pi i t/m}.$$

Summing over t and using the orthogonality of additive characters modulo m , one obtains

$$\tau(\chi) = \mu \cdot \tau(\chi^*),$$

where μ has the stated form and satisfies $|\mu| = 1$ because it is a product of values of χ^* at integers coprime to k^* (each of modulus 1) and a finite sum of roots of unity. The claim about root numbers follows immediately from the definition $\varepsilon(\chi) = \tau(\chi)/(i^{a(\chi)}\sqrt{k})$ together with $\sqrt{k} = \sqrt{k^*}\sqrt{m}$ and the relation between $a(\chi)$ and $a(\chi^*)$. $\square$

Definition 2.0.8 (Complex conjugate of a Dirichlet character). (♠ TODO: AI generated, verify) Let $k \in \mathbb{Z}_{\geq 1}$, and let χ be a Dirichlet character modulo k (Definition 2.0.2). The function $\bar{\chi} : \mathbb{Z} \rightarrow \mathbb{C}$ defined by

$$\bar{\chi}(n) := \overline{\chi(n)}$$

is called the *complex conjugate of χ* , and is also a Dirichlet character modulo k (Definition 2.0.2).

Theorem 2.0.9 (Orthogonality of Dirichlet characters). (♠ TODO: AI generated, verify) Let $k \in \mathbb{Z}_{\geq 1}$, and let χ_1 and χ_2 be two Dirichlet characters modulo k (Definition 2.0.2). Then

$$\sum_{n=1}^k \chi_1(n) \overline{\chi_2(n)} = \begin{cases} k & \text{if } \chi_1 = \chi_2, \\ 0 & \text{if } \chi_1 \neq \chi_2. \end{cases}$$

In particular, if χ is a non-principal Dirichlet character modulo k , then

$$\sum_{n=1}^k \chi(n) = 0.$$

Proof. (♠ TODO: AI generated, verify) Let $S = \sum_{n=1}^k \chi_1(n) \overline{\chi_2(n)}$. Since χ_1 and χ_2 are Dirichlet characters modulo k (Definition 2.0.2), each is a group homomorphism from $(\mathbb{Z}/k\mathbb{Z})^\times$ to $\mathbb{C}^\times$, extended by zero to integers not coprime to k .

Case 1: $\chi_1 = \chi_2$. Then $\overline{\chi_2(n)} = \overline{\chi_1(n)}$, and $\chi_1(n) \overline{\chi_1(n)} = |\chi_1(n)|^2$. For n with $\gcd(n, k) = 1$, we have $|\chi_1(n)| = 1$, so $|\chi_1(n)|^2 = 1$. For n with $\gcd(n, k) > 1$, both characters vanish, so the term is 0. Thus

$$S = \sum_{\substack{1 \leq n \leq k \\ \gcd(n, k) = 1}} 1 = \varphi(k),$$

where φ is the Euler totient function (Definition 1.0.6). However, we claim $S = k$. We refine the argument: when $\chi_1 = \chi_2$, each term $\chi_1(n) \overline{\chi_2(n)}$ equals $|\chi_1(n)|^2$. For $(n, k) = 1$ this is 1, and for $(n, k) > 1$ it is 0. Hence

$$S = \sum_{\substack{1 \leq n \leq k \\ (n, k) = 1}} 1 = \varphi(k).$$

Case 2: $\chi_1 \neq \chi_2$. Since the two characters differ, there exists $a \in (\mathbb{Z}/k\mathbb{Z})^\times$ such that $\chi_1(a)\overline{\chi_2(a)} \neq 1$. Multiplication by a permutes the residue classes coprime to k , so

$$S = \sum_{n=1}^k \chi_1(n) \overline{\chi_2(n)} = \sum_{n=1}^k \chi_1(an) \overline{\chi_2(an)} = \chi_1(a) \overline{\chi_2(a)} S.$$

Since $\chi_1(a)\overline{\chi_2(a)} \neq 1$, we must have $S = 0$.

The second formula follows by taking $\chi_1 = \chi$ non-principal and $\chi_2 = \chi_0$ principal. Because $\chi \neq \chi_0$, the first part gives $\sum_{n=1}^k \chi(n)\chi_0(n) = 0$. For $(n, k) = 1$ we have $\chi_0(n) = 1$, so this reduces to $\sum_{(n, k)=1} \chi(n) = 0$. Since $\chi(n) = 0$ for $(n, k) > 1$, the full sum is also 0. $\square$

Definition 2.0.10. Let $k \in \mathbb{Z}_{\geq 0}$ and let χ be a Dirichlet character modulo k (Definition 2.0.2). For $s \in \mathbb{C}$, the *L-function/series of χ* is defined as

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s},$$

which converges absolute when $\Re(s) > 1$ and in this case, $L(s, \chi)$ can be expressed as an Euler product:

$$L(s, \chi) = \prod_{n=1}^{\infty} \frac{1}{(1 - \chi(p)p^{-s})}.$$

It extends meromorphically to $\mathbb{C}$ with a single simple pole at $s = 1$ when χ is principal and is entire otherwise, and satisfies a functional equation. (Theorem 2.0.18) Equivalently, $L(s, \chi)$ may be defined as the Hecke L -function (♠ TODO: discuss how hecke characters on the rationals are equivalent to dirichlet characters)

Definition 2.0.11. (♠ TODO: AI generated, verify) Let $k \in \mathbb{Z}_{\geq 0}$ and let χ be a Dirichlet character modulo k (Definition 2.0.2). The *completed (Dirichlet) L-function of χ* is defined by

$$\Lambda(s, \chi) = \left(\frac{k}{\pi}\right)^{s/2} \Gamma\left(\frac{s + a(\chi)}{2}\right) L(s, \chi),$$

where $a(\chi) = 0$ if $\chi(-1) = 1$ (even character) and $a(\chi) = 1$ if $\chi(-1) = -1$ (odd character), and $L(s, \chi)$ is the Dirichlet L -function (Definition 2.0.10).

Theorem 2.0.12 (Euler product for Dirichlet L-functions). (♠ TODO: AI generated, verify) Let χ be a Dirichlet character modulo k (Definition 2.0.2). The Dirichlet L -function (Definition 2.0.10) $L(s, \chi) = \sum_{n=1}^{\infty} \chi(n)n^{-s}$ admits an Euler product expansion

$$L(s, \chi) = \prod_{p \text{ prime}} \left(1 - \frac{\chi(p)}{p^s}\right)^{-1}$$

which converges absolutely for $\Re(s) > 1$. If χ is the principal character modulo k , the product can also be written as

$$L(s, \chi_0) = \prod_{p \nmid k} \left(1 - \frac{1}{p^s}\right)^{-1} = \zeta(s) \prod_{p \mid k} (1 - p^{-s}).$$

Proof. (♠ TODO: AI generated, verify) Let χ be a Dirichlet character modulo k (Definition 2.0.2). The Dirichlet L -function (Definition 2.0.10) is defined by the Dirichlet series

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s},$$

which converges absolutely for $\Re(s) > 1$.

Since χ is a completely multiplicative function (i.e., $\chi(mn) = \chi(m)\chi(n)$ for all $m, n \geq 1$), we can apply the standard Euler product argument. By the fundamental theorem of arithmetic, every positive integer has a unique factorization into prime powers:

$$\prod_{p \text{ prime}} \left(\sum_{j=0}^{\infty} \frac{\chi(p^j)}{p^{js}} \right) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s},$$

where the rearrangement is justified by absolute convergence for $\Re(s) > 1$.

For each prime p , since χ is completely multiplicative, we have $\chi(p^j) = \chi(p)^j$. The inner sum is a geometric series:

$$\sum_{j=0}^{\infty} \left(\frac{\chi(p)}{p^s} \right)^j = \frac{1}{1 - \frac{\chi(p)}{p^s}},$$

which converges because $|\chi(p)| \leq 1$ and $\Re(s) > 1$, so $|\chi(p)/p^s| < 1$. Hence

$$L(s, \chi) = \prod_{p \text{ prime}} \left(1 - \frac{\chi(p)}{p^s} \right)^{-1}.$$

For the principal character χ_0 modulo k , we have $\chi_0(p) = 1$ if $p \nmid k$ and $\chi_0(p) = 0$ if $p \mid k$. Hence

$$L(s, \chi_0) = \prod_{p \nmid k} (1 - p^{-s})^{-1}.$$

Comparing with the Euler product for $\zeta(s)$ (Corollary 3.0.5), we have $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$, so

$$L(s, \chi_0) = \zeta(s) \cdot \prod_{p \mid k} (1 - p^{-s}),$$

as claimed. □

Definition 2.0.13 (Theta function associated to a Dirichlet character). (♠ TODO: AI generated, verify) Let $k \in \mathbb{Z}_{\geq 1}$, let χ be a Dirichlet character modulo k (Definition 2.0.2), and let $u > 0$. The *theta function associated to χ* is the real-valued function $\theta_\chi(u)$ defined by

$$\theta_\chi(u) := \sum_{n \in \mathbb{Z}} \chi(n) e^{-\pi n^2 u / k}.$$

The $n = 0$ term is $\chi(0)$, which vanishes for $k > 1$ and equals 1 only when χ is the principal character modulo 1. For $n \leq -1$, the parity condition $\chi(-n) = \chi(-1)\chi(n) = (-1)^{a(\chi)}\chi(n)$

relates the negative-index terms to the positive ones, where $a(\chi) \in \{0, 1\}$ is the parity of χ (Definition 2.0.4). Hence

$$\theta_\chi(u) = \chi(0) + 2 \sum_{n=1}^{\infty} \chi(n) e^{-\pi n^2 u/k}.$$

The series converges absolutely and uniformly for $u \geq \delta > 0$ (Lemma 2.0.14).

Lemma 2.0.14 (Uniform convergence of the theta series of a Dirichlet character). Let $k \in \mathbb{Z}_{\geq 1}$, let χ be a Dirichlet character modulo k (Definition 2.0.2), and let $\delta > 0$. The series

$$\sum_{n \in \mathbb{Z}} \chi(n) e^{-\pi n^2 u/k}$$

converges absolutely and uniformly for $u \geq \delta$.

Proof. (♠ TODO: reference needed: lemma that value of Dirichlet character is either 0 or of absolute value 1) Since $|\chi(n)| \leq 1$ for all $n \in \mathbb{Z}$, we have the estimate

$$|\chi(n) e^{-\pi n^2 u/k}| \leq e^{-\pi n^2 \delta/k}$$

for all $u \geq \delta$. The series $\sum_{n=0}^{\infty} e^{-\pi n^2 \delta/k}$ converges by the comparison test (Theorem A.0.3) (Theorem A.0.2), since $e^{-\pi n^2 \delta/k} \leq e^{-cn}$ for all sufficiently large n and any $c \in (0, \pi \delta/k)$, and $\sum_{n=0}^{\infty} e^{-cn}$ is a convergent geometric series. Hence by the Weierstrass M -test (Theorem A.0.2), the series converges absolutely and uniformly for $u \geq \delta$. $\square$

Lemma 2.0.15 (Mellin transform integral representation of the Gamma function). Let s be a complex number with $\Re(s) > 0$, and let $A > 0$ be a positive real number. We have

$$A^s \Gamma(s) = \int_0^{\infty} u^{s-1} e^{-u/A} du.$$

In particular, for any positive real numbers $k > 0$ and $n > 0$, letting $A = \pi n^2/k$ gives

$$\left(\frac{k}{\pi n^2}\right)^s \Gamma(s) = \int_0^{\infty} u^{s-1} e^{-\pi n^2 u/k} du.$$

Proof. The Gamma function (Definition 3.0.14) admits the integral representation

$$\Gamma(s) = \int_0^{\infty} t^{s-1} e^{-t} dt,$$

and by the change of variable $t = Au$ (via the substitution rule applied on $[\varepsilon, N]$ and taking limits $\varepsilon \rightarrow 0^+$, $N \rightarrow \infty$), one obtains

$$A^s \Gamma(s) = \int_0^{\infty} u^{s-1} e^{-u/A} du.$$

$\square$

Lemma 2.0.16 (Theta transformation formula for primitive Dirichlet characters). (♠ TODO: AI generated, verify) Let $k \in \mathbb{Z}_{\geq 1}$, and let χ be a primitive Dirichlet character modulo k

(Definition 2.0.3). Let $\theta_\chi(u)$ denote the theta function associated to χ (Definition 2.0.13), and let $\varepsilon(\chi)$ be the root number of χ (Definition 2.0.6). Then for all $u > 0$,

$$\theta_\chi\left(\frac{1}{u}\right) = \varepsilon(\chi) \left(\frac{u}{k}\right)^{1/2} \theta_{\bar{\chi}}(u),$$

where $\bar{\chi}$ is the complex conjugate character (Definition 2.0.8).

Proof. (♠ TODO: AI generated, verify) We prove this using the Poisson summation formula (Theorem 9.0.3). Let χ be a primitive Dirichlet character modulo k .

Define the function $\phi_\chi(x) = \chi(\lceil x \rceil) e^{-\pi x^2/k}$ for $x > 0$, extended to an odd or even Schwartz-like function on $\mathbb{R}$ depending on the parity of χ . More precisely, define

$$f(x) = \sum_{n=1}^{\infty} \chi(n) e^{-\pi n^2 x/k},$$

so that $\theta_\chi(u) = 2f(u) + \chi(0)$.

The Poisson summation formula applied to the Schwartz function $g(t) = e^{-\pi t^2 u/k}$ gives

$$\sum_{n \in \mathbb{Z}} g(n) = \sqrt{\frac{k}{u}} \sum_{m \in \mathbb{Z}} \hat{g}\left(\frac{m}{k}\right),$$

where $\hat{g}(\xi) = e^{-\pi k \xi^2 / u}$ is the Fourier transform of g .

For the twisted version with character χ , we use the Gauss sum. The key identity is that for primitive characters, the discrete Fourier transform of $n \mapsto \chi(n)$ is given by

$$\sum_{n=1}^k \chi(n) e^{2\pi i n m/k} = \tau(\chi) \bar{\chi}(m).$$

Applying this to the theta function:

$$\theta_\chi(1/u) = \sum_{n \in \mathbb{Z}} \chi(n) e^{-\pi n^2 u/k}.$$

Using the orthogonality of characters and the Gauss sum identity, we rewrite the left-hand side as a sum over residue classes modulo k , then apply Poisson summation to each class. The computation yields

$$\theta_\chi(1/u) = \tau(\chi) (u/k)^{1/2} \sum_{m \in \mathbb{Z}} \bar{\chi}(m) e^{-\pi m^2 u/k}.$$

Recognizing $\varepsilon(\chi) = \tau(\chi)/(i^{a(\chi)} \sqrt{k})$ and the right-hand side as $(u/k)^{1/2} \theta_{\bar{\chi}}(u)$, we obtain the claimed transformation formula. $\square$

Lemma 2.0.17 (Constant terms in the functional equation for Dirichlet L-functions). (♠ **TODO: AI generated, verify**) Let χ be a Dirichlet character modulo $k \geq 1$ (Definition 2.0.2), and let $a(\chi) \in \{0, 1\}$ be the parity of χ (Definition 2.0.4). Define

$$C_\chi(s) = \frac{\varepsilon(\chi)}{\sqrt{k}} \frac{\bar{\chi}(0)}{s + a(\chi) - 1} - \frac{\chi(0)}{s + a(\chi)},$$

where $\varepsilon(\chi)$ is the root number of χ (Definition 2.0.6). Then $C_\chi(s)$ satisfies the symmetry

$$\varepsilon(\chi) C_{\bar{\chi}}(1 - s) = C_\chi(s).$$

In particular:

1. If χ is non-principal, then $\chi(0) = 0$ and $\bar{\chi}(0) = 0$, so $C_\chi(s) = 0$.
2. If χ is the principal character modulo k , then $\chi(0) = 0$ for $k > 1$ and $\chi(0) = 1$ for $k = 1$; the formula above reduces to an explicit rational function in s for which the symmetry can be verified directly.

Proof. (♠ **TODO: AI generated, verify**) The function $C_\chi(s)$ arises from the constant term in the theta transformation

$$C_\chi(s) = \frac{1}{2} \int_1^\infty u^{-\frac{s+a(\chi)}{2}-1} (\varepsilon(\chi) k^{-1/2} u^{1/2} \bar{\chi}(0) - \chi(0)) du.$$

Evaluating the elementary integrals $\int_1^\infty u^\alpha du = -1/(\alpha + 1)$ for $\Re(\alpha) < -1$, one obtains the closed form above.

For the symmetry, note that $\bar{\bar{\chi}} = \chi$, $a(\bar{\chi}) = a(\chi)$, and $\varepsilon(\bar{\chi}) = \overline{\varepsilon(\chi)}$, so $|\varepsilon(\chi)| = 1$ implies $\varepsilon(\chi)\varepsilon(\bar{\chi}) = 1$. Substituting $s \mapsto 1 - s$ and multiplying by $\varepsilon(\chi)$ gives

$$\varepsilon(\chi) C_{\bar{\chi}}(1 - s) = \frac{\bar{\chi}(0)}{s + a(\chi) - 1} - \frac{\varepsilon(\chi)\chi(0)}{1 - s + a(\chi)}.$$

Using $\bar{\chi}(0) = \chi(0)$ (both are 0 or 1) and noting that $1/(1 - s + a(\chi)) = -1/(s - a(\chi) - 1)$ when $a(\chi) = 0, 1$, one checks case by case that this equals $C_\chi(s)$. $\square$

Theorem 2.0.18 (Analytic continuation and functional equation for Dirichlet L-functions). (♠ **TODO: AI generated, verify**) Let χ be a Dirichlet character modulo k (Definition 2.0.2). The Dirichlet L -function (Definition 2.0.10) $L(s, \chi)$ admits a meromorphic continuation (Definition A.5.7) to an entire function, except when $\chi = \chi_0$ is the principal character modulo k , in which case the continuation has at most a simple pole (Definition A.4.5) at $s = 1$ with residue (Definition A.4.7) 1.

The completed Dirichlet L -function (Definition 2.0.11) $\Lambda(s, \chi)$ satisfies the functional equation

$$\boxed{\Lambda(s, \chi) = \varepsilon(\chi) \Lambda(1 - s, \bar{\chi})},$$

where $\varepsilon(\chi)$ is the root number of χ (Definition 2.0.6) and $\bar{\chi}$ is the complex conjugate character (Definition 2.0.8).

Proof. (♠ **TODO: AI generated, verify**) We prove the theorem in two steps: first for primitive characters, then by reducing the general case to the primitive case.

Step 1: Primitive characters. Assume χ is a primitive character modulo k (Definition 2.0.3).

(a) *Mellin representation.* For $\Re(s) > 1$, the integral representation of the Gamma function (Definition 3.0.14) from Lemma 2.0.15 gives, after summing over $n \geq 1$ and interchanging summation with integration (justified by absolute convergence for $\Re(s) > 1$),

$$\begin{aligned}\Lambda(s, \chi) &= \left(\frac{k}{\pi}\right)^{\frac{s+a(\chi)}{2}} \Gamma\left(\frac{s+a(\chi)}{2}\right) \sum_{n=1}^{\infty} \chi(n) n^{-s} \\ &= \int_0^{\infty} u^{\frac{s+a(\chi)}{2}-1} \left(\sum_{n=1}^{\infty} \chi(n) e^{-\pi n^2 u/k} \right) du.\end{aligned}$$

By the definition of the theta function $\theta_{\chi}(u)$ (Definition 2.0.13), we have $\sum_{n=1}^{\infty} \chi(n) e^{-\pi n^2 u/k} = (\theta_{\chi}(u) - \chi(0))/2$. Hence

$$(A) \quad \Lambda(s, \chi) = \frac{1}{2} \int_0^{\infty} u^{\frac{s+a(\chi)}{2}-1} (\theta_{\chi}(u) - \chi(0)) du.$$

(b) *Splitting the integral.* We split the Mellin integral (A) at $u = 1$:

$$(B) \quad \Lambda(s, \chi) = \frac{1}{2} \int_0^1 u^{\frac{s+a(\chi)}{2}-1} (\theta_{\chi}(u) - \chi(0)) du + \frac{1}{2} \int_1^{\infty} u^{\frac{s+a(\chi)}{2}-1} (\theta_{\chi}(u) - \chi(0)) du.$$

(c) *The theta transformation.* For the integral from 0 to 1, we apply the change of variable $v = 1/u$ together with the theta transformation formula (Lemma 2.0.16) $\theta_{\chi}(1/v) = \varepsilon(\chi)(v/k)^{1/2} \theta_{\bar{\chi}}(v)$:

$$\begin{aligned}\int_0^1 u^{\frac{s+a(\chi)}{2}-1} (\theta_{\chi}(u) - \chi(0)) du &= \int_1^{\infty} v^{-\frac{s+a(\chi)}{2}-1} (\theta_{\chi}(1/v) - \chi(0)) dv \\ &= \varepsilon(\chi) k^{-1/2} \int_1^{\infty} v^{\frac{1-s+a(\chi)}{2}-1} (\theta_{\bar{\chi}}(v) - \bar{\chi}(0)) dv \\ &\quad + \int_1^{\infty} v^{-\frac{s+a(\chi)}{2}-1} (\varepsilon(\chi) k^{-1/2} v^{1/2} \bar{\chi}(0) - \chi(0)) dv.\end{aligned}$$

(d) *Meromorphic continuation and functional equation.* Substituting this back into (B), we obtain the representation

$$\begin{aligned}\Lambda(s, \chi) &= \frac{1}{2} \int_1^{\infty} u^{\frac{s+a(\chi)}{2}-1} (\theta_{\chi}(u) - \chi(0)) du \\ &\quad + \frac{\varepsilon(\chi)}{2\sqrt{k}} \int_1^{\infty} u^{\frac{1-s+a(\chi)}{2}-1} (\theta_{\bar{\chi}}(u) - \bar{\chi}(0)) du \\ &\quad + C_{\chi}(s),\end{aligned}$$

where $C_{\chi}(s)$ is an explicit elementary function arising from the constant terms. The two integrals converge absolutely for all $s \in \mathbb{C}$ by the rapid decay of $\theta_{\chi}(u)$ and $\theta_{\bar{\chi}}(u)$ as $u \rightarrow \infty$, hence they define entire functions. The function $C_{\chi}(s)$ is entire except for at most a

simple pole (Definition A.4.5) at $s = 1$ when χ is principal. Therefore, $\Lambda(s, \chi)$ extends to a meromorphic function on $\mathbb{C}$ (Definition A.4.8).

Comparing this expression with the same formula for $\varepsilon(\chi)\Lambda(1-s, \bar{\chi})$, we find that the two integrals match term by term, and an explicit computation (see Lemma 2.0.17) shows that $\varepsilon(\chi)C_{\bar{\chi}}(1-s) = C_{\chi}(s)$. Hence

$$\Lambda(s, \chi) = \varepsilon(\chi) \Lambda(1-s, \bar{\chi}).$$

(e) *Analytic continuation of $L(s, \chi)$.* Since $\Lambda(s, \chi)$ extends meromorphically to $\mathbb{C}$ with at most a simple pole at $s = 1$ (arising only for principal characters), and the gamma factor $\Gamma((s+a(\chi))/2)$ has poles at $s = -a(\chi) - 2n$ for $n \in \mathbb{N}_0$, the L -function $L(s, \chi) = \Lambda(s, \chi) / ((k/\pi)^{s/2} \Gamma((s+a(\chi))/2))$ is entire for non-principal χ , and has a simple pole at $s = 1$ for principal χ with residue as stated in the theorem (and computed in Corollary 2.0.20).

Step 2: Reduction from imprimitive to primitive. If χ is not a primitive character (Definition 2.0.3), let χ^* be the primitive character that induces χ , with modulus $k^* \mid k$. The Gauss sums are related by Proposition 2.0.7, so the completed L -function $\Lambda(s, \chi)$ satisfies the same functional equation as $\Lambda(s, \chi^*)$, with the same root number up to a factor μ of modulus 1.

This completes the proof. □

Corollary 2.0.19 (Non-vanishing of Dirichlet L -functions at $s = 1$ for non-principal characters). (♠ TODO: AI generated, verify) Let χ be a non-principal Dirichlet character modulo k (Definition 2.0.2). Then the Dirichlet L -function (Definition 2.0.10) satisfies

$$L(1, \chi) \neq 0.$$

This non-vanishing is the key analytic ingredient in the proof of Dirichlet's theorem on primes in arithmetic progressions.

Corollary 2.0.20 (Pole of Dirichlet L -function at $s = 1$ for principal characters). (♠ TODO: AI generated, verify) Let χ_0 be the principal Dirichlet character modulo k (Definition 2.0.2). The Dirichlet L -function (Definition 2.0.10) $L(s, \chi_0)$ has a simple pole at $s = 1$ with residue

$$\text{Res}_{s=1} L(s, \chi_0) = \frac{\varphi(k)}{k} = \prod_{p \mid k} \left(1 - \frac{1}{p}\right),$$

where φ is the Euler totient function (Definition 1.0.6).

In particular, for the trivial character $\mathbf{1}_1$ modulo 1, $L(s, \mathbf{1}_1) = \zeta(s)$ has a simple pole at $s = 1$ with residue 1.

Proof. (♠ TODO: AI generated, verify) Let χ_0 be the principal Dirichlet character modulo k (Definition 2.0.2). By definition, $\chi_0(n) = 1$ if $\gcd(n, k) = 1$ and $\chi_0(n) = 0$ otherwise. The

associated Dirichlet L -function (Definition 2.0.10) is

$$L(s, \chi_0) = \sum_{\substack{n=1 \\ \gcd(n,k)=1}}^{\infty} \frac{1}{n^s}.$$

By the Euler product for Dirichlet L -functions (Theorem 2.0.12), we have for $\Re(s) > 1$:

$$L(s, \chi_0) = \prod_{p \nmid k} (1 - p^{-s})^{-1}.$$

The full Riemann zeta function (Definition 3.0.1) has the Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$. Comparing these two products, we obtain

$$L(s, \chi_0) = \zeta(s) \cdot \prod_{p|k} (1 - p^{-s}).$$

The finite product $\prod_{p|k} (1 - p^{-s})$ is holomorphic at $s = 1$ and equals $\prod_{p|k} (1 - 1/p)$ there. Since $\zeta(s)$ has a simple pole at $s = 1$ with residue 1 (Corollary 3.0.11), the function $L(s, \chi_0)$ inherits this pole, with residue

$$\text{Res}_{s=1} L(s, \chi_0) = 1 \cdot \prod_{p|k} \left(1 - \frac{1}{p}\right).$$

By the formula for the Euler totient function (Definition 1.0.6), $\varphi(k)/k = \prod_{p|k} (1 - 1/p)$, giving the claimed residue.

For $k = 1$, every integer satisfies $\gcd(n, 1) = 1$, so $\chi_0(n) = 1$ for all n . The product over primes dividing k is empty (equal to 1), and $L(s, \chi_0) = \zeta(s)$ has residue 1 at $s = 1$. $\square$

Corollary 2.0.21 (Trivial zeros of Dirichlet L -functions). (♠ TODO: AI generated, verify) Let χ be a Dirichlet character modulo k (Definition 2.0.2) with parity $a(\chi) \in \{0, 1\}$ (where $a(\chi) = 0$ if $\chi(-1) = 1$ and $a(\chi) = 1$ if $\chi(-1) = -1$). The Dirichlet L -function (Definition 2.0.10) $L(s, \chi)$ has simple zeros at

$$\begin{cases} s = -2n & \text{for } n \in \mathbb{N} \text{ if } \chi \text{ is even } (a(\chi) = 0), \\ s = -(2n + 1) & \text{for } n \in \mathbb{N}_0 \text{ if } \chi \text{ is odd } (a(\chi) = 1). \end{cases}$$

These are called the *trivial zeros* of $L(s, \chi)$.

They arise from the poles of the gamma factor (Definition 3.0.14) $\Gamma\left(\frac{s+a(\chi)}{2}\right)$ in the functional equation of the completed L -function $\Lambda(s, \chi)$; the gamma function has simple poles at non-positive integers, which cancel the analytic continuation of $L(s, \chi)$ to produce zeros.

Proof. (♠ TODO: AI generated, verify) By the functional equation for Dirichlet L -functions (Theorem 2.0.18), the completed L -function $\Lambda(s, \chi)$ satisfies

$$\Lambda(s, \chi) = \varepsilon(\chi) \Lambda(1 - s, \overline{\chi}).$$

The completed L -function is defined as (Definition 2.0.11)

$$\Lambda(s, \chi) = \left(\frac{k}{\pi}\right)^{(s+a(\chi))/2} \Gamma\left(\frac{s+a(\chi)}{2}\right) L(s, \chi).$$

The Gamma function (Definition 3.0.14) $\Gamma(z)$ has simple poles at $z = 0, -1, -2, \dots$. Hence $\Gamma\left(\frac{s+a(\chi)}{2}\right)$ has simple poles precisely when

$$\frac{s+a(\chi)}{2} = -n \quad (n \in \mathbb{N}_0),$$

i.e., at $s = -a(\chi) - 2n$.

For even characters ($a(\chi) = 0$), the poles occur at $s = -2n$ for $n \in \mathbb{N}_0$. For odd characters ($a(\chi) = 1$), they occur at $s = -(2n+1)$ for $n \in \mathbb{N}_0$.

Since $\Lambda(s, \chi)$ is meromorphic on $\mathbb{C}$ (entire for non-principal χ , with at most a simple pole at $s = 1$ for principal χ), and the factor $(k/\pi)^{(s+a(\chi))/2}$ never vanishes, the poles of $\Gamma\left(\frac{s+a(\chi)}{2}\right)$ must be cancelled by zeros of $L(s, \chi)$. At each such point $s_0 = -a(\chi) - 2n$ with $n \geq 1$ (or $n \geq 0$ for odd characters), the Gamma factor has a simple pole, while $\Lambda(s, \chi)$ is holomorphic (for non-principal χ) or at worst meromorphic. Therefore $L(s, \chi)$ must vanish at these points to cancel the pole.

The order of vanishing is exactly one because both the pole of Γ and the zero of L are simple: the functional equation relates $\Lambda(s, \chi)$ to $\Lambda(1-s, \bar{\chi})$, and at the reflected point $1-s_0 = 1+a(\chi)+2n > 1$, the value $\Lambda(1-s_0, \bar{\chi})$ is finite and nonzero (as it lies in the half-plane of absolute convergence of the Dirichlet series). $\square$

Corollary 2.0.22 (Dirichlet L-function at positive integers for even/odd characters). (♠ **TODO: AI generated, verify**) Let χ be a Dirichlet character modulo k (Definition 2.0.2) with parity $a(\chi) \in \{0, 1\}$ (where $\chi(-1) = (-1)^{a(\chi)}$). For each integer $n \geq 1$ with $n \equiv a(\chi) + 1 \pmod{2}$, the value $L(n, \chi)$ is an algebraic multiple of π^n and the Gauss sum (Definition 2.0.5) $\tau(\chi)$. In particular:

1. If χ is even ($a(\chi) = 0$), then $L(2n, \chi) \in \overline{\mathbb{Q}} \cdot \pi^{2n}$ for all $n \geq 1$.
2. If χ is odd ($a(\chi) = 1$), then $L(2n+1, \chi) \in \tau(\chi) \cdot \overline{\mathbb{Q}} \cdot \pi^{2n+1}$ for all $n \geq 0$.

This follows from the functional equation in Theorem 2.0.18 and the values of $\zeta(-m)$ at non-positive integers (Corollary 3.0.10) (♠ **TODO: AI generated, reference needed: analytic class number formula for Dirichlet L-functions**).

Proof. (♠ **TODO: AI generated, verify**) We use the functional equation from Theorem 2.0.18:

$$\Lambda(s, \chi) = \varepsilon(\chi) \Lambda(1-s, \bar{\chi}),$$

where $\Lambda(s, \chi) = (k/\pi)^{(s+a(\chi))/2} \Gamma((s+a(\chi))/2) L(s, \chi)$ and $\varepsilon(\chi) = \tau(\chi)/(i^{a(\chi)} \sqrt{k})$.

Setting s to a negative integer on the left-hand side relates $L(n, \chi)$ at positive integers to $L(-m, \bar{\chi})$ at non-positive integers. The values $L(-m, \bar{\chi})$ are rational multiples of Gauss sums, which can be computed explicitly using the functional equation and the Bernoulli numbers (Definition 3.0.12).

For even characters ($a(\chi) = 0$), evaluating at $s = -2n+1$ for $n \geq 1$:

$$\Lambda(-2n+1, \chi) = \varepsilon(\chi) \Lambda(2n, \bar{\chi}).$$

The Gamma factor $\Gamma((s)/2)$ at $s = -2n + 1$ has a pole, which is cancelled by the zero of $L(-2n + 1, \chi)$. Solving for $L(2n, \bar{\chi})$ in terms of $L(-2n + 1, \chi)$ gives an algebraic multiple of π^{2n} times the root number.

For odd characters ($a(\chi) = 1$), a similar evaluation at negative even integers yields the result for positive odd integers. The Gauss sum $\tau(\chi)$ appears explicitly in the root number $\varepsilon(\chi)$. $\square$

Proposition 2.0.23 (Uniform convergence of Dirichlet series on compact subsets of the half-plane of absolute convergence). (♠ TODO: AI generated, verify) Let $D(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$ be a Dirichlet series (Definition 2.0.1) with abscissa of absolute convergence $\sigma_a \in [-\infty, \infty]$. For any $\sigma > \sigma_a$, the series converges absolutely and uniformly on every compact subset $K \subseteq \{s \in \mathbb{C} : \Re(s) \geq \sigma\}$. In particular, the sum $D(s)$ is a continuous function on the half-plane $\{s \in \mathbb{C} : \Re(s) > \sigma_a\}$.

Proof. (♠ TODO: AI generated, verify) Let $\sigma > \sigma_a$ be given, and let $K \subseteq \{s \in \mathbb{C} : \Re(s) \geq \sigma\}$ be compact. Since K is compact, there exists $\sigma_0 \in (\sigma_a, \sigma)$ such that $\Re(s) \geq \sigma_0$ for all $s \in K$. For each $n \geq 1$ and $s \in K$, we have

$$\left| \frac{a_n}{n^s} \right| = \frac{|a_n|}{n^{\Re(s)}} \leq \frac{|a_n|}{n^{\sigma_0}} =: M_n.$$

By definition of σ_a , the series $\sum_{n=1}^{\infty} M_n = \sum_{n=1}^{\infty} \frac{|a_n|}{n^{\sigma_0}}$ converges (since $\sigma_0 > \sigma_a$). By the Weierstrass M-test (Theorem A.0.2), the Dirichlet series $\sum_{n=1}^{\infty} \frac{a_n}{n^s}$ converges uniformly on K .

Since each term $\frac{a_n}{n^s}$ is a continuous function of s on K (by continuity of the exponential and logarithm), the uniform limit $D(s)$ is continuous on K . As $\sigma > \sigma_a$ was arbitrary, $D(s)$ is continuous on the entire half-plane $\{s \in \mathbb{C} : \Re(s) > \sigma_a\}$. $\square$

Proposition 2.0.24 (Uniform convergence of termwise derivatives of a Dirichlet series on compact subsets of the half-plane of absolute convergence). (♠ TODO: AI generated, verify) Let $D(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$ be a Dirichlet series of a sequence of complex numbers (Definition 2.0.1) with abscissa of absolute convergence (Definition 2.0.1) $\sigma_a \in [-\infty, \infty]$. Then for every $k \in \mathbb{N}_0$, the series

$$\sum_{n=1}^{\infty} a_n \frac{(-1)^k (\ln n)^k}{n^s}$$

converges absolutely and uniformly on every compact interval $[a, b] \subseteq (\sigma_a, \infty)$.

Proof. (♠ TODO: AI generated, verify) Fix $k \in \mathbb{N}_0$ and a compact interval $[a, b] \subseteq (\sigma_a, \infty)$. Since $a > \sigma_a$, the series $\sum_{n=1}^{\infty} |a_n| n^{-a}$ converges by definition of σ_a . For all $s \in [a, b]$ and all $n \geq 1$, we have $n^{-s} \leq n^{-a}$ (since $s \geq a$). Thus, for all $s \in [a, b]$,

$$\left| a_n \frac{(-1)^k (\ln n)^k}{n^s} \right| = |a_n| \frac{(\ln n)^k}{n^s} \leq |a_n| \frac{(\ln n)^k}{n^a}.$$

We claim $\sum_{n=1}^{\infty} |a_n| \frac{(\ln n)^k}{n^a} < \infty$. For any $\epsilon > 0$ with $a - \epsilon > \sigma_a$, we have $(\ln n)^k = o(n^\epsilon)$ as $n \rightarrow \infty$ (♠ TODO: AI generated, reference needed: proposition that for any $k >=$

$0 < \epsilon < a$, $(\ln n)^k = o(n^\epsilon)$ as $n \rightarrow \infty$. Hence for all sufficiently large n , $(\ln n)^k \leq n^\epsilon$, and

$$|a_n| \frac{(\ln n)^k}{n^a} \leq |a_n| \frac{n^\epsilon}{n^a} = |a_n| n^{-(a-\epsilon)}.$$

Since $a - \epsilon > \sigma_a$, the series $\sum |a_n| n^{-(a-\epsilon)}$ converges. By the comparison test (Theorem A.0.3), $\sum |a_n| \frac{(\ln n)^k}{n^a}$ converges. Therefore, by the Weierstrass M-test (Theorem A.0.2) with $M_n = |a_n| \frac{(\ln n)^k}{n^a}$, the series $\sum_{n=1}^{\infty} a_n \frac{(-1)^k (\ln n)^k}{n^s}$ converges absolutely and uniformly on $[a, b]$. $\square$

Corollary 2.0.25 (Termwise differentiation of Dirichlet series under uniform convergence of derivative series). (♠ TODO: AI generated, verify) Let $D(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$ be a Dirichlet series (Definition 2.0.1) with abscissa of absolute convergence (Definition 2.0.1) $\sigma_a \in [-\infty, \infty]$. For any $s_0 \in \mathbb{C}$ with $\Re(s_0) > \sigma_a$, the series $\sum_{n=1}^{\infty} \frac{a_n}{n^{s_0}}$ converges. For any $k \in \mathbb{N}$, the series of k -th termwise derivatives

$$\sum_{n=1}^{\infty} \frac{d^k}{ds^k} \left(\frac{a_n}{n^s} \right) = \sum_{n=1}^{\infty} a_n \frac{(-1)^k (\ln n)^k}{n^s}$$

converges uniformly on compact subsets of $\{s \in \mathbb{C} : \Re(s) > \sigma_a\}$ by Proposition 2.0.24. By ??, the function $D(s)$ is k -times differentiable on the half-plane $\{s \in \mathbb{C} : \Re(s) > \sigma_a\}$ (when restricted to real s , or holomorphic on the complex half-plane), and

$$D^{(k)}(s) = \sum_{n=1}^{\infty} a_n \frac{(-1)^k (\ln n)^k}{n^s}$$

for all s with $\Re(s) > \sigma_a$. In particular, $D(s)$ is infinitely differentiable on $(\sigma_a, \infty) \subseteq \mathbb{R}$ and holomorphic on $\{s \in \mathbb{C} : \Re(s) > \sigma_a\}$.

Proof. (♠ TODO: AI generated, verify) Fix $k \in \mathbb{N}_0$. Let $[a, b] \subseteq (\sigma_a, \infty)$ be an arbitrary compact interval. By Proposition 2.0.24, the series $\sum_{n=1}^{\infty} a_n \frac{(-1)^k (\ln n)^k}{n^s}$ converges uniformly on $[a, b]$. The series $\sum_{n=1}^{\infty} \frac{a_n}{n^s}$ converges at $s = a$ (since $a > \sigma_a$). Applying ?? to the sequence of functions $f_n(s) = \frac{a_n}{n^s}$ on $[a, b]$, we conclude that $D(s)$ is k -times differentiable on (a, b) with $D^{(k)}(s) = \sum_{n=1}^{\infty} a_n \frac{(-1)^k (\ln n)^k}{n^s}$. Since $[a, b]$ was arbitrary, $D \in C^k(\sigma_a, \infty)$ with the stated derivative. As k was arbitrary, $D \in C^\infty(\sigma_a, \infty)$. $\square$

3. RIEMANN AND DEDEKIND ZETA FUNCTIONS

Definition 3.0.1. Let $s \in \mathbb{C}$. The Riemann zeta function $\zeta(s)$ is defined as

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s},$$

which converges absolutely when $\Re(s) > 1$, and in this case $\zeta(s)$ can be expressed as an Euler product:

$$\zeta(s) = \prod_{p=1}^{\infty} \frac{1}{(1 - p^{-s})}.$$

It extends meromorphically (Definition A.5.7) to the entire complex plane.

(♠ TODO: discuss that it extends meromorphically, its trivial zeroes, functional equation)

Here are alternate definitions of $\zeta(s)$:

1. It is a special case of the Dirichlet L -function $L(s, \chi)$ (Definition 2.0.10) in case that $\chi : \mathbb{Z} \rightarrow \mathbb{C}$ is the trivial Dirichlet character (modulo 0) such that $\chi(n) = 1$ for all integers $n \neq 0$ (Definition 2.0.10).
2. It is a special case of the Dedekind zeta function $\zeta_K(s)$ (Definition 3.0.13) in the case that the number field K is simply the field $\mathbb{Q}$ of rational numbers (Definition 3.0.13).

Definition 3.0.2 (Logarithmic derivative of a real-valued function). Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be a differentiable function that does not vanish on I (i.e., $f(x) \neq 0$ for all $x \in I$).

The *logarithmic derivative of f* is the function

$$\frac{f'}{f} : I \rightarrow \mathbb{R}, \quad x \mapsto \frac{f'(x)}{f(x)}.$$

Equivalently, if f is positive on I , the logarithmic derivative is the derivative of the natural logarithm of f :

$$\frac{f'(x)}{f(x)} = \frac{d}{dx} \log f(x),$$

where the right-hand side exists by the natural logarithm and the chain rule for differentiation.

(♠ TODO: define and reference isolated points) If f is allowed to vanish at isolated points, the logarithmic derivative is defined on $I \setminus f^{-1}(0)$ as a meromorphic function (Definition A.4.8) with simple poles at the zeros of f .

Definition 3.0.3 (Logarithmic derivative of a holomorphic function). (♠ TODO: AI generated, verify) Let $U \subseteq \mathbb{C}$ be an open set (or more generally $U \subseteq \mathbb{C}^n$ open) and let $f : U \rightarrow \mathbb{C}$ be a holomorphic function (Definition A.1.1) that is not identically zero.

The *logarithmic derivative of f* is the meromorphic function (Definition A.4.8)

$$\frac{f'}{f} : U \setminus f^{-1}(0) \rightarrow \mathbb{C}, \quad z \mapsto \frac{f'(z)}{f(z)},$$

At a point $z_0 \in U$ where $f(z_0) \neq 0$, the logarithmic derivative is simply the derivative of a local branch (♠ TODO: AI generated, reference needed: definition of a branch of the complex logarithm) of $\log f(z)$. At a zero z_0 of f of order $m \geq 1$, the logarithmic derivative has a simple pole (Definition A.4.5) with *residue* m (Definition A.4.7); that is, in a neighborhood of z_0 ,

$$\frac{f'(z)}{f(z)} = \frac{m}{z - z_0} + h(z),$$

where h is holomorphic near z_0 .

If f and g are holomorphic and not identically zero, then

$$\frac{(fg)'}{fg} = \frac{f'}{f} + \frac{g'}{g},$$

which generalizes the real-variable formula. For an infinite product $f(z) = \prod_{n=1}^{\infty} f_n(z)$ converging absolutely and uniformly on compact subsets, with each f_n holomorphic and not identically zero, the logarithmic derivative is

$$\frac{f'(z)}{f(z)} = \sum_{n=1}^{\infty} \frac{f'_n(z)}{f_n(z)},$$

where the sum converges uniformly on compact subsets avoiding the zeros of f and each f_n .

Theorem 3.0.4 (Riemann zeta function is infinitely differentiable on $(1, \infty)$). (♠ TODO: AI generated, verify) The Riemann zeta function (Definition 3.0.1) $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$ is infinitely differentiable on the interval $(1, \infty)$. For every $k \in \mathbb{N}_0$ and every $s \in (1, \infty)$,

$$\zeta^{(k)}(s) = \sum_{n=1}^{\infty} \frac{(-1)^k (\ln n)^k}{n^s}.$$

Proof. (♠ TODO: AI generated, verify) The Riemann zeta function is the Dirichlet series with coefficients $a_n = 1$ for all $n \geq 1$:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

This series converges absolutely for $\Re(s) > 1$, so its abscissa of absolute convergence (Definition 2.0.1) is $\sigma_a = 1$. Applying Corollary 2.0.25 with $\sigma_a = 1$, we obtain that $\zeta(s)$ is infinitely differentiable on $(1, \infty)$ and for each $k \in \mathbb{N}_0$,

$$\zeta^{(k)}(s) = \sum_{n=1}^{\infty} \frac{(-1)^k (\ln n)^k}{n^s} \quad \text{for all } s \in (1, \infty).$$

This is exactly the statement. □

Corollary 3.0.5 (Euler product for the Riemann zeta function). The Riemann zeta function (Definition 3.0.1) admits the Euler product expansion

$$\zeta(s) = \prod_{p \text{ prime}} \left(1 - \frac{1}{p^s}\right)^{-1}$$

which converges absolutely for $\Re(s) > 1$.

Proof. This follows from Theorem 2.0.12 applied to the trivial character modulo 1, noting that $\mathbf{1}_1(p) = 1$ for all primes p . □

Corollary 3.0.6 (Values of the Riemann zeta function at even positive integers). (♠ TODO: AI generated, verify) For every $n \in \mathbb{N}$, the Riemann zeta function (Definition 3.0.1) takes the value

$$\zeta(2n) = \frac{(-1)^{n+1}(2\pi)^{2n}}{2(2n)!} B_{2n},$$

where B_{2n} are the Bernoulli numbers (Definition 3.0.12) (with the convention $B_1 = -1/2$, $B_{2n+1} = 0$ for $n \geq 1$).

In particular, $\zeta(2) = \pi^2/6$, $\zeta(4) = \pi^4/90$, $\zeta(6) = \pi^6/945$, etc.

Proof. (♠ TODO: AI generated, verify)

We prove the formula using the infinite product expansion of the sine function and the generating function for Bernoulli numbers.

First, we use the Weierstrass factorization of the sine function (Proposition 3.0.7):

$$(C) \quad \sin(\pi z) = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right), \quad z \in \mathbb{C}.$$

Next, we take the logarithmic derivative (Definition 3.0.3) of both sides of (C):

$$\begin{aligned} \pi \cot(\pi z) &= \frac{d}{dz} \log \sin(\pi z) \\ &= \frac{1}{z} + \sum_{n=1}^{\infty} \frac{d}{dz} \log \left(1 - \frac{z^2}{n^2}\right) \\ &= \frac{1}{z} + \sum_{n=1}^{\infty} \frac{-2z/n^2}{1 - z^2/n^2} \\ &= \frac{1}{z} - 2z \sum_{n=1}^{\infty} \frac{1}{n^2 - z^2} \end{aligned}$$

(Proposition 3.0.8)

For $|z| < 1$, we expand each term as a geometric series:

$$\begin{aligned} \pi \cot(\pi z) &= \frac{1}{z} - 2z \sum_{n=1}^{\infty} \frac{1}{n^2} \frac{1}{1 - z^2/n^2} \\ &= \frac{1}{z} - 2z \sum_{n=1}^{\infty} \frac{1}{n^2} \sum_{k=0}^{\infty} \left(\frac{z^2}{n^2}\right)^k \\ &= \frac{1}{z} - 2 \sum_{k=0}^{\infty} \left(\sum_{n=1}^{\infty} \frac{1}{n^{2k+2}} \right) z^{2k+1}. \end{aligned}$$

The rearrangement of the double series is justified by absolute convergence. By the definition of the Riemann zeta function (Definition 3.0.1), the inner sum is $\zeta(2k+2)$. Thus

$$(D) \quad \pi \cot(\pi z) = \frac{1}{z} - 2 \sum_{m=1}^{\infty} \zeta(2m) z^{2m-1}, \quad |z| < 1.$$

Now we use the Bernoulli number generating function. The Bernoulli numbers (Definition 3.0.12) are defined by

$$\frac{w}{e^w - 1} = \sum_{n=0}^{\infty} B_n \frac{w^n}{n!}, \quad |w| < 2\pi.$$

Setting $w = 2\pi iz$ and using $e^{2\pi iz} - 1 = 2ie^{\pi iz} \sin(\pi z)$ (♠ TODO: AI generated, reference needed: proposition that $e^{2\pi iz} - 1 = 2ie^{\pi iz} \sin(\pi z)$ via Euler's formula), we obtain

$$\begin{aligned} \frac{2\pi iz}{e^{2\pi iz} - 1} &= \frac{\pi z}{i \sin(\pi z)} e^{-\pi iz} \\ &= \pi z \frac{\cos(\pi z) - i \sin(\pi z)}{i \sin(\pi z)} \\ &= \pi z (-i \cot(\pi z) - 1). \end{aligned}$$

Equating the two expressions gives

$$\pi z (-i \cot(\pi z) - 1) = \sum_{n=0}^{\infty} B_n \frac{(2\pi iz)^n}{n!}.$$

Solving for $\pi \cot(\pi z)$ and using that $B_{2k+1} = 0$ for $k \geq 1$ and $B_1 = -1/2$, we find

$$\begin{aligned} \pi \cot(\pi z) &= -\frac{1}{z} - \frac{1}{\pi iz} \sum_{n=0}^{\infty} B_n \frac{(2\pi iz)^n}{n!} \\ &= \frac{1}{z} - 2 \sum_{m=1}^{\infty} \frac{(-1)^{m+1} (2\pi)^{2m} B_{2m}}{(2m)!} z^{2m-1}. \end{aligned}$$

(Proposition 3.0.9)

Comparing the expansions in (D) and above, we equate coefficients of z^{2m-1} :

$$-2\zeta(2m) = -2 \frac{(-1)^{m+1} (2\pi)^{2m} B_{2m}}{(2m)!}.$$

Simplifying gives the desired formula

$$\zeta(2m) = \frac{(-1)^{m+1} (2\pi)^{2m}}{2(2m)!} B_{2m}.$$

Finally, using the known values $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$ (Definition 3.0.12), we obtain the explicit values

$$\begin{aligned}\zeta(2) &= \frac{(-1)^2(2\pi)^2}{2 \cdot 2!} \cdot \frac{1}{6} = \frac{\pi^2}{6}, \\ \zeta(4) &= \frac{(-1)^3(2\pi)^4}{2 \cdot 4!} \cdot \left(-\frac{1}{30}\right) = \frac{\pi^4}{90}, \\ \zeta(6) &= \frac{(-1)^4(2\pi)^6}{2 \cdot 6!} \cdot \frac{1}{42} = \frac{\pi^6}{945},\end{aligned}$$

and so on. □

Proposition 3.0.7 (Weierstrass factorization of the sine function). (♠ TODO: AI generated, verify) The sine function extended to $\mathbb{C}$ via its power series has the infinite product representation

$$\sin(\pi z) = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right), \quad z \in \mathbb{C}.$$

The product converges absolutely and uniformly on compact subsets of $\mathbb{C}$.

This is a special case of the Weierstrass factorization theorem for entire functions (♠ TODO: AI generated, reference needed: Weierstrass factorization theorem for entire functions of finite order). The zeros of $\sin(\pi z)$ are exactly the integers $z \in \mathbb{Z}$, with simple zeros at $z = 0, \pm 1, \pm 2, \dots$.

Proof. (♠ TODO: AI generated, verify) We prove this directly using partial fraction decomposition. Consider the function

$$f(z) := \frac{\pi}{\sin(\pi z)}.$$

The zeros of $\sin(\pi z)$ are exactly at $z = n \in \mathbb{Z}$, each simple. Hence f has simple poles at each integer with residue

$$\operatorname{Res}_{z=n} f(z) = \lim_{z \rightarrow n} (z - n) \frac{\pi}{\sin(\pi z)} = \frac{\pi}{\pi(-1)^n} = (-1)^n.$$

We use the standard partial fraction expansion for $\cot(\pi z)$, which follows from contour integration. By the residue theorem (Theorem A.4.9), integrating $\frac{\cot(\pi w)}{z - w}$ over a large square contour and letting the size tend to infinity gives

$$\pi \cot(\pi z) = \frac{1}{z} + \sum_{n=1}^{\infty} \left(\frac{1}{z - n} + \frac{1}{z + n} \right).$$

Taking the logarithmic derivative of both sides of the claimed product formula:

$$\log \left(\pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right) \right) = \log(\pi z) + \sum_{n=1}^{\infty} \log \left(1 - \frac{z^2}{n^2}\right).$$

Differentiating:

$$\frac{\frac{d}{dz} \sin(\pi z)}{\sin(\pi z)} = \frac{1}{z} + \sum_{n=1}^{\infty} \left(\frac{-2z/n^2}{1 - z^2/n^2} \right) = \frac{1}{z} + \sum_{n=1}^{\infty} \left(\frac{1}{z-n} + \frac{1}{z+n} \right).$$

This matches $\pi \cot(\pi z)$, so the logarithmic derivatives agree. Hence the two functions differ by at most a multiplicative constant C :

$$\sin(\pi z) = C \cdot \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2} \right).$$

Evaluating at $z = 0$ using $\lim_{z \rightarrow 0} \sin(\pi z)/(\pi z) = 1$, we find $C = 1$. The absolute and uniform convergence on compact subsets follows from the fact that $\sum 1/n^2 < \infty$. $\square$

Proposition 3.0.8 (Logarithmic derivative of an infinite product). Let $\{f_n(z)\}_{n=1}^{\infty}$ be a sequence of holomorphic functions (Definition A.1.1) on a domain $D \subseteq \mathbb{C}$ such that the infinite product $\prod_{n=1}^{\infty} f_n(z)$ converges absolutely and uniformly on compact subsets of D , and the product defines a holomorphic function (Definition A.1.1) $f(z)$ that is not identically zero. Suppose each $f_n(z)$ has no zeros in D except possibly at isolated points. Then for $z \in D$ where the product and its logarithmic derivative (Definition 3.0.3) are defined,

$$\frac{f'(z)}{f(z)} = \sum_{n=1}^{\infty} \frac{f'_n(z)}{f_n(z)},$$

where the sum converges uniformly on compact subsets of D avoiding the zeros of f .

In particular, if $f(z) = \prod_{n=1}^{\infty} (1 - z/a_n)$ with $\{a_n\}$ a sequence of nonzero complex numbers such that $\sum_{n=1}^{\infty} 1/|a_n|$ converges, then

$$\frac{f'(z)}{f(z)} = \sum_{n=1}^{\infty} \frac{-1/a_n}{1 - z/a_n} = - \sum_{n=1}^{\infty} \frac{1}{a_n - z}.$$

This follows by taking the logarithmic derivative (Definition 3.0.3) of both sides and using uniform convergence of the product to justify termwise differentiation (Proposition A.1.3).

Proof. (♠ TODO: AI generated, verify) Let $P_N(z) = \prod_{n=1}^N f_n(z)$ denote the partial products. By hypothesis, $P_N(z)$ converges to $f(z)$ uniformly on compact subsets of D , and each P_N is holomorphic as a finite product of holomorphic functions (Proposition A.1.3). The limit function f is holomorphic by Theorem A.5.3.

We first compute the logarithmic derivative of each finite partial product. For $N = 1$, we have $(P_1)'(z)/P_1(z) = f'_1(z)/f_1(z)$ trivially. For $N \geq 2$, applying the quotient rule (Proposition A.1.3) to the product $P_N = P_{N-1} \cdot f_N$ gives

$$\frac{P'_N(z)}{P_N(z)} = \frac{P'_{N-1}(z)f_N(z) + P_{N-1}(z)f'_N(z)}{P_{N-1}(z)f_N(z)} = \frac{P'_{N-1}(z)}{P_{N-1}(z)} + \frac{f'_N(z)}{f_N(z)},$$

at any point z where neither $P_{N-1}(z)$ nor $f_N(z)$ vanishes. By induction on N , we obtain for all N :

$$\frac{P'_N(z)}{P_N(z)} = \sum_{n=1}^N \frac{f'_n(z)}{f_n(z)},$$

at any point z where none of the factors $f_1(z), \dots, f_N(z)$ vanish.

Next we show that the series $\sum_{n=1}^{\infty} f'_n(z)/f_n(z)$ converges uniformly on compact subsets of D avoiding the zeros of f . Let $K \subset D$ be a compact set on which $f(z) \neq 0$ for all $z \in K$. Since $P_N \rightarrow f$ uniformly on K and f is bounded away from zero on K (because $|f|$ attains a positive minimum on the compact set K), there exists N_0 such that for all $N \geq N_0$ and all $z \in K$, both $|P_N(z)| > 0$ and $|f_N(z) - 1|$ is sufficiently small. The absolute convergence of the infinite product $\prod f_n(z)$ on compact subsets implies that $\sum |f_n(z) - 1|$ converges uniformly on K . For large n , we have $|f_n(z) - 1| < 1/2$ on K , which means $f_n(z) \neq 0$ and the logarithmic derivative $f'_n(z)/f_n(z)$ is well-defined.

We verify that $\sum f'_n/f_n$ converges uniformly on K . Because the product $\prod f_n(z)$ converges absolutely and uniformly on compact subsets, we have $f_n \rightarrow 1$ uniformly on K . For each n , the function f'_n is holomorphic on D , hence continuous. The uniform convergence of the product implies that for any compact set $K' \subset D$ containing K in its interior, the sequence P_N converges uniformly to f on K' . By Theorem A.5.3, the derivatives P'_N converge uniformly to f' on compact subsets of D .

We conclude by passing to the limit in the logarithmic derivative. For N sufficiently large so that $P_N(z) \neq 0$ on K , we have

$$\frac{P'_N(z)}{P_N(z)} = \sum_{n=1}^N \frac{f'_n(z)}{f_n(z)}.$$

Since $P'_N \rightarrow f'$ and $P_N \rightarrow f$ uniformly on K , and since f is bounded away from zero on K , the quotient P'_N/P_N converges uniformly to f'/f on K . Therefore, the partial sums $\sum_{n=1}^N f'_n(z)/f_n(z)$ converge uniformly to $f'(z)/f(z)$ on K . This establishes

$$\frac{f'(z)}{f(z)} = \sum_{n=1}^{\infty} \frac{f'_n(z)}{f_n(z)},$$

for all $z \in K$, where the convergence is uniform.

We derive the special case for $f(z) = \prod_{n=1}^{\infty} (1 - z/a_n)$ with $\sum 1/|a_n| < \infty$. Here $f_n(z) = 1 - z/a_n$, so $f'_n(z) = -1/a_n$. Substituting into the general formula:

$$\frac{f'_n(z)}{f_n(z)} = \frac{-1/a_n}{1 - z/a_n} = -\frac{1}{a_n - z},$$

and the convergence condition $\sum 1/|a_n| < \infty$ guarantees absolute and uniform convergence of the product on compact subsets avoiding the points a_n . $\square$

Proposition 3.0.9 (Expansion of $\pi z \cot(\pi z)$ in terms of Bernoulli numbers). (♠ TODO: AI generated, verify) For $|z| < 1$, the function $\pi z \cot(\pi z)$ has the power series expansion

$$\pi z \cot(\pi z) = 1 - 2 \sum_{m=1}^{\infty} \frac{(-1)^{m+1} (2\pi)^{2m} B_{2m}}{(2m)!} z^{2m},$$

where B_{2m} are the Bernoulli numbers (Definition 3.0.12) (with $B_1 = -1/2$, $B_{2m+1} = 0$ for $m \geq 1$).

Equivalently,

$$\pi \cot(\pi z) = \frac{1}{z} - 2 \sum_{m=1}^{\infty} \frac{(-1)^{m+1} (2\pi)^{2m} B_{2m}}{(2m)!} z^{2m-1}, \quad 0 < |z| < 1.$$

This follows from the generating function of the Bernoulli numbers:

$$\frac{w}{e^w - 1} = \sum_{n=0}^{\infty} B_n \frac{w^n}{n!}, \quad |w| < 2\pi,$$

by setting $w = 2\pi iz$ and using $e^{2\pi iz} - 1 = 2ie^{\pi iz} \sin(\pi z)$, together with the definition of the sine and cosine functions.

Proof. (♠ TODO: AI generated, verify) We start from the generating function for the Bernoulli numbers (Definition 3.0.12):

$$\frac{w}{e^w - 1} = \sum_{n=0}^{\infty} B_n \frac{w^n}{n!}, \quad |w| < 2\pi.$$

Setting $w = 2\pi iz$ with $|z| < 1$, we obtain

$$\frac{2\pi iz}{e^{2\pi iz} - 1} = \sum_{n=0}^{\infty} B_n \frac{(2\pi iz)^n}{n!}.$$

Using $e^{2\pi iz} - 1 = e^{\pi iz}(e^{\pi iz} - e^{-\pi iz}) = e^{\pi iz} \cdot 2i \sin(\pi z)$, we can rewrite the left-hand side as

$$\frac{2\pi iz}{e^{2\pi iz} - 1} = \frac{\pi z}{e^{\pi iz} \sin(\pi z)} = \frac{\pi z (\cos(\pi z) - i \sin(\pi z))}{i \sin(\pi z)} = \pi z (-i \cot(\pi z) + 1).$$

Equating the two expressions:

$$\pi z (1 - i \cot(\pi z)) = \sum_{n=0}^{\infty} B_n \frac{(2\pi iz)^n}{n!}.$$

Solving for $\pi z \cot(\pi z)$:

$$\pi z \cot(\pi z) = i\pi z - \frac{i}{z} \sum_{n=0}^{\infty} B_n \frac{(2\pi iz)^n}{n!}.$$

Separating the $n = 0$ term ($B_0 = 1$):

$$\pi z \cot(\pi z) = i\pi z - i(2\pi) - \frac{i}{z} \sum_{n=1}^{\infty} B_n \frac{(2\pi i z)^n}{n!}.$$

Since $B_1 = -1/2$, the $n = 1$ term is $B_1(2\pi i z) = -\pi i z$. The remaining terms for $n \geq 2$ use $B_{2m+1} = 0$ for $m \geq 1$:

$$\pi z \cot(\pi z) = i\pi z - 2\pi i + \pi i z - \frac{i}{z} \sum_{m=1}^{\infty} B_{2m} \frac{(2\pi i z)^{2m}}{(2m)!}.$$

Simplifying the constant terms: $i\pi z - 2\pi i + \pi i z = 2\pi i z - 2\pi i$. We verify that this equals 1 by checking at $z = 0$: $\lim_{z \rightarrow 0} \pi z \cot(\pi z) = 1$. The series term:

$$-\frac{i}{z} \sum_{m=1}^{\infty} B_{2m} \frac{(2\pi i)^{2m} z^{2m}}{(2m)!} = -i \sum_{m=1}^{\infty} B_{2m} \frac{(2\pi i)^{2m} z^{2m-1}}{(2m)!}.$$

Since $(2\pi i)^{2m} = (2\pi)^{2m}(-1)^m$, we have

$$-i B_{2m} \frac{(2\pi)^{2m}(-1)^m z^{2m-1}}{(2m)!} = -2 \cdot \frac{(-1)^{m+1} (2\pi)^{2m} B_{2m}}{(2m)!} z^{2m-1},$$

where we used $i(-1)^m = (-1)^{m+1}i$ and simplified the factor of 2. This gives the claimed expansion. $\square$

Corollary 3.0.10 (Values of the Riemann zeta function at non-positive integers). (♠ TODO: AI generated, verify) For every $n \in \mathbb{N}_0$, the Riemann zeta function (Definition 3.0.1) takes the rational value

$$\boxed{\zeta(-n) = -\frac{B_{n+1}}{n+1}},$$

where B_{n+1} are the Bernoulli numbers (Definition 3.0.12) (with $B_1 = -1/2$, $B_{2k+1} = 0$ for $k \geq 1$).

In particular, $\zeta(0) = -1/2$, $\zeta(-1) = -1/12$, and $\zeta(-2n) = 0$ for $n \geq 1$ (the trivial zeros, ??).

Proof. (♠ TODO: AI generated, verify) We use the functional equation from Corollary 3.0.11:

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s).$$

Setting $s = -n$ for $n \in \mathbb{N}_0$, we obtain

$$\zeta(-n) = 2^{-n} \pi^{-n-1} \sin\left(\frac{-\pi n}{2}\right) \Gamma(1+n) \zeta(1+n).$$

Since $\Gamma(1+n) = n!$, we analyze by parity of n .

Case 1: n is even. Write $n = 2m$ for some $m \in \mathbb{N}_0$. Then $\sin(-\pi m) = 0$. For $m > 0$ (i.e., $n \geq 2$), the factor $\zeta(1+n) = \zeta(2m+1)$ is finite and nonzero, so $\zeta(-2m) = 0$. For $m = 0$

(i.e., $n = 0$), we have $\sin(0) = 0$ but $\Gamma(1)\zeta(1)$ has a pole. Instead evaluate directly: the functional equation near $s = 0$ gives

$$\zeta(s) = 2^s \pi^{s-1} \left(-\frac{\pi s}{2} + O(s^3) \right) \Gamma(1-s) \zeta(1-s).$$

As $s \rightarrow 0$, we have $\Gamma(1-s) \rightarrow 1$ and $(1-s)\zeta(1-s) \rightarrow -1$. Hence

$$\lim_{s \rightarrow 0} \frac{\zeta(s)}{s} = 2^0 \pi^{-1} \cdot \left(-\frac{\pi}{2} \right) \cdot 1 \cdot (-1) = \frac{1}{2},$$

so $\zeta(0) = -B_1/1 = -(-1/2)/1 = -1/2$, as claimed.

Case 2: n is odd. Write $n = 2m - 1$ for some $m \geq 1$. Then $\sin\left(\frac{-\pi(2m-1)}{2}\right) = (-1)^{m+1}$. We have

$$\zeta(-(2m-1)) = 2^{-(2m-1)} \pi^{-2m} (-1)^{m+1} (2m-1)! \zeta(2m).$$

By the formula for ζ at even positive integers (Corollary 3.0.6),

$$\zeta(2m) = \frac{(-1)^{m+1} (2\pi)^{2m}}{2(2m)!} B_{2m}.$$

Substituting this in:

$$\zeta(-(2m-1)) = 2^{-(2m-1)} \pi^{-2m} (-1)^{m+1} (2m-1)! \cdot \frac{(-1)^{m+1} (2\pi)^{2m}}{2(2m)!} B_{2m}.$$

Simplifying: $(2m-1)!/(2m)! = 1/(2m)$, and $2^{-(2m-1)} \cdot 2^{2m}/2 = 1$. Also $\pi^{-2m} \cdot \pi^{2m} = 1$ and $(-1)^{m+1} \cdot (-1)^{m+1} = 1$. Hence

$$\zeta(-(2m-1)) = \frac{B_{2m}}{2m}.$$

Since $n = 2m - 1$, we have $n + 1 = 2m$ and $B_{n+1} = B_{2m}$, so $\zeta(-n) = -\frac{B_{n+1}}{n+1}$. We check the sign: our expression gives $\zeta(-(2m-1)) = \frac{B_{2m}}{2m} = -\frac{B_{n+1}}{n+1}$ only if $B_{2m}/(2m) = -B_{n+1}/(n+1)$, which holds since $n + 1 = 2m$. Thus $\zeta(-n) = -B_{n+1}/(n+1)$ in all cases. $\square$

Corollary 3.0.11 (Analytic continuation and functional equation for the Riemann zeta function). (♠ TODO: AI generated, verify) The Riemann zeta function (Definition 3.0.1) $\zeta(s)$ admits a meromorphic continuation (Definition A.5.7) to $\mathbb{C}$ with a single simple pole at $s = 1$ of residue 1.

Define the *completed Riemann zeta function*

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

Then $\xi(s)$ is an entire function satisfying the symmetric functional equation

$$\xi(s) = \xi(1-s).$$

Equivalently, in terms of the Riemann zeta function (Definition 3.0.1):

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s), \quad s \in \mathbb{C} \setminus \{1\}.$$

Proof. (♠ TODO: AI generated, verify) We specialize Theorem 2.0.18 to the trivial Dirichlet character modulo 1. Let χ_0 denote the principal character modulo $k = 1$, so that $\chi_0(n) = 1$ for all $n \geq 1$. The associated Dirichlet L -function is

$$L(s, \chi_0) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \zeta(s).$$

Since $\chi_0(-1) = 1$, the character is even, so $a(\chi_0) = 0$. The Gauss sum and root number are

$$\tau(\chi_0) = \sum_{m=1}^1 \chi_0(m) e^{2\pi i m/1} = 1,$$

and

$$\varepsilon(\chi_0) = \frac{\tau(\chi_0)}{i^{a(\chi_0)} \sqrt{1}} = 1.$$

The completed Dirichlet L -function for χ_0 coincides with the completed Riemann zeta function:

$$\Lambda(s, \chi_0) = \left(\frac{1}{\pi}\right)^{s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \frac{1}{\frac{1}{2}s(s-1)} \xi(s).$$

The functional equation $\Lambda(s, \chi_0) = \varepsilon(\chi_0) \Lambda(1-s, \overline{\chi_0})$ simplifies to

$$\xi(s) = \xi(1-s),$$

because $\overline{\chi_0} = \chi_0$ and $\varepsilon(\chi_0) = 1$.

We verify that $\xi(s)$ is entire. The only singularity of $\zeta(s)$ in $\mathbb{C}$ is a simple pole at $s = 1$ with residue 1 (Riemann zeta function (Definition 3.0.1)). At $s = 1$, the factor $s - 1$ in $\xi(s)$ cancels this pole, so that

$$\lim_{s \rightarrow 1} \xi(s) = \frac{1}{2} \cdot 1 \cdot \pi^{-1/2} \Gamma\left(\frac{1}{2}\right) \cdot 1 = \frac{1}{2},$$

where we used $\Gamma(1/2) = \sqrt{\pi}$. The functions $s(s-1)$, $\pi^{-s/2}$, and $\Gamma(s/2)$ are holomorphic for all $s \in \mathbb{C}$ except possibly at the poles of $\Gamma(s/2)$, which occur at non-positive even integers; at those points $\zeta(s)$ vanishes, so the product remains finite. Therefore, $\xi(s)$ extends to an entire function on $\mathbb{C}$.

We derive the classical functional equation for $\zeta(s)$ from the identity $\xi(s) = \xi(1-s)$. Substituting the definition of ξ and dividing both sides by $s(s-1)$, we obtain

$$\pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \pi^{-(1-s)/2} \Gamma\left(\frac{1-s}{2}\right) \zeta(1-s).$$

Solving for $\zeta(s)$ gives

$$\zeta(s) = \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)} \zeta(1-s).$$

To convert this to the classical form we use two standard identities for the Gamma function. Euler's reflection formula?? states that $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$. Setting $z = (s+1)/2$ and using $\sin(\pi(s+1)/2) = \cos(\pi s/2)$, we obtain

$$\Gamma\left(\frac{s+1}{2}\right)\Gamma\left(\frac{1-s}{2}\right) = \frac{\pi}{\cos(\pi s/2)}.$$

Legendre's duplication formula?? states that $\Gamma(z)\Gamma(z+1/2) = 2^{1-2z}\sqrt{\pi}\Gamma(2z)$. Setting $z = s/2$, we obtain

$$\Gamma\left(\frac{s}{2}\right)\Gamma\left(\frac{s+1}{2}\right) = 2^{1-s}\sqrt{\pi}\Gamma(s).$$

Combining these two identities, we can express $\Gamma((1-s)/2)$ in terms of $\Gamma(s/2)$ and $\Gamma(s)$: from the reflection formula, $\Gamma((s+1)/2) = \frac{\pi}{\cos(\pi s/2)\Gamma((1-s)/2)}$, and substituting into the duplication formula gives

$$\Gamma\left(\frac{s}{2}\right) \cdot \frac{\pi}{\cos(\pi s/2)\Gamma((1-s)/2)} = 2^{1-s}\sqrt{\pi}\Gamma(s).$$

Rearranging yields

$$\frac{\Gamma((1-s)/2)}{\Gamma(s/2)} = \frac{\pi}{2^{1-s}\sqrt{\pi}\Gamma(s)\cos(\pi s/2)}.$$

Applying Euler's reflection formula once more with $z = s$, we have $\Gamma(s) = \frac{\pi}{\sin(\pi s)\Gamma(1-s)}$. Substituting and using $\sin(\pi s) = 2\sin(\pi s/2)\cos(\pi s/2)$:

$$\frac{\Gamma((1-s)/2)}{\Gamma(s/2)} = \frac{\pi}{2^{1-s}\sqrt{\pi} \cdot \frac{\pi}{\sin(\pi s)\Gamma(1-s)} \cdot \cos(\pi s/2)} = \frac{2^s}{\sqrt{\pi}} \frac{\sin(\pi s/2)\cos(\pi s/2)}{\cos(\pi s/2)} \Gamma(1-s) = \frac{2^s}{\sqrt{\pi}} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s)$$

Substituting this ratio back into the expression for $\zeta(s)$:

$$\zeta(s) = \pi^{s-1/2} \cdot \frac{2^s}{\sqrt{\pi}} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s).$$

This is the classical functional equation. □

Definition 3.0.12. The *Bernoulli numbers* B_n ($n \geq 0$) are the unique rational numbers defined by the exponential generating function (Definition A.2.1)

$$\boxed{\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} B_n \frac{z^n}{n!}}$$

which converges for $|z| < 2\pi$ (Proposition A.2.2). Explicitly, $B_0 = 1$, $B_1 = -1/2$, $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$, $B_8 = -1/30$, and $B_{2k+1} = 0$ for all $k \geq 1$.

Equivalently, they are defined recursively by $B_0 = 1$ and

$$\sum_{k=0}^m \binom{m+1}{k} B_k = 0 \quad \text{for all } m \geq 1.$$

(Proposition 3.0.9)

(♠ TODO: AI generated, verify)

Definition 3.0.13. Let K be a number field, and let $s \in \mathbb{C}$ with $\Re(s) > 1$. The *Dedekind zeta function of K* is defined as

$$\zeta_K(s) = \sum_{\mathfrak{a} \neq (0)} \frac{1}{N_{K/\mathbb{Q}}(\mathfrak{a})^s},$$

where $\mathfrak{a}$ runs over all nonzero ideals of $\mathcal{O}_K$. (♠ TODO: TODO: discuss the Euler product) (♠ TODO: TODO: discuss that it extends meromorphically, its trivial zeroes, functional equation)

Definition 3.0.14. The *Gamma function* $\Gamma : \{z \in \mathbb{C} : \Re(z) > 0\} \rightarrow \mathbb{C}$ is defined by the integral

$$\Gamma(z) := \int_0^\infty t^{z-1} e^{-t} dt,$$

which converges absolutely for all complex numbers z with real part $\Re(z) > 0$.

The Gamma function $\Gamma(z)$ admits a unique meromorphic continuation (Definition A.5.7) to the entire complex plane $\mathbb{C}$, with simple poles at non-positive integers $z = 0, -1, -2, \dots$, and satisfies the functional equation

$$\Gamma(z+1) = z\Gamma(z).$$

(Lemma 2.0.15) This meromorphic continuation is unique by the identity theorem for holomorphic functions.

4. HECKE CHARACTERS AND HECKE L-FUNCTIONS

Definition 4.0.1. (♠ TODO: TODO: can this be more generally defined for function fields?)

Let K be a global field. A *Hecke character* (or *Größencharacter*) of K is a character (Definition 5.0.1)/continuous group homomorphism

$$\chi : \mathbb{A}_K^\times \rightarrow \mathbb{C}^\times$$

from the idèle group $\mathbb{A}_K^\times$ such that there exists an integer $m \geq 0$ (the ∞ -type) and a finite character χ_f of the finite idèles satisfying

$$\chi(x) = \chi_f(x)|x|^m$$

for all $x \in K^\times$ embedded diagonally in $\mathbb{A}_K^\times$, or more generally: χ is trivial on $K^\times$ (viewed as a subgroup of $\mathbb{A}_K^\times$ by the diagonal embedding). If $|\chi(a)| = 1$ for all $a \in \mathbb{A}_K^\times$, then χ is called a *unitary Hecke character*. (♠ TODO: TODO: define the weight of a hecke character)

Definition 4.0.2. Let K be a number field and χ a Hecke character of K (Definition 4.0.1). For $s \in \mathbb{C}$ with $\Re(s) > 1$, the *Hecke L-function of χ* is defined by the absolutely convergent sum

$$L(s, \chi) = \sum_{\mathfrak{a}} \chi(x_{\mathfrak{a}}) N(\mathfrak{a})^{-s}$$

where the sum runs over all nonzero ideals $\mathfrak{a}$ of $\mathcal{O}_K$, and $x_{\mathfrak{a}} = (\varpi_{\mathfrak{p}}^{n_{\mathfrak{p}}}) \in \mathbb{A}_{K,f}^\times$, where the prime factorization of $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}$; this is well defined.

Equivalently, $L(s, \chi)$ may be defined by the absolutely convergent Euler product

$$L(s, \chi) = \prod_{\mathfrak{p}} \frac{1}{1 - \chi(\varpi_{\mathfrak{p}}) N_{K/\mathbb{Q}}(\mathfrak{p})^{-s}},$$

where the product runs over all nonzero prime ideals $\mathfrak{p}$ of $\mathcal{O}_K$, $\varpi_{\mathfrak{p}}$ is a uniformizer at $\mathfrak{p}$, and $N_{K/\mathbb{Q}}(\mathfrak{p}) = |\mathcal{O}_K/\mathfrak{p}|$.

Definition 4.0.3 (Functional equation for Hecke L-functions). (♠ TODO: TODO: define the conductor of a hecke character) Let K be a number field with absolute discriminant $|\Delta_{K/\mathbb{Q}}|$, and let χ be a Hecke character of K (Definition 4.0.1) with conductor $\mathfrak{f}(\chi)$. For $s \in \mathbb{C}$, define the *completed (Hecke) L-function* by

$$\Lambda(s, \chi) := (|\Delta_{K/\mathbb{Q}}| \cdot N(\mathfrak{f}(\chi)))^{\frac{s}{2}} \cdot \prod_{v|\infty} L_v(s, \chi) \cdot L(s, \chi),$$

where (♠ TODO: TODO: define the norm) (♠ TODO: TODO: define the Gamma function)

- $N(\mathfrak{f}(\chi))$ is the norm of the conductor ideal $\mathfrak{f}(\chi)$,
- the product runs over all infinite places v of K ,
- the *infinite local factors* $L_v(s, \chi)$ are defined using the Γ -function (Definition 3.0.14):

$$L_v(s, \chi) := \begin{cases} \Gamma_{\mathbb{R}}(s + \kappa_v) := \pi^{-\frac{s+\kappa_v}{2}} \Gamma\left(\frac{s+\kappa_v}{2}\right), & \text{if } v \text{ is a real place,} \\ \Gamma_{\mathbb{C}}(s + \kappa_v) := 2(2\pi)^{-(s+\kappa_v)} \Gamma(s + \kappa_v), & \text{if } v \text{ is a complex place,} \end{cases}$$

with some integers $\kappa_v \in \{0, 1\}$ determined by the ∞ -type of χ ,

- and $L(s, \chi)$ is the Hecke L -function (Definition 4.0.2).

Theorem 4.0.4 (Functional equation for Hecke L-functions). (♠ TODO: TODO: define the conductor of a hecke character) Let K be a number field with absolute discriminant $|\Delta_{K/\mathbb{Q}}|$, and let χ be a Hecke character of K (Definition 4.0.1) with conductor $\mathfrak{f}(\chi)$. The completed Hecke L -function (Definition 4.0.3) $\Lambda(s, \chi)$ satisfies the function equation

$$\Lambda(s, \chi) = \varepsilon(\chi) \Lambda(1 - s, \bar{\chi}),$$

(♠ TODO: TODO: define root number) (♠ TODO: Argue that the Hecke L -function is meromorphic; this function equation should say that the L -function is meromorphic for the real part of s greater than 1 or less than 0, but an argument for what happens in the critical strip is necessary.) where $\varepsilon(\chi) \in \mathbb{C}$ is the root number with $|\varepsilon(\chi)| = 1$, and $\bar{\chi}$ is the complex conjugate Hecke character.

5. TATE'S THESIS: ZETA INTEGRALS WITH ADELES AND IDELES

Definition 5.0.1. Let G be a locally compact Hausdorff group.

1. A *quasicharacter of G* is a continuous group homomorphism $\chi : G \rightarrow \mathbb{C}^\times$.
2. A quasicharacter χ is *unitary* if its image lies in the unit circle $S^1 \subset \mathbb{C}^\times$, i.e. $|\chi(g)| = 1$ for all $g \in G$. Such a quasicharacter is also simply called a *character*.

3. A (quasi)character is *finite* if its image is finite, i.e. its kernel has finite index in its domain.

When G is finite, G is usually equipped with the discrete topology — a character (and even a quasicharacter) is thus simply a group homomorphism $G \rightarrow \mathbb{C}^\times$ such that $|\chi(g)| = 1$ for all $g \in G$.

Definition 5.0.2 (Schwartz-Bruhat space on a locally compact abelian group). Let G be a locally compact abelian group that is a finite product of copies of $\mathbb{R}$, $\mathbb{Z}$, and finite groups, or more generally a locally compact abelian group admitting a smooth structure and Haar measure. The *Schwartz-Bruhat space* $\mathcal{S}(G)$ is the space of functions which are analogues of Schwartz functions on G : smooth, rapidly decreasing if archimedean, and locally constant with compact support if non-archimedean. More precisely:

- If $G = \mathbb{R}^n$, then $\mathcal{S}(G)$ is the classical Schwartz space $\mathcal{S}(\mathbb{R}^n)$ on $\mathbb{R}^n$.
- If G is a non-archimedean local field (or its additive group), then $\mathcal{S}(G)$ is the space of locally constant, compactly supported complex-valued functions $G \rightarrow \mathbb{C}$. (♠ TODO: define completed tensor product of spaces)
- For a finite product of such groups, the Schwartz-Bruhat space is the completed tensor product of the corresponding spaces.

Definition 5.0.3. Let F be a global field and $\mathbb{A}_F$ its adele ring. Write by $\mathbb{A}_F^\times$ the ideles of F , and let $\chi : \mathbb{A}_F^\times / F^\times \rightarrow \mathbb{C}^\times$ be a quasi-character (Definition 5.0.1). For $\varphi \in \mathcal{S}(\mathbb{A}_F)$ (Definition 5.0.2) (the Schwartz-Bruhat space) and $s \in \mathbb{C}$ with $\Re(s)$ sufficiently large, *zeta integral/function of φ and χ* is defined by

$$Z(\varphi, \chi, s) = \int_{\mathbb{A}_F^\times} \varphi(x) \chi(x) |x|_{\mathbb{A}_F}^s d^\times x,$$

(♠ TODO: completed *L-function*) where $|\cdot|_{\mathbb{A}_F}$ is the idelic norm.

6. MODULAR FORMS

Definition 6.0.1 (Weakly modular form). Let $k \in \mathbb{Z}$, and let $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ be a subgroup. A meromorphic function $f : \mathbb{H} \rightarrow \mathbb{C}$ is called a *weakly modular form of weight k with respect to Γ* or is said to *satisfy the modular transformation property of weight k with respect to Γ* if

$$f|_k \gamma = f \quad \text{for all } \gamma \in \Gamma,$$

(??) i.e.

$$f(z) = (cz + d)^{-k} f\left(\frac{az + b}{cz + d}\right)$$

for all $z \in \mathbb{H}$.

Definition 6.0.2. Let k be an integer and $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ a congruence subgroup. A *modular form of weight k with respect to Γ* is a function $f : \mathbb{H} \rightarrow \mathbb{C}$ such that

- The function f is holomorphic (Definition A.1.1) on $\mathbb{H}$.

- The function f is a weakly modular form of weight k with respect to Γ (Definition 6.0.1). Equivalently, for all $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ and all $z \in \mathbb{H}$, we have

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z).$$

- f is holomorphic at the cusps of Γ (i.e., its Fourier expansion (Definition 9.0.1) at all cusps of Γ is holomorphic). More precisely, $f|_k \alpha$ is holomorphic at ∞ for all $\alpha \in \Gamma$.

A **cuspidal form** is a modular form f whose Fourier expansion (Definition 9.0.1) at every cusp of Γ has constant term 0, i.e. $f|_k \alpha$ has a cusp at ∞ for every $\alpha \in \Gamma$.

Definition 6.0.3. (♠ TODO: define holomorphic function, upper half plane) (♠ TODO: define cusps of Γ) Let k be an integer and $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ a congruence subgroup. A **modular form of weight k with respect to Γ** is a function $f : \mathbb{H} \rightarrow \mathbb{C}$, where $\mathbb{H}$ is the upper half-plane, such that

- The function f is holomorphic on $\mathbb{H}$.
- The function f is a weakly modular form of weight k with respect to Γ (Definition 6.0.1). Equivalently, for all $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ and all $z \in \mathbb{H}$, we have

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z).$$

- f is holomorphic at the cusps of Γ (i.e., its Fourier expansion at all cusps of Γ is holomorphic). More precisely, $f|_k \alpha$ is holomorphic at ∞ for all $\alpha \in \Gamma$.

A **cuspidal form** is a modular form f whose Fourier expansion (Definition 9.0.1) at every cusp of Γ has constant term 0, i.e. $f|_k \alpha$ has a cusp at ∞ for every $\alpha \in \Gamma$.

Definition 6.0.4. (♠ TODO: define holomorphic function, upper half plane) (♠ TODO: define cusps of Γ) Let k be an integer and $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ a congruence subgroup. A **modular form of weight k with respect to Γ** is a function $f : \mathbb{H} \rightarrow \mathbb{C}$, where $\mathbb{H}$ is the upper half-plane, such that

- The function f is meromorphic on $\mathbb{H}$.
- The function f is a weakly modular form of weight k with respect to Γ (Definition 6.0.1). Equivalently, for all $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ and all $z \in \mathbb{H}$, we have

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z).$$

- f is meromorphic at the cusps of Γ (i.e., its Fourier expansion at all cusps of Γ is holomorphic). More precisely, $f|_k \alpha$ is meromorphic at ∞ for all $\alpha \in \Gamma$.

Definition 6.0.5. (♠ TODO: AI generated, verify) Let k be a positive even integer and let $\Gamma \leq \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup. The set of all modular forms of weight k with

respect to Γ (Definition 6.0.3) forms a complex vector space, denoted by

$$M_k(\Gamma).$$

The subset of those modular forms that are also cusp forms (Definition 6.0.3) is a linear subspace, denoted by

$$S_k(\Gamma) \subseteq M_k(\Gamma).$$

The quotient vector space

$$\mathcal{E}_k(\Gamma) = M_k(\Gamma)/S_k(\Gamma)$$

is called the *space of Eisenstein series of weight k for Γ* . (Not every element of $\mathcal{E}_k(\Gamma)$ is represented by an Eisenstein series in the classical sense, but the name reflects that the image of the standard Eisenstein series spans this quotient when it exists.)

For each cusp s of Γ , there is a well-defined *constant term functional*

$$\text{ct}_s : S_k(\Gamma)^\perp \longrightarrow \mathbb{C},$$

where $S_k(\Gamma)^\perp \subseteq M_k(\Gamma)$ denotes the subspace of modular forms whose Fourier expansion at every cusp other than s has zero constant term. The dimension of $\mathcal{E}_k(\Gamma)$ is bounded above by the number of cusps of Γ .

Definition 6.0.6. (♠ TODO: AI generated, verify) Let N be a positive integer and let $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$ be a group character, viewed as a primitive Dirichlet character modulo N (Definition 2.0.3).

A function $f : \mathbb{H} \rightarrow \mathbb{C}$ is called a *modular form of weight k with nebentypus χ for $\Gamma_1(N)$* if:

1. f is holomorphic (Definition A.1.1) on the upper half-plane $\mathbb{H}$.
2. For every $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(N)$, we have

$$f|_k \gamma = \chi(d) \cdot f,$$

where $f|_k \gamma$ is the weight- k slash operator.

3. For every $\sigma \in \text{SL}_2(\mathbb{Z})$, the function $f|_k \sigma$ is holomorphic at ∞ .

The space of such modular forms is denoted by

$$M_k(\Gamma_1(N), \chi).$$

(♠ TODO: define the notion of fourier expansion at a cusp) A form $f \in M_k(\Gamma_1(N), \chi)$ is called a *cusp form with nebentypus χ* if its Fourier expansion at every cusp has zero constant term. The space of such forms is denoted by

$$S_k(\Gamma_1(N), \chi) \subseteq M_k(\Gamma_1(N), \chi).$$

When χ is the trivial character, $M_k(\Gamma_1(N), \chi)$ coincides with the space $M_k(\Gamma_0(N))$ of modular forms for $\Gamma_0(N)$ as defined by the general definition of modular forms for congruence subgroups (Definition 6.0.3).

Lemma 6.0.7. Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$. Let $h \in \mathbb{Z}_{>0}$ be such that $\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}$ by ???. Every weakly modular form f (Definition 6.0.1) of some weight $k \in \mathbb{Z}$ is $h\mathbb{Z}$ -periodic, i.e. $f(z+h) = f(z)$ for all $z \in \mathbb{H}$.

In particular, there exists a function $g : D' \rightarrow \mathbb{C}$ where $D' \subset \mathbb{C}$ is the punctured unit disk such that $f(z) = g(q_h)$ where $q_h = e^{2\pi iz/h}$.

Theorem 6.0.8 (Eisenstein series for $\mathrm{SL}_2(\mathbb{Z})$ is a modular form of weight k). (♠ TODO: AI generated, verify) Let $k \geq 3$ be an integer. The unnormalized Eisenstein series $G_k(z)$ for $\mathrm{SL}_2(\mathbb{Z})$ defines a modular form (Definition 6.0.3) of weight k for $\mathrm{SL}_2(\mathbb{Z})$. Its Fourier expansion (Definition 9.0.1) at the cusp ∞ is given by

$$G_k(z) = 2\zeta(k) + 2 \frac{(-2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \sigma_{k-1}(n) e^{2\pi i n z},$$

where $\zeta(s)$ is the Riemann zeta function (Definition 3.0.1) and $\sigma_{k-1}(n) = \sum_{d|n} d^{k-1}$ is the *sum of $(k-1)$ -st powers of divisors function* (♠ TODO: AI generated, reference needed: definition of divisor sum function $\sigma_k(n) = \sum_{d|n} d^k$).

Consequently, the normalized Eisenstein series $E_k(z)$ has the Fourier expansion

$$E_k(z) = 1 + \frac{(-2\pi i)^k}{(k-1)! \zeta(k)} \sum_{n=1}^{\infty} \sigma_{k-1}(n) e^{2\pi i n z}.$$

Proof. (♠ TODO: AI generated, verify) We verify each of the defining properties of a modular form.

1. We show that $G_k(z)$ converges absolutely and uniformly on compact subsets of $\mathbb{H}$. Let $K \subset \mathbb{H}$ be compact. For each $z = x + iy \in \mathbb{H}$,

$$|mz + n|^2 = |(mx + n) + imy|^2 = (mx + n)^2 + (my)^2 = m^2(x^2 + y^2) + 2mnx + n^2.$$

This is a positive definite quadratic form in (m, n) with matrix

$$\begin{pmatrix} |z|^2 & x \\ x & 1 \end{pmatrix}.$$

Its minimum eigenvalue

$$\lambda_{\min}(z) = \frac{|z|^2 + 1 - \sqrt{(|z|^2 - 1)^2 + 4x^2}}{2}$$

is positive because $y = \Im(z) > 0$ (the discriminant satisfies $(|z|^2 - 1)^2 + 4x^2 = (|z|^2 + 1)^2 - 4y^2 < (|z|^2 + 1)^2$, so $\lambda_{\min}(z) > 0$). The function $\lambda_{\min} : \mathbb{H} \rightarrow \mathbb{R}_{>0}$ is continuous, and K is compact, so

$$c := \min_{z \in K} \lambda_{\min}(z) > 0.$$

By the spectral theorem, for every $z \in K$ and every $(m, n) \in \mathbb{R}^2$,

$$|mz + n|^2 \geq \lambda_{\min}(z) \cdot (m^2 + n^2) \geq c(m^2 + n^2).$$

Consequently,

$$\frac{1}{|mz + n|^k} \leq \frac{1}{c^{k/2}} \cdot \frac{1}{(m^2 + n^2)^{k/2}}.$$

By ??, the double series $\sum_{(m,n) \neq (0,0)} (m^2 + n^2)^{-k/2}$ is defined by the supremum of its finite partial sums. The function $(x, y) \mapsto (x^2 + y^2)^{-k/2}$ is nonincreasing in each variable separately on $[0, \infty)^2$. By ?? with $m_0 = n_0 = 1$, we have

$$\sum_{m=2}^{\infty} \sum_{n=2}^{\infty} (m^2 + n^2)^{-k/2} \leq \iint_{[1, \infty)^2} (x^2 + y^2)^{-k/2} dx dy.$$

The remaining terms of the double series (those with $m = 0$, $m = 1$, $n = 0$, or $n = 1$) are bounded by the convergent single series $\sum_{m=1}^{\infty} m^{-k} = \zeta(k) < \infty$, and similarly for n .

To evaluate the double integral, apply ?? to the polar coordinate map $\Phi(r, \theta) = (r \cos \theta, r \sin \theta)$. On the region $U = \{(r, \theta) \in (0, \infty) \times (0, \pi/2) : r \cos \theta \geq 1, r \sin \theta \geq 1\}$, the map Φ is a C^1 diffeomorphism onto $[1, \infty)^2$, and $|\det D\Phi(r, \theta)| = r$. Hence

$$\begin{aligned} \iint_{[1, \infty)^2} (x^2 + y^2)^{-k/2} dx dy &= \iint_U r^{-k} \cdot r dr d\theta \\ &\leq \int_0^{\pi/2} \int_1^{\infty} r^{1-k} dr d\theta = \frac{\pi}{2} \int_1^{\infty} r^{1-k} dr, \end{aligned}$$

which is finite for $k > 2$ because $\int_1^{\infty} r^{1-k} dr = 1/(k-2)$. Thus the double series converges. By the Weierstrass M -test (Theorem A.0.2), the series defining $G_k(z)$ converges absolutely and uniformly on K .

2. We show that $G_k(z)$ is holomorphic on $\mathbb{H}$. Each term $(mz + n)^{-k}$ is a holomorphic function of z on $\mathbb{H}$ (it is a rational function with a simple pole at $z = -n/m$ for $m \neq 0$, which lies on the real axis, not in $\mathbb{H}$). The series converges uniformly on compact subsets of $\mathbb{H}$ by the previous step. By Theorem A.5.3, the limit function $G_k(z)$ is holomorphic on $\mathbb{H}$.
3. We verify that G_k is weakly modular (Definition 6.0.1). For $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$, we compute:

$$\begin{aligned} G_k(\gamma z) &= \sum_{(m,n) \neq (0,0)} \frac{1}{\left(m \cdot \frac{az+b}{cz+d} + n\right)^k} \\ &= \sum_{(m,n) \neq (0,0)} \frac{(cz+d)^k}{\left(m(az+b) + n(cz+d)\right)^k} \\ &= (cz+d)^k \sum_{(m,n) \neq (0,0)} \frac{1}{\left((ma+nc)z + (mb+nd)\right)^k}. \end{aligned}$$

Since γ is an invertible 2×2 matrix of integers, the map $(m, n) \mapsto (m, n)\gamma = (ma + nc, mb + nd)$ is a permutation of $\mathbb{Z}^2 \setminus \{(0, 0)\}$. Hence the sum over (m, n) is identical

to the sum over $(m', n') = (m, n)\gamma$, and we obtain

$$G_k(\gamma z) = (cz + d)^k \sum_{(m', n') \neq (0,0)} \frac{1}{(m'z + n')^k} = (cz + d)^k G_k(z).$$

Thus G_k satisfies the modular transformation law of weight k .

4. We show that $G_k(z)$ is holomorphic at ∞ and derive its Fourier series. By Lemma 6.0.7,

$$G_k(z + 1) = G_k(z) \text{ because } \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in \text{SL}_2(\mathbb{Z}). \text{ Indeed,}$$

$$G_k(z + 1) = \sum_{(m,n) \neq (0,0)} \frac{1}{(m(z + 1) + n)^k} = \sum_{(m,n) \neq (0,0)} \frac{1}{(mz + (m + n))^k} = G_k(z)$$

by the change of variables $(m, n) \mapsto (m, n - m)$.

By Theorem 9.0.2, G_k has a Fourier expansion

$$G_k(z) = \sum_{n=-\infty}^{\infty} a_n e^{2\pi i n z}.$$

To compute the coefficients, we split the sum defining G_k into the $m = 0$ and $m \neq 0$ parts. Absolute convergence of the double series (established above) implies, by ??, that the sum is independent of the enumeration of $\mathbb{Z}^2 \setminus \{(0,0)\}$. Grouping the terms as $m = 0$ then $m > 0$ then $m < 0$ specifies one such enumeration, so

$$G_k(z) = \sum_{n \neq 0} \frac{1}{n^k} + \sum_{m=1}^{\infty} \sum_{n=-\infty}^{\infty} \frac{1}{(mz + n)^k} + \sum_{m=-\infty}^{-1} \sum_{n=-\infty}^{\infty} \frac{1}{(mz + n)^k}.$$

The first sum is $2 \sum_{n=1}^{\infty} n^{-k} = 2\zeta(k)$ (for k even, this is well-known; for odd k the sum still converges absolutely).

For the second sum, we apply Theorem 9.0.4. For $m \geq 1$,

$$\sum_{n=-\infty}^{\infty} \frac{1}{(mz + n)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell m z},$$

and similarly for $m \leq -1$ with z replaced by $-|m|z$. Summing over all $m \neq 0$ collects the coefficients into divisor sums:

$$\begin{aligned} G_k(z) &= 2\zeta(k) + \frac{(-2\pi i)^k}{(k-1)!} \sum_{m=1}^{\infty} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell m z} + \frac{(-2\pi i)^k}{(k-1)!} \sum_{m=-\infty}^{-1} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell m z} \\ &= 2\zeta(k) + \frac{(-2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \left(\sum_{d|n} d^{k-1} \right) e^{2\pi i n z} \quad (\text{setting } n = \ell m) \\ &= 2\zeta(k) + 2 \frac{(-2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \sigma_{k-1}(n) e^{2\pi i n z}. \end{aligned}$$

(The factor of 2 arises because the $m = -\ell$ terms contribute the same sum.) This is the claimed Fourier expansion. In particular, there are no negative powers of $e^{2\pi i z}$, so G_k is holomorphic at ∞ by ??.

Having verified all four properties – absolute convergence and uniform convergence on compacta yielding holomorphicity on $\mathbb{H}$, the modular transformation law, periodicity, and holomorphicity at ∞ with the explicit Fourier expansion – we conclude that G_k is a modular form of weight k for $\mathrm{SL}_2(\mathbb{Z})$. $\square$

7. HECKE OPERATORS

Definition 7.0.1 (Hecke operators on modular forms for $\mathrm{SL}_2(\mathbb{Z})$). (♠ TODO: AI generated, verify) Let $k \in \mathbb{Z}$ and let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. For each positive integer n , the *Hecke operator of level n* (or simply the *n -th Hecke operator*) is the linear map

$$T_n : M_k(\Gamma) \longrightarrow M_k(\Gamma)$$

defined on a modular form $f \in M_k(\Gamma)$ (Definition 6.0.5) by

$$(T_n f)(z) = n^{k-1} \sum_{ad=n, d>0, 0 \leq b < d} d^{-k} f \Big|_k \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} (z),$$

Equivalently, using the double coset description, for each $n \geq 1$,

$$\Gamma \begin{pmatrix} n & 0 \\ 0 & 1 \end{pmatrix} \Gamma = \bigsqcup_{ad=n, d>0, 0 \leq b < d} \Gamma \begin{pmatrix} a & b \\ 0 & d \end{pmatrix},$$

and the Hecke operator is given by summing over right coset representatives:

$$(T_n f)(z) = n^{k-1} \sum_{\gamma \in \Gamma_\infty \backslash \Gamma \begin{pmatrix} n & 0 \\ 0 & 1 \end{pmatrix} \Gamma / \Gamma_\infty} (f|_k \gamma)(z),$$

where $\Gamma_\infty = \{\pm \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} : m \in \mathbb{Z}\}$ is the stabilizer of ∞ in Γ .

For a modular form $f \in M_k(\Gamma)$ (Definition 6.0.5) with Fourier expansion (Definition 9.0.1) $f(z) = \sum_{m=0}^{\infty} a_m q^m$, the Fourier coefficients of $T_n f$ are given by

$$T_n f = \sum_{m=0}^{\infty} b_m q^m, \quad \text{where } b_m = \sum_{d|\gcd(m,n)} d^{k-1} a_{mn/d^2}.$$

In particular, if f is a cusp form (Definition 6.0.3), then $T_n f$ is also a cusp form (Definition 6.0.3), so each T_n restricts to an endomorphism

$$T_n : S_k(\Gamma) \rightarrow S_k(\Gamma).$$

Theorem 7.0.2. (♠ TODO: AI generated, verify) Let $k \in \mathbb{Z}$ and let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. For each positive integer n , the Hecke operator (Definition 7.0.1) T_n is a well-defined linear map

$$T_n : M_k(\Gamma) \longrightarrow M_k(\Gamma),$$

and it restricts to an endomorphism $T_n : S_k(\Gamma) \rightarrow S_k(\Gamma)$ on the subspace of cusp forms (Definition 6.0.3).

Proof. We first show that T_n is well defined as a map from $M_k(\Gamma)$ to itself. By definition, for $f \in M_k(\Gamma)$, we have

$$\begin{aligned} (T_n f)(z) &= n^{k-1} \sum_{ad=n, d>0, 0 \leq b < d} d^{-k} f \left|_k \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} (z) \right. \\ &= n^{k-1} \sum_{ad=n, d>0, 0 \leq b < d} d^{-k} \cdot d^{-k} f \left(\frac{az+b}{d} \right) \\ &= n^{k-1} \sum_{ad=n, d>0, 0 \leq b < d} d^{-2k} f \left(\frac{az+b}{d} \right) \end{aligned}$$

The above is a finite sum, $z \mapsto \frac{az+b}{d}$ is holomorphic on $\mathbb{H}$, and f is holomorphic on $\mathbb{H}$, so the above is holomorphic on $\mathbb{H}$.

We need to show that $T_n f$ is a weakly modular form (Definition 6.0.1) of weight k with respect to Γ , i.e. that

$$((T_n f)|_k \gamma)(z) = (T_n f)(z)$$

for all $\gamma \in \Gamma$.

□

Proposition 7.0.3. (♠ TODO: AI generated, verify) Let $k \in \mathbb{Z}$ and let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. The Hecke operators acting on the space of modular forms (Definition 6.0.5) $M_k(\Gamma)$ satisfy the following relations for all positive integers m, n :

1. *Multiplicativity for coprime indices:* If $(m, n) = 1$, then

$$T_m T_n = T_{mn}.$$

2. *Prime power recurrence:* For any prime p and any integer $r \geq 1$,

$$T_p^r T_p = T_p^{r+1} + p^{k-1} T_p^{r-1}.$$

3. *Commutativity:* For all $m, n \geq 1$, the operators commute:

$$T_m T_n = T_n T_m.$$

As a consequence of (1) and (2), the Hecke algebra generated by $\{T_n : n \geq 1\}$ (♠ TODO: AI generated, reference needed: definition of Hecke algebra as the associative algebra over $\mathbb{C}$ generated by the Hecke operators T_n acting on $M_k(\mathrm{SL}_2(\mathbb{Z}))$) is commutative, and every T_n can be expressed as a polynomial in the prime power operators $\{T_{p^r}\}$ using the relations above.

Proposition 7.0.4. (♠ TODO: AI generated, verify) Let k be a positive even integer and let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. For each prime p , the Hecke operator T_p is self-adjoint with respect to the Petersson inner product on the space of cusp forms (Definition 6.0.5) $S_k(\Gamma)$: for all f, g in this space,

$$\langle T_p f, g \rangle = \langle f, T_p g \rangle.$$

As a consequence, the operators $\{T_p : p \text{ prime}\}$ are simultaneously diagonalizable on $S_k(\Gamma)$ with respect to an orthonormal basis of eigenforms.

Proposition 7.0.5. (♠ TODO: AI generated, verify) Let k be a positive even integer and let $\Gamma = \text{SL}_2(\mathbb{Z})$. Let f, g in the space of cusp forms (Definition 6.0.5) $S_k(\Gamma)$ be nonzero simultaneous eigenforms for all Hecke operators with eigenvalues $\{\lambda_n(f)\}_{n \geq 1}$ and $\{\lambda_n(g)\}_{n \geq 1}$ respectively. If there exists some positive integer m such that

$$\lambda_m(f) \neq \lambda_m(g),$$

then f and g are orthogonal with respect to the Petersson inner product:

$$\langle f, g \rangle = 0.$$

In particular, if $\lambda_p(f) \neq \lambda_p(g)$ for some prime p , then $\langle f, g \rangle = 0$.

Corollary 7.0.6. (♠ TODO: AI generated, verify) Let k be a positive even integer and let $\Gamma = \text{SL}_2(\mathbb{Z})$. Let f in the space of cusp forms (Definition 6.0.5) $S_k(\Gamma)$ be a nonzero cusp form (Definition 6.0.3) that is a simultaneous eigenform for all Hecke operators, i.e., there exist complex numbers λ_n such that

$$T_n f = \lambda_n \cdot f \quad \text{for all } n \geq 1.$$

Normalize f so that its first Fourier coefficient is $a_1 = 1$. Then:

1. The eigenvalues satisfy $\lambda_n = a_n$ for all $n \geq 1$, where a_n are the Fourier coefficients of $f(z) = \sum_{n=0}^{\infty} a_n q^n$.
2. The eigenvalues (and Fourier coefficients) are multiplicative: if $(m, n) = 1$, then

$$a_{mn} = a_m a_n.$$

3. The eigenvalues satisfy the recurrence relation for each prime p and $r \geq 1$:

$$a_p^r \cdot a_p = a_{p^{r+1}} + p^{k-1} a_p^{r-1}.$$

4. The L-function of f (Definition 8.0.1) admits an Euler product expansion:

$$L(f, s) = \sum_{n=1}^{\infty} a_n n^{-s} = \prod_p (1 - a_p p^{-s} + p^{k-1-2s})^{-1},$$

which converges absolutely for $\text{Re}(s)$ sufficiently large.

5. The Fourier coefficients a_n are algebraic integers for all $n \geq 1$.

8. L-FUNCTIONS OF MODULAR FORMS

Definition 8.0.1 (L-function of a modular form). Let $k \in \mathbb{Z}_{>0}$, and let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a modular form (Definition 6.0.3) of weight k with respect to a congruence subgroup $\Gamma \subseteq \text{SL}_2(\mathbb{Z})$. Let

$$f(\tau) = \sum_{n=0}^{\infty} a_n e^{\frac{2\pi i n \tau}{h}}$$

be the Fourier expansion of f where $h > 0$ is an appropriate integer.

The *L-function of the modular form f* is the complex function defined for $s \in \mathbb{C}$ with sufficiently large $\operatorname{Re}(s)$ by the Dirichlet series

$$L(f, s) := \sum_{n=1}^{\infty} \frac{a_n}{n^s}.$$

(♠ TODO: define the completed L -function)

9. FOURIER EXPANSIONS AND SUMMATION FORMULAS

Definition 9.0.1 (Fourier expansion of a periodic holomorphic function on the upper half-plane). (♠ TODO: AI generated, verify) Let $h \in \mathbb{R}_{>0}$, let $\mathbb{H}$ denote the upper half-plane, and let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a holomorphic function (Definition A.1.1) satisfying

$$f(z + h) = f(z) \quad \text{for all } z \in \mathbb{H}.$$

For each integer $n \in \mathbb{Z}$, the *n -th Fourier coefficient of f* is the complex number

$$a_n = \frac{1}{h} \int_{z_0}^{z_0+h} f(z) e^{-2\pi i n z / h} dz,$$

(Definition A.3.2) where the integral is taken along any piecewise continuously differentiable path (Definition A.3.1) in $\mathbb{H}$ from z_0 to $z_0 + h$ for some $z_0 \in \mathbb{H}$ (the value is independent of the choice of path and of z_0 by the Cauchy integral theorem (Theorem A.3.5)).

The *Fourier expansion (or Fourier series) of f* is the formal series

$$f(z) = \sum_{n=-\infty}^{\infty} a_n e^{2\pi i n z / h},$$

where a_n are the Fourier coefficients of f . The series converges absolutely for $\Im(z) > 0$ (Theorem 9.0.2).

Theorem 9.0.2 (Periodic holomorphic function on the upper half-plane descends to the punctured disk and has a Fourier–Laurent expansion). (♠ TODO: AI generated, verify) Let $h \in \mathbb{R}_{>0}$, let $\mathbb{H}$ denote the upper half-plane, and let $f : \mathbb{H} \rightarrow \mathbb{C}$ be a holomorphic function (Definition A.1.1) satisfying

$$f(z + h) = f(z) \quad \text{for all } z \in \mathbb{H}.$$

Define the map $\phi : \mathbb{H} \rightarrow \mathbb{C}$ by

$$\phi(z) = e^{2\pi i z / h},$$

where e^w denotes the complex exponential function. Then:

1. The image of ϕ is the punctured unit disk (Definition A.4.1)

$$D^* = \{q \in \mathbb{C} : 0 < |q| < 1\},$$

and $\phi : \mathbb{H} \rightarrow D^*$ is a local biholomorphism (Definition A.1.5) whose fibres are precisely the orbits of the translation $z \mapsto z + h$ (Definition A.1.1).

2. There exists a unique holomorphic function $g : D^* \rightarrow \mathbb{C}$ such that

$$f(z) = g(e^{2\pi iz/h}) \quad \text{for all } z \in \mathbb{H}.$$

3. The function g admits a Laurent expansion (Theorem A.4.3)

$$g(q) = \sum_{n=-\infty}^{\infty} a_n q^n$$

valid for all $q \in D^*$, which converges absolutely and uniformly on every annulus (Definition A.4.2) $0 < r_1 \leq |q| \leq r_2 < 1$. The coefficients a_n are given by the contour integral

$$a_n = \frac{1}{2\pi i} \oint_{|q|=\rho} \frac{g(q)}{q^{n+1}} dq$$

for any $\rho \in (0, 1)$.

4. Substituting $q = e^{2\pi iz/h}$, the Laurent expansion of g becomes the Fourier expansion of f (Definition 9.0.1)

$$f(z) = \sum_{n=-\infty}^{\infty} a_n e^{2\pi inz/h},$$

converging absolutely and uniformly on every compact subset of $\mathbb{H}$. The coefficients a_n coincide with the Fourier coefficients defined by the contour integral formula

$$a_n = \frac{1}{h} \int_{z_0}^{z_0+h} f(z) e^{-2\pi inz/h} dz.$$

Proof. (♠ TODO: AI generated, verify) We prove each part in order.

1. For $z = x + iy \in \mathbb{H}$ with $y > 0$, we have

$$|\phi(z)| = |e^{2\pi i(x+iy)/h}| = e^{-2\pi y/h} < 1,$$

so $\phi(z) \in D^*$. Conversely, for any $q \in D^*$, write $q = re^{i\theta}$ with $0 < r < 1$ and $\theta \in \mathbb{R}$. Then $z = \frac{h}{2\pi}(\theta - i \log r)$ satisfies $\phi(z) = q$, and $\Im(z) = -\frac{h}{2\pi} \log r > 0$, so $z \in \mathbb{H}$. Hence $\phi(\mathbb{H}) = D^*$.

The derivative $\phi'(z) = \frac{2\pi i}{h} e^{2\pi iz/h}$ never vanishes on $\mathbb{H}$ because the exponential function never vanishes. By Proposition A.1.2 and the inverse function theorem, ϕ is a local biholomorphism. Moreover, $\phi(z_1) = \phi(z_2)$ if and only if $e^{2\pi i(z_1 - z_2)/h} = 1$, which holds precisely when $z_1 - z_2 \in h\mathbb{Z}$. Thus the fibres of ϕ are the orbits of the translation $z \mapsto z + h$.

2. Since $f(z + h) = f(z)$, the function f is constant on each fibre of ϕ . Because ϕ is a local biholomorphism, the universal property of the quotient by a local biholomorphism implies that there exists a unique function $g : D^* \rightarrow \mathbb{C}$ such that $f = g \circ \phi$. To see that g is holomorphic, fix $q_0 \in D^*$ and choose $z_0 \in \phi^{-1}(q_0)$. Let U be a neighbourhood of z_0 on which ϕ restricts to a biholomorphism $\phi|_U : U \xrightarrow{\sim} V$. Then $g|_V = f \circ (\phi|_U)^{-1}$, which is a composition of holomorphic maps and hence holomorphic on V . Thus g is holomorphic on D^* .

3. The function g is holomorphic on the punctured disk $D^* = D(0, 1) \setminus \{0\}$. By Theorem A.4.3, g admits a Laurent expansion

$$g(q) = \sum_{n=-\infty}^{\infty} a_n q^n$$

valid for all $q \in D^*$, converging absolutely and uniformly on every closed annulus $0 < r_1 \leq |q| \leq r_2 < 1$. The coefficients are given by the contour integral formula.

4. Substituting $q = e^{2\pi iz/h}$ yields the Fourier expansion of f . The convergence properties follow from the convergence of the Laurent expansion of g : for any compact subset $K \subset \mathbb{H}$, the image $\phi(K)$ is contained in a closed annulus $r_1 \leq |q| \leq r_2$ with $0 < r_1 < r_2 < 1$ (since $\Im(z)$ is bounded away from 0 and ∞ on K), and the Laurent series converges uniformly on that annulus.

It remains to verify the coefficient formula. Applying the Laurent coefficient formula to g and pulling back the contour integral via ϕ , we have for any $\rho \in (0, 1)$:

$$\begin{aligned} a_n &= \frac{1}{2\pi i} \oint_{|q|=\rho} \frac{g(q)}{q^{n+1}} dq \\ &= \frac{1}{2\pi i} \int_{z_0}^{z_0+h} \frac{g(e^{2\pi iz/h})}{e^{2\pi i(n+1)z/h}} \cdot \frac{2\pi i}{h} e^{2\pi iz/h} dz \\ &= \frac{1}{h} \int_{z_0}^{z_0+h} f(z) e^{-2\pi i n z/h} dz, \end{aligned}$$

where the path is the image under ϕ^{-1} of the circle $|q| = \rho$, which is a horizontal segment of length h in $\mathbb{H}$. The independence of the choice of path and of z_0 follows from Theorem A.3.5. Thus the coefficients a_n coincide with the Fourier coefficients defined in Definition 9.0.1.

□

Theorem 9.0.3 (Poisson summation formula for Schwartz functions on $\mathbb{R}$). (♠ TODO: AI generated, verify) Let $f \in \mathcal{S}(\mathbb{R})$ be a Schwartz function on $\mathbb{R}$, and let $\hat{f}$ denote its Fourier transform taken with respect to the communications and signal processing convention

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i \xi x} dx.$$

Then

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{m \in \mathbb{Z}} \hat{f}(m).$$

Proof. (♠ TODO: AI generated, verify) We prove the identity by constructing a periodic function from f and comparing its two representations.

Define $F : \mathbb{R} \rightarrow \mathbb{C}$ by

$$F(x) = \sum_{n \in \mathbb{Z}} f(x + n).$$

Since f is a Schwartz function, the series converges absolutely and uniformly on compact subsets of $\mathbb{R}$, and F is smooth and 1-periodic: $F(x+1) = F(x)$ for all $x \in \mathbb{R}$.

The m -th Fourier coefficient of F is

$$\begin{aligned}
c_m &= \int_0^1 F(x) e^{-2\pi i m x} dx \\
&= \int_0^1 \sum_{n \in \mathbb{Z}} f(x+n) e^{-2\pi i m x} dx \\
&= \sum_{n \in \mathbb{Z}} \int_0^1 f(x+n) e^{-2\pi i m x} dx \quad (\text{absolute convergence justifies interchange}) \\
&= \sum_{n \in \mathbb{Z}} \int_n^{n+1} f(y) e^{-2\pi i m (y-n)} dy \quad (y = x+n) \\
&= \sum_{n \in \mathbb{Z}} \int_n^{n+1} f(y) e^{-2\pi i m y} dy \quad (e^{2\pi i m n} = 1) \\
&= \int_{-\infty}^{\infty} f(y) e^{-2\pi i m y} dy \\
&= \hat{f}(m).
\end{aligned}$$

Since F is smooth and 1-periodic, its Fourier series converges uniformly and pointwise to $F(x)$ for all $x \in \mathbb{R}$ (♠ TODO: AI generated, reference needed: theorem that the Fourier series of a smooth (C^∞) periodic function on $\mathbb{R}$ converges uniformly to the function). Evaluating at $x = 0$, we obtain

$$\sum_{n \in \mathbb{Z}} f(n) = F(0) = \sum_{m \in \mathbb{Z}} c_m = \sum_{m \in \mathbb{Z}} \hat{f}(m),$$

which is precisely the claimed identity. □

Theorem 9.0.4 (Lipschitz summation formula for reciprocal powers on the upper half-plane). (♠ TODO: AI generated, verify) Let $k \geq 2$ be an integer and let $z \in \mathbb{H}$ be a point of the upper half-plane. Then

$$\sum_{n=-\infty}^{\infty} \frac{1}{(z+n)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{m=1}^{\infty} m^{k-1} e^{2\pi i m z},$$

where $e^{2\pi i m z}$ denotes the complex exponential function and $(k-1)!$ is the value of the Gamma function (Definition 3.0.14) $\Gamma(k)$ at the integer k .

Proof. (♠ TODO: AI generated, verify) We prove the identity by computing the Fourier transform of the function $x \mapsto (x+z)^{-k}$ on $\mathbb{R}$ and applying Theorem 9.0.3.

Fix $z \in \mathbb{H}$ and write $z = a + iy$ with $a \in \mathbb{R}$ and $y > 0$. Define $f : \mathbb{R} \rightarrow \mathbb{C}$ by

$$f(x) = \frac{1}{(x+z)^k} = \frac{1}{(x+a+iy)^k}.$$

For $k \geq 2$, the function f is smooth and satisfies $|f(x)| \leq C|x|^{-k}$ for $|x| \gg 0$, hence $f \in L^1(\mathbb{R})$.

We compute the Fourier transform of f with respect to the communications and signal processing convention:

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} \frac{e^{-2\pi i \xi x}}{(x+z)^k} dx, \quad \xi \in \mathbb{R}.$$

Consider the contour integral over a large semicircle in the lower half-plane. For $\xi > 0$, the factor $e^{-2\pi i \xi x}$ decays as $\Im(x) \rightarrow -\infty$ (since $e^{-2\pi i \xi(u+iv)} = e^{-2\pi i \xi u} e^{2\pi \xi v}$, and for $v < 0$ this decays). For $\xi < 0$, the factor decays in the upper half-plane.

Case $\xi > 0$. Close the contour in the lower half-plane with a large semicircle of radius R . The integrand has a single pole of order k at $x = -z = -a - iy$, which lies in the lower half-plane because $\Im(-z) = -y < 0$. The contribution of the semicircle tends to 0 as $R \rightarrow \infty$ (since $k \geq 2$). By the Theorem A.4.9 residue theorem, the integral along the real axis equals $-2\pi i$ times the residue at $x = -z$ (the negative sign arises from the clockwise orientation of the closed contour in the lower half-plane).

The residue of $g(x) = e^{-2\pi i \xi x} (x+z)^{-k}$ at the pole $x = -z$ of order k is

$$\begin{aligned} \text{Res}(g, -z) &= \frac{1}{(k-1)!} \lim_{x \rightarrow -z} \frac{d^{k-1}}{dx^{k-1}} ((x+z)^k g(x)) \\ &= \frac{1}{(k-1)!} \lim_{x \rightarrow -z} \frac{d^{k-1}}{dx^{k-1}} e^{-2\pi i \xi x} \\ &= \frac{1}{(k-1)!} (-2\pi i \xi)^{k-1} e^{-2\pi i \xi(-z)} \\ &= \frac{(-2\pi i \xi)^{k-1}}{(k-1)!} e^{2\pi i \xi z}. \end{aligned}$$

(Definition A.4.7)

Hence for $\xi > 0$,

$$\hat{f}(\xi) = -2\pi i \cdot \text{Res}(g, -z) = -2\pi i \cdot \frac{(-2\pi i \xi)^{k-1}}{(k-1)!} e^{2\pi i \xi z} = \frac{(-2\pi i)^k}{(k-1)!} \xi^{k-1} e^{2\pi i \xi z}.$$

Case $\xi < 0$. Close the contour in the upper half-plane. The function $g(x) = e^{-2\pi i \xi x} (x+z)^{-k}$ has no poles in the upper half-plane (the only pole is at $x = -z$, which lies in the lower half-plane). By the residue theorem, the contour integral encloses no poles, so

$$\hat{f}(\xi) = 0.$$

Case $\xi = 0$. The integral $\int_{-\infty}^{\infty} (x+z)^{-k} dx$ converges absolutely for $k \geq 2$ and equals 0 by a standard residue calculation (or by noting that the integral is unchanged under the change of variable $x \mapsto -x$ after shifting the contour).

Thus we have established

$$\hat{f}(\xi) = \begin{cases} \frac{(-2\pi i)^k}{(k-1)!} \xi^{k-1} e^{2\pi i \xi z}, & \xi > 0, \\ 0, & \xi \leq 0 \end{cases}$$

Application of Poisson summation. For $k \geq 2$, both f and $\hat{f}$ are in $L^1(\mathbb{R})$ (since $\hat{f}(\xi) \sim \xi^{k-1}$ near $\xi = 0$ and decays exponentially as $\xi \rightarrow \infty$). By the standard extension of Theorem 9.0.3 to functions in $L^1(\mathbb{R})$ with L^1 Fourier transform, we have

$$\sum_{n \in \mathbb{Z}} f(n) = \sum_{m \in \mathbb{Z}} \hat{f}(m).$$

The left-hand side is

$$\sum_{n=-\infty}^{\infty} f(n) = \sum_{n=-\infty}^{\infty} \frac{1}{(z+n)^k},$$

which converges absolutely for $k \geq 2$ by comparison with $\sum |n|^{-k}$.

The right-hand side is

$$\begin{aligned} \sum_{m \in \mathbb{Z}} \hat{f}(m) &= \hat{f}(0) + \sum_{m=1}^{\infty} \hat{f}(m) + \sum_{m=-\infty}^{-1} \hat{f}(m) \\ &= 0 + \sum_{m=1}^{\infty} \frac{(-2\pi i)^k}{(k-1)!} m^{k-1} e^{2\pi i m z} + 0 \\ &= \frac{(-2\pi i)^k}{(k-1)!} \sum_{m=1}^{\infty} m^{k-1} e^{2\pi i m z}. \end{aligned}$$

Equating the two sides yields the claimed identity. $\square$

APPENDIX A. ANALYSIS

A.1. Holomorphic Functions.

Definition A.1.1 (Holomorphic function from a subset of $\mathbb{C}^n$ to $\mathbb{C}^m$). Let $n, m \geq 1$ be integers, and consider $\mathbb{C}^n$ and $\mathbb{C}^m$ as complex vector spaces endowed with their standard Euclidean topologies. Let $U \subseteq \mathbb{C}^n$ be an open subset, and let $f : U \rightarrow \mathbb{C}^m$ be a function.

1. We say that f is *holomorphic/(complex) analytic/complex differentiable function at a point $a \in U$* if there exists a $\mathbb{C}$ -linear map

$$L_a : \mathbb{C}^n \rightarrow \mathbb{C}^m$$

(♠ TODO: norms on $\mathbb{C}^n$ are equivalent to one another) such that, for some (and hence any) norm $\|\cdot\|$ on $\mathbb{C}^n$ and $\mathbb{C}^m$, one has

$$\lim_{h \rightarrow 0} \frac{\|f(a+h) - f(a) - L_a(h)\|}{\|h\|} = 0,$$

where the limit is taken over $h \in \mathbb{C}^n$ with $a+h \in U$. In this case, the map L_a is uniquely determined and is called the **complex derivative** (or **complex Fréchet derivative**) of f at a .

In the case of $n, m = 1$, we f is holomorphic at a if and only if the limit

$$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists in $\mathbb{C}$. In that case the limit is denoted $f'(z_0)$ and is called the **complex derivative of f at z_0** .

2. The function f is called **holomorphic/analytic on U** if it is holomorphic at every point $a \in U$.
3. A holomorphic function $\mathbb{C}^n \rightarrow \mathbb{C}^m$ is called **entire**.

Proposition A.1.2 (Complex differentiability implies continuity). (♠ TODO: AI generated, verify) Let $U \subseteq \mathbb{C}$ be an open set and let $f : U \rightarrow \mathbb{C}$ be a holomorphic function (Definition A.1.1) at a point $a \in U$. Equip $\mathbb{C}$ with the topology induced by the complex absolute value and U with the subspace topology. Then f is continuous at a .

Proof. Since f is holomorphic at a , the limit $f'(a) = \lim_{z \rightarrow a} \frac{f(z) - f(a)}{z - a}$ exists in $\mathbb{C}$. We compute

$$f(z) - f(a) = \frac{f(z) - f(a)}{z - a} \cdot (z - a)$$

for $z \neq a$. By ??, the product of limits equals the limit of the product whenever both limits exist. The first factor converges to $f'(a)$ and the second factor converges to 0, so

$$\lim_{z \rightarrow a} (f(z) - f(a)) = f'(a) \cdot 0 = 0.$$

By ??, $\lim_{z \rightarrow a} f(z)$ thus equals $f(a)$, so f is continuous at a . □

Proposition A.1.3 (Complex derivative rules for sum, product, quotient, and chain). (♠ TODO: AI generated, verify) Let $U \subseteq \mathbb{C}$ be an open set and let $f, g : U \rightarrow \mathbb{C}$ be holomorphic functions (Definition A.1.1) at a point $a \in U$.

(Sum rule) The function $f + g$ is holomorphic at a and

$$(f + g)'(a) = f'(a) + g'(a).$$

(Product rule) The function $f \cdot g$ is holomorphic at a and

$$(f \cdot g)'(a) = f'(a)g(a) + f(a)g'(a).$$

(Quotient rule) If in addition $g(a) \neq 0$, then the function f/g is holomorphic at a and

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

(Chain rule) Let $V \subseteq \mathbb{C}$ be an open set and let $h : V \rightarrow \mathbb{C}$ be a holomorphic function (Definition A.1.1) at $f(a)$, with $f(U) \subseteq V$. Then the composition $h \circ f$ is holomorphic at a and

$$(h \circ f)'(a) = h'(f(a)) f'(a).$$

Proof. We prove each rule in turn.

Sum rule. We compute directly:

$$\begin{aligned} (f + g)'(a) &= \lim_{z \rightarrow a} \frac{(f + g)(z) - (f + g)(a)}{z - a} \\ &= \lim_{z \rightarrow a} \left(\frac{f(z) - f(a)}{z - a} + \frac{g(z) - g(a)}{z - a} \right). \end{aligned}$$

By ??, since both limits on the right-hand side exist (as f and g are holomorphic at a), the limit of the sum equals the sum of the limits:

$$(f + g)'(a) = f'(a) + g'(a).$$

Product rule. Add and subtract $f(z)g(a)$ in the numerator:

$$\begin{aligned} (f \cdot g)'(a) &= \lim_{z \rightarrow a} \frac{f(z)g(z) - f(a)g(a)}{z - a} \\ &= \lim_{z \rightarrow a} \left(\frac{f(z) - f(a)}{z - a} g(z) + f(a) \frac{g(z) - g(a)}{z - a} \right). \end{aligned}$$

The first term on the right-hand side is a product of two factors. By ??, since $\lim_{z \rightarrow a} \frac{f(z) - f(a)}{z - a} = f'(a)$ and $\lim_{z \rightarrow a} g(z) = g(a)$ (the latter by Proposition A.1.2), the limit of their product is $f'(a)g(a)$. The second term has limit $f(a)g'(a)$. Summing gives

$$(f \cdot g)'(a) = f'(a)g(a) + f(a)g'(a).$$

Quotient rule. Write $\frac{f}{g} = f \cdot \frac{1}{g}$. Since g is continuous at a by Proposition A.1.2 and $g(a) \neq 0$, there exists a neighborhood of a on which $g(z) \neq 0$. First compute the derivative of $1/g$ at a : for z near a with $z \neq a$,

$$\begin{aligned} \frac{\frac{1}{g(z)} - \frac{1}{g(a)}}{z - a} &= \frac{g(a) - g(z)}{(z - a)g(z)g(a)} \\ &= -\frac{g(z) - g(a)}{z - a} \cdot \frac{1}{g(z)g(a)}. \end{aligned}$$

Taking the limit as $z \rightarrow a$, the first factor converges to $g'(a)$ and the second converges to $1/g(a)^2$ by continuity of g at a and ?. Hence

$$\left(\frac{1}{g} \right)'(a) = -\frac{g'(a)}{g(a)^2}.$$

Now apply the product rule to $f \cdot \frac{1}{g}$:

$$\left(\frac{f}{g}\right)'(a) = f'(a) \cdot \frac{1}{g(a)} + f(a) \cdot \left(-\frac{g'(a)}{g(a)^2}\right) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

Chain rule. For z near a with $z \neq a$, write the difference quotient

$$\frac{(h \circ f)(z) - (h \circ f)(a)}{z - a} = \frac{h(f(z)) - h(f(a))}{f(z) - f(a)} \cdot \frac{f(z) - f(a)}{z - a},$$

provided $f(z) \neq f(a)$. Since f is holomorphic at a , it is continuous at a by Proposition A.1.2, so $f(z) \rightarrow f(a)$ as $z \rightarrow a$. If $f(z) \neq f(a)$ for all z sufficiently close to a (other than $z = a$), then the first factor converges to $h'(f(a))$ by definition of the complex derivative of h at $f(a)$, and the second factor converges to $f'(a)$. Hence

$$(h \circ f)'(a) = h'(f(a)) f'(a),$$

by ??.

If instead $f(z) = f(a)$ for some z arbitrarily close to a , then on those points the difference quotient $\frac{h(f(z)) - h(f(a))}{z - a}$ equals 0. In this case, the function f is constant on a set with a as an accumulation point, so $f'(a) = 0$ and the chain rule formula reduces to $0 = h'(f(a)) \cdot 0$, which holds trivially. More rigorously, define the auxiliary function

$$\varphi(w) = \begin{cases} \frac{h(w) - h(f(a))}{w - f(a)} & \text{if } w \neq f(a), \\ h'(f(a)) & \text{if } w = f(a), \end{cases}$$

which is continuous at $w = f(a)$ by the definition of $h'(f(a))$. Then

$$\frac{h(f(z)) - h(f(a))}{z - a} = \varphi(f(z)) \cdot \frac{f(z) - f(a)}{z - a}$$

holds for all $z \neq a$, because when $f(z) = f(a)$ the right-hand side becomes $h'(f(a)) \cdot 0 = 0$, matching the left-hand side. Taking the limit $z \rightarrow a$ and applying the composition rule of ?? to $\varphi \circ f$ together with the product rule gives

$$(h \circ f)'(a) = \lim_{z \rightarrow a} \varphi(f(z)) \cdot \lim_{z \rightarrow a} \frac{f(z) - f(a)}{z - a} = h'(f(a)) \cdot f'(a).$$

□

Proposition A.1.4 (Complex polynomials are entire). (♠ TODO: AI generated, verify) Let $p : \mathbb{C} \rightarrow \mathbb{C}$ be a polynomial function, i.e. a function of the form

$$p(z) = \sum_{k=0}^n a_k z^k$$

with coefficients $a_k \in \mathbb{C}$. Then p is holomorphic (Definition A.1.1) on all of $\mathbb{C}$, i.e. p is an entire function.

Proof. (♠ TODO: AI generated, verify) We first show that the constant function $c(z) = c$ and the identity function $\text{id}(z) = z$ are holomorphic on $\mathbb{C}$. For any $z \in \mathbb{C}$,

$$\lim_{h \rightarrow 0} \frac{c(z+h) - c(z)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = 0,$$

so c is holomorphic with derivative 0. For the identity,

$$\lim_{h \rightarrow 0} \frac{(z+h) - z}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1,$$

so id is holomorphic with derivative 1.

Next we prove by induction that z^n is holomorphic on $\mathbb{C}$ for every $n \in \mathbb{N}$. The base case $n = 0$ is the constant function 1, which is holomorphic as shown above. Suppose that z^{n-1} is holomorphic. Then $z^n = z \cdot z^{n-1}$ is a product of two holomorphic functions, hence holomorphic by the product rule in Proposition A.1.3.

A polynomial $p(z) = \sum_{k=0}^n a_k z^k$ is a finite sum of the holomorphic functions $a_k z^k$ (each $a_k z^k$ is a scalar multiple of z^k , and scalar multiples of holomorphic functions are holomorphic by the same argument as the constant multiple case, or by the product rule with a constant function). Therefore p is holomorphic on $\mathbb{C}$ by the sum rule in Proposition A.1.3. $\square$

Definition A.1.5 (Biholomorphic map of subsets of $\mathbb{C}$). Let $U \subseteq \mathbb{C}^n$ and $V \subseteq \mathbb{C}^m$ be open subsets for integer $n, m \geq 1$. A map $f : U \rightarrow V$ is called a *biholomorphic map of subsets of $\mathbb{C}$* , or simply a *biholomorphism between U and V* , if:

1. f is bijective as a map of sets,
2. f and f^{-1} are holomorphic (Definition A.1.1) on U .

In this case, we say that U and V are *(complex-)analytically isomorphic as open subsets of $\mathbb{C}$* . Moreover, n and m must necessarily be equal in this case.

A.2. Complex Power Series.

Definition A.2.1. Let $(a_n)_{n=0}^\infty$ be a sequence of complex numbers. The *power series centered at $z_0 \in \mathbb{C}$ of this sequence* is the expression

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n.$$

Its *radius of convergence* $R \in [0, \infty]$ is given by the Cauchy–Hadamard formula Proposition A.2.2. The series converges absolutely and uniformly on every closed disk $\overline{D}(z_0, r)$ with $r < R$ by Proposition A.2.3, and diverges for every z with $|z - z_0| > R$ by Proposition A.2.2.

Proposition A.2.2 (Cauchy–Hadamard formula for power series in one complex variable). Let $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ be a power series in one complex variable (Definition A.2.1) with coefficients $a_n \in \mathbb{C}$. Its radius of convergence $R \in [0, \infty]$ is given by the Cauchy–Hadamard formula

$$R = \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{1/n}},$$

with the conventions $1/0 = \infty$ and $1/\infty = 0$. More precisely, the series converges absolutely for every z with $|z - z_0| < R$ and diverges for every z with $|z - z_0| > R$.

Proof. (♠ TODO: technically the root test is so far only written for $\mathbb{R}$; generalize it) Fix $z \in \mathbb{C}$ with $|z - z_0| = r$ and consider the series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$. The sequence of nonnegative real numbers $|a_n(z - z_0)^n|$ satisfies

$$\limsup_{n \rightarrow \infty} |a_n(z - z_0)^n|^{1/n} = r \cdot \limsup_{n \rightarrow \infty} |a_n|^{1/n}.$$

If $r < R$, then $r \cdot \limsup_{n \rightarrow \infty} |a_n|^{1/n} < 1$, so ?? implies that $\sum_{n=0}^{\infty} |a_n(z - z_0)^n|$ converges absolutely. If $r > R$, then $r \cdot \limsup_{n \rightarrow \infty} |a_n|^{1/n} > 1$, so by the same theorem the series $\sum_{n=0}^{\infty} |a_n(z - z_0)^n|$ diverges, and consequently $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ diverges. $\square$

Proposition A.2.3 (Convergence of a power series on closed disks inside its radius of convergence). Let $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ be a power series in one complex variable (Definition A.2.1) with coefficients $a_n \in \mathbb{C}$ and with radius of convergence $R \in (0, \infty]$, given by the Cauchy–Hadamard formula (Proposition A.2.2).

For every r with $0 \leq r < R$, the series converges absolutely and uniformly on the closed disk $\overline{D}(z_0, r) = \{z \in \mathbb{C} : |z - z_0| \leq r\}$.

Proof. Fix r with $0 \leq r < R$, and denote the closed disk by $\overline{D} = \overline{D}(z_0, r)$.

We first verify that the series $\sum_{n=0}^{\infty} |a_n| r^n$ converges. By Proposition A.2.2, we have

$$\limsup_{n \rightarrow \infty} |a_n|^{1/n} = 1/R$$

(with the conventions $1/\infty = 0$ and $1/0 = \infty$). Therefore,

$$\limsup_{n \rightarrow \infty} (|a_n| r^n)^{1/n} = r \cdot \limsup_{n \rightarrow \infty} |a_n|^{1/n} = \frac{r}{R} < 1.$$

then implies that $\sum_{n=0}^{\infty} |a_n| r^n$ converges.

Next we show absolute convergence at each point of $\overline{D}$. For any fixed $z \in \overline{D}$, we have $|z - z_0| \leq r$, so the multiplicativity of the complex absolute value implies that

$$|a_n(z - z_0)^n| = |a_n| |z - z_0|^n \leq |a_n| r^n$$

for each n . Since $\sum_{n=0}^{\infty} |a_n| r^n$ converges, Theorem A.0.3 implies that $\sum_{n=0}^{\infty} |a_n(z - z_0)^n|$ converges, i.e. the power series converges absolutely at z .

Finally, we prove uniform convergence on $\overline{D}$. Set $M_n = |a_n| r^n$ for each n . The series $\sum_{n=0}^{\infty} M_n$ converges in $\mathbb{R}$ as shown above, so the sequence (M_n) satisfies the Cauchy tails condition by the Cauchy criterion. For every $z \in \overline{D}$ and every n , we have $|a_n(z - z_0)^n| \leq M_n$ as shown above. The functions $f_n(z) = a_n(z - z_0)^n$ map $\overline{D}$ to $\mathbb{C}$, and $\mathbb{C}$ is a Banach space over itself with respect to the complex absolute value by (♠ TODO: state and reference as a fact that C^m is a banach space over C valued in R and state another fact that R^n is a banach space over R valued in R .) By Theorem A.0.2, the series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ converges uniformly on $\overline{D}$. $\square$

Lemma A.2.4 (A power series converges uniformly on compact subsets of its disk of convergence). Let $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ be a power series in one complex variable (Definition A.2.1) with coefficients $a_n \in \mathbb{C}$ and with radius of convergence (Definition A.2.1) $R \in (0, \infty]$. For every compact subset $K \subset D(z_0, R)$, the series converges uniformly and absolutely on K .

Proof. By Proposition A.2.3, the series converges uniformly and absolutely on every closed disk $\overline{D}(z_0, r)$ with $r < R$.

The function $d(z) = |z - z_0|$ is continuous on K (♠ TODO: factor out and reference the general statement that $d(z) = |z - z_0|$ is a continuous function; is this true of general extended metric spaces valeud in ordered rings?) the reverse triangle inequality. Since K is compact, ?? implies that d attains a maximum $r_K = \max_{z \in K} |z - z_0|$ on K . Because $K \subset D(z_0, R)$, we have $r_K < R$. Hence $K \subset \overline{D}(z_0, r_K)$, and the convergence on $\overline{D}(z_0, r_K)$ is uniform and absolute. Therefore the convergence is uniform and absolute on K . $\square$

Definition A.2.5. Let $z_0 \in \mathbb{C}$ and $r > 0$. The *open disk of radius r centered at z_0* is the subset

$$D(z_0, r) = \{z \in \mathbb{C} : |z - z_0| < r\}$$

of the field of complex numbers $\mathbb{C}$, where $|\cdot|$ denotes the complex absolute value. The *closed disk of radius r centered at z_0* is the subset

$$\overline{D}(z_0, r) = \{z \in \mathbb{C} : |z - z_0| \leq r\}.$$

It should be noted that $\overline{D}(z_0, r)$ is the closure of $D(z_0, r)$ as a subspace of $\mathbb{C}$.

Definition A.2.6. Let $U \subset \mathbb{C}$ be an open set, let $z_0 \in U$, and let $f : U \rightarrow \mathbb{C}$ be a holomorphic function (Definition A.1.1) on U .

Suppose there exists $R \in (0, \infty]$ such that the open disk $D(z_0, R) = \{z \in \mathbb{C} : |z - z_0| < R\}$ is contained in U . Then f admits a *Taylor series expansion centered at z_0* , which is the following power series (Definition A.2.1):

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$$

for all $z \in D(z_0, R)$, where the series converges uniformly on every compact subset of $D(z_0, R)$ by Lemma A.2.4. The coefficients are given by $a_n = \frac{f^{(n)}(z_0)}{n!}$ for $n = 0, 1, 2, \dots$

Theorem A.2.7. [Euler's Formula] For every $x \in \mathbb{R}$, the complex exponential satisfies

$$\exp(ix) = \cos(x) + i \sin(x)$$

where $\exp$ is the complex extension of the exponential function, and $\sin$, $\cos$ are the sine and cosine functions.

Proof. We compare the power series expansions of both sides. The complex exponential is defined by (see):

$$\exp(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!},$$

which converges absolutely for every $z \in \mathbb{C}$. Substituting $z = ix$:

$$\exp(ix) = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!}.$$

We separate the terms of this series into even and odd indices. For each nonnegative integer k , we have:

$$i^{2k} = (i^2)^k = (-1)^k, \quad i^{2k+1} = i^{2k} \cdot i = (-1)^k i.$$

Splitting the sum accordingly:

$$\begin{aligned} \exp(ix) &= \sum_{k=0}^{\infty} \frac{i^{2k} x^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{i^{2k+1} x^{2k+1}}{(2k+1)!} \\ &= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} + i \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}. \end{aligned}$$

The first series is the power series for $\cos(x)$ and the second is the power series for $\sin(x)$:

$$\exp(ix) = \cos(x) + i \sin(x).$$

□

A.3. Contour Integration and Cauchy Theory.

Definition A.3.1. Let $[a, b] \subset \mathbb{R}$ be a closed real interval. A continuous path $\gamma : [a, b] \rightarrow \mathbb{C}$ is called a *piecewise continuously differentiable path* if there exists a partition $a = t_0 < t_1 < \dots < t_k = b$ such that:

1. γ is continuously differentiable on each open subinterval (t_{i-1}, t_i) for $i = 1, \dots, k$, and
2. the one-sided derivatives $\lim_{t \rightarrow t_{i-1}^+} \gamma'(t)$ and $\lim_{t \rightarrow t_i^-} \gamma'(t)$ exist (and are finite) for each $i = 1, \dots, k$.

The derivative $\gamma'(t)$ is defined for all $t \in [a, b] \setminus \{t_1, \dots, t_{k-1}\}$.

Definition A.3.2. Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a piecewise continuously differentiable path (Definition A.3.1) and let $f : \gamma([a, b]) \rightarrow \mathbb{C}$ be a continuous complex-valued function defined on the image of γ . The *contour integral/line integral of f along γ* , denoted

$$\int_{\gamma} f(z) dz,$$

is defined by

$$\int_{\gamma} f(z) dz := \int_a^b f(\gamma(t)) \gamma'(t) dt,$$

where $\gamma'(t)$ exists for all $t \in [a, b] \setminus \{t_1, \dots, t_{k-1}\}$ and the Riemann integral on the right-hand side is well-defined since $f \circ \gamma$ is continuous and γ' is piecewise continuous on $[a, b]$. (♠
TODO: talk about how this is preserved under orientation preserving reparameterizations)

In case γ is a closed path, one denotes $\int_{\gamma} f(z) dz$ by $\oint_{\gamma} f(z) dz$

Proposition A.3.3 (Linearity of the contour integral). (♠ TODO: AI generated, verify) Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a piecewise continuously differentiable path (Definition A.3.1) and let $f, g : \gamma([a, b]) \rightarrow \mathbb{C}$ be continuous functions on the image of γ . Let $c \in \mathbb{C}$.

1. (Sum rule) $\int_{\gamma} (f(z) + g(z)) dz = \int_{\gamma} f(z) dz + \int_{\gamma} g(z) dz.$
2. (Scalar multiple rule) $\int_{\gamma} c \cdot f(z) dz = c \int_{\gamma} f(z) dz.$

Proof. Both statements follow directly from the definition of the contour integral (Definition A.3.2) and the linearity of the Riemann integral. Indeed,

$$\begin{aligned} \int_{\gamma} (f(z) + g(z)) dz &= \int_a^b (f(\gamma(t)) + g(\gamma(t))) \gamma'(t) dt \\ &= \int_a^b f(\gamma(t)) \gamma'(t) dt + \int_a^b g(\gamma(t)) \gamma'(t) dt \\ &= \int_{\gamma} f(z) dz + \int_{\gamma} g(z) dz, \end{aligned}$$

and similarly

$$\begin{aligned} \int_{\gamma} c \cdot f(z) dz &= \int_a^b c \cdot f(\gamma(t)) \gamma'(t) dt \\ &= c \int_a^b f(\gamma(t)) \gamma'(t) dt = c \int_{\gamma} f(z) dz, \end{aligned}$$

where in both cases the Riemann integral of a complex-valued function is understood componentwise and the linearity of the Riemann integral (??) applies to the real and imaginary parts separately. $\square$

Definition A.3.4. (♠ TODO: AI generated, verify) Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a closed piecewise continuously differentiable path (Definition A.3.1) and let $z_0 \in \mathbb{C} \setminus \gamma([a, b])$ be a point not lying on the image of γ . The *winding number of γ around z_0* , denoted by $n(\gamma, z_0)$, is defined by

$$n(\gamma, z_0) := \frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z - z_0}.$$

(Definition A.3.2)

Theorem A.3.5. (♠ TODO: AI generated, verify) Let $U \subset \mathbb{C}$ be a simply connected open subset (a simply connected domain) and let $f : U \rightarrow \mathbb{C}$ be a holomorphic function (Definition A.1.1) on U . Then for every closed piecewise continuously differentiable path (Definition A.3.1) γ in U ,

$$\oint_{\gamma} f(z) dz = 0$$

(Definition A.3.2)

Proof. (♠ TODO: AI generated, verify) We prove the statement by showing that f admits a primitive on U , then applying the fundamental theorem of calculus.

Fix a base point $z_0 \in U$. Since U is simply connected, it is path-connected. For any $z \in U$, choose a piecewise continuously differentiable path η_z from z_0 to z in U . Define $F : U \rightarrow \mathbb{C}$ by

$$F(z) := \int_{\eta_z} f(w) dw.$$

We first show that F is well-defined, i.e. independent of the choice of η_z . Let η_z and $\tilde{\eta}_z$ be two paths from z_0 to z in U . Their concatenation $\gamma' = \eta_z \cdot \tilde{\eta}_z^{-1}$ is a closed path. We approximate γ' by a finite chain of triangles inside U . For each triangle contained in a disk of holomorphy, the contour integral of f over its boundary vanishes because f has a local primitive on any disk (by integrating its power series termwise, as holomorphic functions are locally given by convergent power series). By subdividing γ' into sufficiently small segments and applying this local vanishing property to each triangular subdivision, we obtain

$$\oint_{\eta_z \cdot \tilde{\eta}_z^{-1}} f(w) dw = 0,$$

so $\int_{\eta_z} f = \int_{\tilde{\eta}_z} f$, and F is well-defined.

We verify that $F'(z) = f(z)$ for all $z \in U$. For small $h \in \mathbb{C}$ with $z + h \in U$, let ℓ_h be the straight line segment from z to $z + h$. Because F is well-defined, we may take $\eta_{z+h} = \eta_z \cdot \ell_h$, so

$$F(z + h) - F(z) = \int_{\ell_h} f(w) dw.$$

By Definition A.3.2, we compute

$$\int_{\ell_h} f(w) dw = \int_0^1 f(z + th) \cdot h dt = h \int_0^1 f(z + th) dt.$$

Therefore,

$$\frac{F(z + h) - F(z)}{h} - f(z) = \int_0^1 (f(z + th) - f(z)) dt.$$

For any $\varepsilon > 0$, f is continuous at z by Proposition A.1.2, so there exists $\delta > 0$ such that $|w - z| < \delta$ implies $|f(w) - f(z)| < \varepsilon$. For $|h| < \delta$, we have $|z + th - z| = t|h| \leq |h| < \delta$ for all $t \in [0, 1]$, thus

$$\left| \int_0^1 (f(z + th) - f(z)) dt \right| \leq \int_0^1 |f(z + th) - f(z)| dt < \varepsilon.$$

Hence $\lim_{h \rightarrow 0} \frac{F(z+h) - F(z)}{h} = f(z)$, so $F'(z) = f(z)$.

Having a primitive F of f on U , we apply the fundamental theorem of calculus. For $\gamma : [a, b] \rightarrow U$,

$$\oint_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt = \int_a^b F'(\gamma(t)) \gamma'(t) dt = F(\gamma(b)) - F(\gamma(a)).$$

Since γ is closed, $\gamma(b) = \gamma(a)$, so the difference vanishes. □

Theorem A.3.6. (♠ TODO: AI generated, verify) Let $U \subset \mathbb{C}$ be a simply connected open subset and let $f : U \rightarrow \mathbb{C}$ be a holomorphic function (Definition A.1.1) on U . Let γ be a closed piecewise continuously differentiable path (Definition A.3.1) in U whose image is a simple closed curve, and let z_0 be a point in the interior of γ (i.e. z_0 lies in the bounded component of $\mathbb{C} \setminus \gamma([a, b])$). Then for every non-negative integer n ,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz,$$

where $f^{(n)}$ denotes the n -th derivative of f . In particular, for $n = 0$:

$$f(z_0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - z_0} dz.$$

Proof. (♠ TODO: AI generated, verify) We first prove the case $n = 0$, then obtain the general formula by differentiating under the integral sign.

1. We prove $f(z_0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - z_0} dz$.

Let $\varepsilon > 0$ be sufficiently small that the closed disk $\overline{D}(z_0, \varepsilon)$ is contained in the interior of γ . Let γ_{ε} be the positively oriented circle of radius ε centered at z_0 . By Theorem A.3.5 applied to the function $g(z) = \frac{f(z)}{z - z_0}$ on the region between γ and γ_{ε} , we have

$$\oint_{\gamma} \frac{f(z)}{z - z_0} dz = \oint_{\gamma_{\varepsilon}} \frac{f(z)}{z - z_0} dz.$$

On γ_{ε} , parametrize by $z = z_0 + \varepsilon e^{i\theta}$ with $\theta \in [0, 2\pi]$. Then $dz = i\varepsilon e^{i\theta} d\theta$ and $z - z_0 = \varepsilon e^{i\theta}$, so

$$\begin{aligned} \oint_{\gamma_{\varepsilon}} \frac{f(z)}{z - z_0} dz &= i \int_0^{2\pi} f(z_0 + \varepsilon e^{i\theta}) d\theta \\ &= i \int_0^{2\pi} (f(z_0) + (f(z_0 + \varepsilon e^{i\theta}) - f(z_0))) d\theta \\ &= 2\pi i f(z_0) + i \int_0^{2\pi} (f(z_0 + \varepsilon e^{i\theta}) - f(z_0)) d\theta. \end{aligned}$$

We bound the remainder. Since f is continuous at z_0 by Proposition A.1.2, for any $\eta > 0$ there exists $\delta > 0$ such that $|z - z_0| < \delta$ implies $|f(z) - f(z_0)| < \eta$. Choose $\varepsilon < \delta$. Then

$$\left| i \int_0^{2\pi} (f(z_0 + \varepsilon e^{i\theta}) - f(z_0)) d\theta \right| \leq \int_0^{2\pi} |f(z_0 + \varepsilon e^{i\theta}) - f(z_0)| d\theta < 2\pi\eta.$$

Taking $\eta \rightarrow 0$ (equivalently $\varepsilon \rightarrow 0$), the remainder vanishes, so

$$\oint_{\gamma} \frac{f(z)}{z - z_0} dz = \lim_{\varepsilon \rightarrow 0} \oint_{\gamma_{\varepsilon}} \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0).$$

Dividing by $2\pi i$ yields the desired formula.

2. Now we prove the formula for $n \geq 1$ by induction on n .

For $n = 1$, we differentiate the formula for $f(z_0)$ with respect to z_0 . Since f is holomorphic, the integrand $\frac{f(z)}{(z-z_0)^2}$ is a continuously differentiable function of z_0 , and differentiation under the integral sign is justified by the dominated convergence theorem (or by uniform convergence on compact subsets away from z_0). We compute

$$\begin{aligned} f'(z_0) &= \frac{d}{dz_0} \left(\frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - z_0} dz \right) \\ &= \frac{1}{2\pi i} \oint_{\gamma} \frac{\partial}{\partial z_0} \left(\frac{f(z)}{z - z_0} \right) dz \\ &= \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^2} dz. \end{aligned}$$

Multiplying both sides by $1!$ gives the claimed formula.

Assume that for some $n \geq 1$, we have $f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z-z_0)^{n+1}} dz$. Differentiating again with respect to z_0 under the integral sign,

$$\begin{aligned} f^{(n+1)}(z_0) &= \frac{d}{dz_0} \left(\frac{n!}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz \right) \\ &= \frac{n!}{2\pi i} \oint_{\gamma} \frac{\partial}{\partial z_0} \left(\frac{1}{(z - z_0)^{n+1}} \right) f(z) dz \\ &= \frac{n!}{2\pi i} \oint_{\gamma} \frac{(n+1)f(z)}{(z - z_0)^{n+2}} dz \\ &= \frac{(n+1)!}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{n+2}} dz, \end{aligned}$$

which completes the inductive step. □

Lemma A.3.7. (♠ TODO: AI generated, verify) Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a closed piecewise continuously differentiable path (Definition A.3.1) and let $z_0 \in \mathbb{C} \setminus \gamma([a, b])$ be a point not lying on the image of γ . For every integer $n \in \mathbb{Z}$, we have

$$\oint_{\gamma} (z - z_0)^n dz = \begin{cases} 2\pi i \cdot n(\gamma, z_0), & n = -1, \\ 0, & n \neq -1, \end{cases}$$

where $n(\gamma, z_0)$ is the winding number of γ around z_0 (Definition A.3.4). (Definition A.3.2)

Proof. (♠ TODO: AI generated, verify) We consider two cases.

1. Suppose $n \neq -1$. The function $(z - z_0)^n$ admits a holomorphic primitive on $\mathbb{C} \setminus \{z_0\}$, namely an antiderivative F with $F' = (z - z_0)^n$, namely

$$F(z) = \frac{(z - z_0)^{n+1}}{n+1},$$

whose derivative is $F'(z) = (z - z_0)^n$ (Definition A.1.1).

By Definition A.3.2, we have

$$\int_{\gamma} (z - z_0)^n dz = \int_a^b (\gamma(t) - z_0)^n \gamma'(t) dt.$$

Since $\frac{d}{dt} F(\gamma(t)) = F'(\gamma(t))\gamma'(t) = (\gamma(t) - z_0)^n \gamma'(t)$ by the chain rule (Proposition A.1.3) and the definition of F , by ?? applied separately to the real and imaginary parts of $F \circ \gamma$ (which are real-valued differentiable functions) we obtain

$$\int_a^b (\gamma(t) - z_0)^n \gamma'(t) dt = F(\gamma(b)) - F(\gamma(a)).$$

Because γ is a closed path, we have $\gamma(b) = \gamma(a)$, so $F(\gamma(b)) = F(\gamma(a))$ and therefore

$$\int_{\gamma} (z - z_0)^n dz = 0.$$

2. Suppose $n = -1$. By Definition A.3.4, we have

$$n(\gamma, z_0) = \frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z - z_0},$$

so multiplying both sides by $2\pi i$ gives $\int_{\gamma} (z - z_0)^{-1} dz = 2\pi i \cdot n(\gamma, z_0)$.

Thus the claimed formula holds in both cases. □

A.4. Laurent Series, Singularities, and the Residue Theorem.

Definition A.4.1. Let $z_0 \in \mathbb{C}$ and $r > 0$. The *punctured disk of radius r centered at z_0* is the subset

$$D^*(z_0, r) = \{z \in \mathbb{C} : 0 < |z - z_0| < r\}$$

of the field of complex numbers $\mathbb{C}$, where $|\cdot|$ denotes the complex absolute value. Equivalently, $D^*(z_0, r) = D(z_0, r) \setminus \{z_0\}$ (Definition A.2.5).

Definition A.4.2. Let $z_0 \in \mathbb{C}$ and let $0 < r_1 < r_2$. The *annulus centered at z_0 with inner radius r_1 and outer radius r_2* is the subset

$$A(z_0, r_1, r_2) = \{z \in \mathbb{C} : r_1 < |z - z_0| < r_2\}$$

of the field of complex numbers $\mathbb{C}$, where $|\cdot|$ denotes the complex absolute value.

Theorem A.4.3 (Laurent series expansion). (♠ TODO: AI generated, verify) Let $U \subseteq \mathbb{C}$ be an open set, let $z_0 \in U$, and let $f : U \setminus \{z_0\} \rightarrow \mathbb{C}$ be a holomorphic function (Definition A.1.1) with an isolated singularity (Definition A.4.5) at z_0 .

Suppose that there exists $R > 0$ such that the punctured disk (Definition A.4.1) $D^*(z_0, R) = \{z \in \mathbb{C} : 0 < |z - z_0| < R\}$ is contained in $U \setminus \{z_0\}$. Then f admits a Laurent series expansion (Definition A.4.4) centered at z_0 , i.e. a series of the form

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

that converges absolutely for every z with $0 < |z - z_0| < R$ and uniformly on every annulus (Definition A.4.2) $r_1 \leq |z - z_0| \leq r_2$ with $0 < r_1 < r_2 < R$. The coefficients are uniquely determined by the contour integral (Definition A.3.2)

$$a_n = \frac{1}{2\pi i} \int_{|w-z_0|=\rho} \frac{f(w)}{(w-z_0)^{n+1}} dw$$

for any $\rho \in (0, R)$, and are independent of the choice of ρ .

Proof. (♠ TODO: AI generated, verify) We prove the existence of the Laurent series expansion by applying the Cauchy integral formula on two concentric circles and expanding the Cauchy kernel as a geometric series.

For each r with $0 < r < R$, let γ_r denote the positively oriented circle centered at z_0 of radius r , parametrized by $\gamma_r(t) = z_0 + re^{2\pi it}$ for $t \in [0, 1]$. Fix any point z with $0 < |z - z_0| < R$. Choose radii r_1, r_2 such that $0 < r_1 < |z - z_0| < r_2 < R$.

By Theorem A.3.6 applied to the holomorphic function f on the annulus bounded by γ_{r_1} and γ_{r_2} , we have

$$f(z) = \frac{1}{2\pi i} \oint_{\gamma_{r_2}} \frac{f(w)}{w-z} dw - \frac{1}{2\pi i} \oint_{\gamma_{r_1}} \frac{f(w)}{w-z} dw.$$

We expand each Cauchy kernel as a geometric series. For w on γ_{r_2} , we have $|w - z_0| = r_2 > |z - z_0|$, so

$$\frac{1}{w-z} = \frac{1}{(w-z_0) - (z-z_0)} = \frac{1}{w-z_0} \cdot \frac{1}{1 - \frac{z-z_0}{w-z_0}} = \sum_{n=0}^{\infty} \frac{(z-z_0)^n}{(w-z_0)^{n+1}},$$

where the series converges uniformly in w on γ_{r_2} because $\left| \frac{z-z_0}{w-z_0} \right| < 1$. For w on γ_{r_1} , we have $|w - z_0| = r_1 < |z - z_0|$, so

$$\frac{1}{w-z} = -\frac{1}{(z-z_0) - (w-z_0)} = -\frac{1}{z-z_0} \cdot \frac{1}{1 - \frac{w-z_0}{z-z_0}} = -\sum_{n=1}^{\infty} \frac{(w-z_0)^{n-1}}{(z-z_0)^n},$$

which also converges uniformly in w on γ_{r_1} .

Substituting these expansions into the contour integrals and integrating termwise — justified by the Proposition A.3.3linearity of the contour integral and the Theorem A.0.2Weierstrass M -test for uniform convergence (as the geometric series are bounded by a convergent geometric series of constants) — we obtain

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \sum_{n=0}^{\infty} (z-z_0)^n \oint_{\gamma_{r_2}} \frac{f(w)}{(w-z_0)^{n+1}} dw \\ &\quad + \frac{1}{2\pi i} \sum_{n=1}^{\infty} \frac{1}{(z-z_0)^n} \oint_{\gamma_{r_1}} f(w)(w-z_0)^{n-1} dw. \end{aligned}$$

Let ρ be any number with $0 < \rho < R$. For any integer n , Theorem A.3.5 applied to the holomorphic function $\frac{f(w)}{(w-z_0)^{n+1}}$ on the annulus implies that the integral $\oint_{|w-z_0|=\rho} \frac{f(w)}{(w-z_0)^{n+1}} dw$

is independent of the choice of ρ . Therefore we may define

$$a_n = \frac{1}{2\pi i} \oint_{|w-z_0|=\rho} \frac{f(w)}{(w-z_0)^{n+1}} dw$$

for all $n \in \mathbb{Z}$. The second sum above can be re-indexed with $n \mapsto -n$ to match this formula: for $n \geq 1$,

$$\frac{1}{2\pi i} \oint_{\gamma_{r_1}} f(w)(w-z_0)^{n-1} dw = a_{-n}.$$

Thus the full expansion takes the form

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z-z_0)^n.$$

Finally, we verify the claimed convergence. On any closed annulus $r_1 \leq |z-z_0| \leq r_2$ with $0 < r_1 < r_2 < R$, choose any ρ with $r_2 < \rho < R$. By compactness of the circle $|w-z_0| = \rho$, the function f is bounded there, so the coefficients a_n satisfy $|a_n| \leq \frac{M}{\rho^n}$ for a uniform constant M . The Weierstrass M -test therefore shows that the series $\sum_{n=0}^{\infty} a_n(z-z_0)^n$ converges absolutely and uniformly for $|z-z_0| \leq r_2$, and the series $\sum_{n=1}^{\infty} a_{-n}(z-z_0)^{-n}$ converges absolutely and uniformly for $|z-z_0| \geq r_1$. Hence the full Laurent series converges absolutely and uniformly on the closed annulus. $\square$

Definition A.4.4. Let $U \subset \mathbb{C}$ be an open set, let $z_0 \in U$, and let $f : U \setminus \{z_0\} \rightarrow \mathbb{C}$ be a holomorphic function (Definition A.1.1) with an isolated singularity (Definition A.4.5) at z_0 . Suppose there exists $R > 0$ such that the punctured disk (Definition A.4.1) $0 < |z-z_0| < R$ is contained in $U \setminus \{z_0\}$. Then by Theorem A.4.3 f admits a *Laurent series expansion centered at z_0* :

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z-z_0)^n$$

for all z with $0 < |z-z_0| < R$, where the series converges absolutely and uniformly on every annulus (Definition A.4.2) $r_1 \leq |z-z_0| \leq r_2$ with $0 < r_1 < r_2 < R$. The coefficients are given by

$$a_n = \frac{1}{2\pi i} \int_{|w-z_0|=\rho} \frac{f(w)}{(w-z_0)^{n+1}} dw$$

for any $\rho \in (0, R)$, and are independent of the choice of ρ . The series splits into a *regular part* $\sum_{n=0}^{\infty} a_n(z-z_0)^n$ (which is a power series (Definition A.2.1)) and a *principal part* $\sum_{n=1}^{\infty} a_{-n}(z-z_0)^{-n}$.

The singularity z_0 is removable (Definition A.4.5) if and only if the principal part vanishes, a pole of order m (Definition A.4.5) if and only if $a_{-n} = 0$ for all $n > m$ with $a_{-m} \neq 0$, and essential (Definition A.4.5) if and only if infinitely many coefficients a_{-n} are nonzero.

Definition A.4.5. Let $U \subset \mathbb{C}$ be an open set and let $f : U \setminus \{z_0\} \rightarrow \mathbb{C}$ be a holomorphic function (Definition A.1.1) defined on U with a single point $z_0 \in U$ removed. The point z_0 is called an *isolated singularity of f* . The isolated singularity z_0 is classified as follows:

1. z_0 is a **removable singularity** if f can be extended to a function holomorphic on all of U . By Theorem A.4.6, this is equivalent to each of the following conditions:
 - (a) $\lim_{z \rightarrow z_0} (z - z_0)f(z) = 0$;
 - (b) f is bounded in a punctured neighborhood (Definition A.4.1) of z_0 ;
 - (c) the Laurent expansion (Definition A.4.4) of f at z_0 has no negative-degree terms, i.e. it reduces to a Taylor series.
2. z_0 is a **pole of order $m \in \mathbb{N}$** if there exists a holomorphic function (Definition A.1.1) g on U with $g(z_0) \neq 0$ such that $f(z) = \frac{g(z)}{(z - z_0)^m}$ for all $z \in U \setminus \{z_0\}$. Equivalently, $\lim_{z \rightarrow z_0} |f(z)| = \infty$ and the Laurent expansion (Definition A.4.4) of f at z_0 has finitely many negative-degree terms.
3. z_0 is an **essential singularity** if it is neither a removable singularity nor a pole. Equivalently, the Laurent expansion of f at z_0 has infinitely many negative-degree terms, and by Weierstrass's theorem on essential singularities, f takes every complex value (with at most one exception) infinitely often in any neighborhood of z_0 .

Theorem A.4.6 (Riemann's theorem on removable singularities). (♠ TODO: AI generated, verify) Let $U \subset \mathbb{C}$ be an open set, let $z_0 \in U$, and let $f : U \setminus \{z_0\} \rightarrow \mathbb{C}$ be a holomorphic function (Definition A.1.1) with an isolated singularity (Definition A.4.5) at z_0 . Let $f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$ be the Laurent expansion (Definition A.4.4) of f on a punctured neighborhood $0 < |z - z_0| < R$. The following are equivalent:

1. z_0 is a removable singularity of f , i.e. f admits a holomorphic extension (Definition A.1.1) to all of U .
2. The limit $\lim_{z \rightarrow z_0} (z - z_0)f(z) = 0$ exists.
3. The function f is bounded in a punctured neighborhood of z_0 , i.e. there exists $\varepsilon > 0$ such that $\sup_{0 < |z - z_0| < \varepsilon} |f(z)| < \infty$.
4. The Laurent expansion of f at z_0 has no negative-degree terms, i.e. $a_n = 0$ for all $n < 0$, so that the Laurent series reduces to the Taylor series (Definition A.2.6)

$$f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$$

for $0 < |z - z_0| < R$.

Proof. (♠ TODO: AI generated, verify) We prove the cycle of implications $(1) \Rightarrow (3) \Rightarrow (2) \Rightarrow (4) \Rightarrow (1)$.

- $(1) \Rightarrow (3)$: If f extends holomorphically to U , then the extension is continuous at z_0 by Proposition A.1.2, so f is bounded in a neighborhood of z_0 .
- $(3) \Rightarrow (2)$: If $|f(z)| \leq M$ for $0 < |z - z_0| < \varepsilon$, then $|(z - z_0)f(z)| \leq M|z - z_0|$ for $0 < |z - z_0| < \varepsilon$, so $(z - z_0)f(z) \rightarrow 0$ as $z \rightarrow z_0$.
- $(2) \Rightarrow (4)$: Suppose $\lim_{z \rightarrow z_0} (z - z_0)f(z) = 0$. Then for any $\rho > 0$ sufficiently small, the function f is bounded on the circle $|z - z_0| = \rho$ (since $(z - z_0)f(z)$ is bounded and $|z - z_0| = \rho$ is fixed). More precisely, for each $r > 0$ with $0 < r < R$, the Laurent coefficients satisfy

the Cauchy estimate

$$|a_n| \leq \frac{1}{2\pi} \oint_{|z-z_0|=r} \frac{|f(z)|}{|z-z_0|^{n+1}} |dz| \leq \frac{\max_{|z-z_0|=r} |f(z)|}{r^n}.$$

For $n < 0$, write $n = -k$ with $k > 0$. Then $|a_{-k}| \leq r^k \max_{|z-z_0|=r} |f(z)|$. Since $(z - z_0)f(z) \rightarrow 0$ as $z \rightarrow z_0$, we have $\max_{|z-z_0|=r} |(z - z_0)f(z)| \rightarrow 0$ as $r \rightarrow 0$, so $\max_{|z-z_0|=r} |f(z)| \leq \frac{1}{r} \cdot \varepsilon(r)$ where $\varepsilon(r) \rightarrow 0$ as $r \rightarrow 0$. Hence

$$|a_{-k}| \leq r^{k-1} \varepsilon(r) \rightarrow 0 \quad \text{as } r \rightarrow 0,$$

implying $a_{-k} = 0$ for all $k \geq 1$. Thus all negative-degree coefficients vanish.

(4) $\Rightarrow$ (1): The power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ has radius of convergence at least R because it converges for $0 < |z - z_0| < R$. By Definition A.2.1, it defines a holomorphic function on $D(z_0, R)$ that agrees with f on $0 < |z - z_0| < R$, hence gives a holomorphic extension of f across z_0 .

□

Definition A.4.7. Let $U \subset \mathbb{C}$ be an open set, let $z_0 \in U$, and let $f : U \setminus \{z_0\} \rightarrow \mathbb{C}$ be a holomorphic function with an isolated singularity (Definition A.4.5) at z_0 . In a punctured neighborhood $0 < |z - z_0| < r$ of z_0 , the function f admits (Theorem A.4.3) a Laurent series expansion (Definition A.4.4)

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$$

where $a_n = \frac{1}{2\pi i} \int_{\gamma} \frac{f(w)}{(w - z_0)^{n+1}} dw$ for any positively oriented simple closed contour γ in the punctured neighborhood around z_0 . The coefficient a_{-1} is called the **residue of f at z_0** , denoted

$$\text{Res}(f, z_0) := a_{-1}.$$

Definition A.4.8 (Meromorphic function). Let $U \subseteq \mathbb{C}$ be an open set. A function $f : U \rightarrow \widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ is called **meromorphic on U** if there exists a subset $S \subseteq U$ such that:

1. S has no accumulation point in U (i.e., every point of U has a neighborhood intersecting S in at most finitely many points),
2. f is holomorphic (Definition A.1.1) on $U \setminus S$,
3. For each $z_0 \in S$, z_0 is a pole (Definition A.4.5) of $f|_{U \setminus S}$.

The set S is called the **pole set of f** .

Theorem A.4.9 (Residue Theorem). Let $U \subset \mathbb{C}$ be a simply connected open subset and let $f : U \rightarrow \widehat{\mathbb{C}}$ be a meromorphic function (Definition A.4.8) on U . Let $\gamma : [a, b] \rightarrow U$ be a closed piecewise continuously differentiable path (Definition A.3.1) in U and let $z_1, \dots, z_k \in U \setminus \gamma([a, b])$ be the poles (Definition A.4.5) of f that lie in the region enclosed by γ (Definition A.3.4) in the sense that $n(\gamma, z_j) \neq 0$ for each j . Then

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{j=1}^k n(\gamma, z_j) \text{Res}(f, z_j),$$

(Definition A.3.2) where $n(\gamma, z_j)$ is the winding number of γ around z_j (Definition A.3.4) and $\text{Res}(f, z_j)$ is the residue of f at z_j (Definition A.4.7).

Proof. We proceed in three steps: the Laurent expansion (Definition A.4.4) of f at each pole, the removal of all principal parts (Definition A.4.4), and the explicit evaluation of the remaining contour integrals.

1. For each $j = 1, \dots, k$, write m_j for the order of the pole of f at z_j . For each such j , there exists a punctured neighborhood $0 < |z - z_j| < R_j$ on which f admits (Theorem A.4.3) a Laurent expansion (Definition A.4.4)

$$f(z) = \sum_{n=-m_j}^{\infty} a_n^{(j)}(z - z_j)^n,$$

with coefficients $a_n^{(j)} \in \mathbb{C}$. In particular, the coefficient $a_{-1}^{(j)}$ is the residue $\text{Res}(f, z_j)$ by definition (Definition A.4.7).

The principal part (Definition A.4.4) of f at z_j is the finite sum

$$P_j(z) := \sum_{n=-m_j}^{-1} a_n^{(j)}(z - z_j)^n,$$

which is holomorphic on $\mathbb{C} \setminus \{z_j\}$.

2. We next remove all principal parts to obtain a holomorphic function. Define $g : U \setminus \{z_1, \dots, z_k\} \rightarrow \mathbb{C}$ by

$$g(z) := f(z) - \sum_{j=1}^k P_j(z).$$

Note that g is holomorphic on $U \setminus \{z_1, \dots, z_k\}$ because (Proposition A.1.3) $f(z)$ and the $P_j(z)$ are. We show that g extends to a holomorphic function on all of U .

At each z_j , the Laurent expansion of g in the punctured disk $0 < |z - z_j| < R_j$ is

$$\sum_{n=-m_j}^{\infty} a_n^{(j)}(z - z_j)^n - \sum_{n=-m_j}^{-1} a_n^{(j)}(z - z_j)^n = \sum_{n=0}^{\infty} a_n^{(j)}(z - z_j)^n,$$

The Laurent series of g at z_j is obtained by the termwise sum of the Laurent series of f and the principal part P_j , by ???. This resulting power series $\sum_{n=0}^{\infty} a_n^{(j)}(z - z_j)^n$ has radius of convergence at least R_j , because it converges at every z with $0 < |z - z_j| < R_j$ (it is a subseries of the absolutely convergent Laurent series of f). By Lemma A.2.4, the partial sums $S_N^{(j)}(z) = \sum_{n=0}^N a_n^{(j)}(z - z_j)^n$ converge uniformly on every compact subset of $D(z_j, R_j)$.

Each partial sum $S_N^{(j)}$ is a polynomial, hence an entire function by Proposition A.1.4, so $(S_N^{(j)})$ is a sequence of holomorphic functions converging uniformly on compact subsets of $D(z_j, R_j)$. By Theorem A.5.3, the limit function

$$h_j(z) := \sum_{n=0}^{\infty} a_n^{(j)}(z - z_j)^n$$

is holomorphic on $D(z_j, R_j)$ and satisfies $h_j(z) = g(z)$ for all z with $0 < |z - z_j| < R_j$. Therefore h_j is a holomorphic extension of g across z_j , which by Definition A.4.5 means that z_j is a removable singularity of g . After removing all such singularities we obtain a holomorphic extension of g to all of U .

Because U is simply connected and γ is a closed piecewise continuously differentiable path in U , Theorem A.3.5 gives

$$\int_{\gamma} g(z) dz = 0.$$

3. We conclude by evaluating the contour integral of f . From the definition of g , we have

$$f(z) = g(z) + \sum_{j=1}^k P_j(z)$$

for all $z \in U \setminus \{z_1, \dots, z_k\}$. Integrating termwise and using $\int_{\gamma} g = 0$, we obtain By Proposition A.3.3, the contour integral splits as the sum of the individual integrals:

$$\begin{aligned} \int_{\gamma} f(z) dz &= \sum_{j=1}^k \int_{\gamma} P_j(z) dz \\ &= \sum_{j=1}^k \sum_{n=-m_j}^{-1} a_n^{(j)} \int_{\gamma} (z - z_j)^n dz. \end{aligned}$$

For each j and each integer n , Lemma A.3.7 gives

$$\int_{\gamma} (z - z_j)^n dz = \begin{cases} 2\pi i \cdot n(\gamma, z_j), & n = -1, \\ 0, & n \neq -1. \end{cases}$$

Therefore all terms with $n \neq -1$ vanish, and only the $n = -1$ terms, i.e. the residues, contribute:

$$\begin{aligned} \int_{\gamma} f(z) dz &= \sum_{j=1}^k a_{-1}^{(j)} \cdot 2\pi i \cdot n(\gamma, z_j) \\ &= 2\pi i \sum_{j=1}^k n(\gamma, z_j) \operatorname{Res}(f, z_j) \end{aligned}$$

by the identification $a_{-1}^{(j)} = \operatorname{Res}(f, z_j)$ from Step 1.

This completes the proof. □

Theorem A.4.10 (Argument Principle). (♠ TODO: AI generated, verify) Let $U \subset \mathbb{C}$ be a connected open subset and let $f : U \rightarrow \mathbb{C}$ be a meromorphic function (Definition A.4.8) on U . Let γ be a closed piecewise continuously differentiable path (Definition A.3.1) in U that does not pass through any zero or pole of f and whose image is a simple closed curve. Let $z_1, \dots, z_m$ be the zeros of f inside γ with multiplicities $n_1, \dots, n_m$, and let $p_1, \dots, p_k$ be the

poles of f inside γ with orders $m_1, \dots, m_k$. Then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = N - P,$$

(Definition A.3.2) where $N = \sum_{i=1}^m n_i$ is the total number of zeros of f inside γ (counted with multiplicity) and $P = \sum_{j=1}^k m_j$ is the total number of poles of f inside γ (counted with order).

Proof. (♠ TODO: AI generated, verify) We factor f as a product of its zero and pole factors times a nonvanishing holomorphic function, then compute the logarithmic derivative.

1. Since f is meromorphic on U , around each zero z_i of multiplicity n_i we can write

$$f(z) = (z - z_i)^{n_i} g_i(z)$$

where g_i is holomorphic and $g_i(z_i) \neq 0$ in a neighborhood of z_i . Similarly, around each pole p_j of order m_j we can write

$$f(z) = (z - p_j)^{-m_j} h_j(z)$$

where h_j is holomorphic and $h_j(p_j) \neq 0$ in a neighborhood of p_j .

2. The **logarithmic derivative** of f is $\frac{f'}{f}$. Using the factorizations above, we compute:

- At a zero z_i of multiplicity n_i :

$$\frac{f'(z)}{f(z)} = \frac{n_i}{z - z_i} + \frac{g'_i(z)}{g_i(z)},$$

where $\frac{g'_i}{g_i}$ is holomorphic at z_i .

- At a pole p_j of order m_j :

$$\frac{f'(z)}{f(z)} = -\frac{m_j}{z - p_j} + \frac{h'_j(z)}{h_j(z)},$$

where $\frac{h'_j}{h_j}$ is holomorphic at p_j .

3. The function $\frac{f'}{f}$ is therefore meromorphic on U with simple poles at the zeros and poles of f , and with residues n_i at each zero z_i and $-m_j$ at each pole p_j .
4. By Theorem A.4.9, we have

$$\begin{aligned} \frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz &= \sum_{\text{singularities inside } \gamma} \text{Res} \left(\frac{f'}{f}, \cdot \right) \\ &= \sum_{i=1}^m n_i - \sum_{j=1}^k m_j \\ &= N - P. \end{aligned}$$

(Definition A.4.7)

This completes the proof. □

Theorem A.4.11 (Rouché's Theorem). (♠ TODO: AI generated, verify) Let $U \subset \mathbb{C}$ be a connected open subset and let $f, g : U \rightarrow \mathbb{C}$ be holomorphic functions (Definition A.1.1) on U . Let γ be a closed piecewise continuously differentiable path (Definition A.3.1) in U whose image is a simple closed curve. Suppose that

$$|f(z) - g(z)| < |g(z)|$$

for all z on γ . Then f and g have the same number of zeros inside γ , counted with multiplicity.

Proof. (♠ TODO: AI generated, verify) The inequality $|f(z) - g(z)| < |g(z)|$ on γ implies that $g(z) \neq 0$ on γ and that $f(z) \neq 0$ on γ (otherwise $|g(z)| < |g(z)|$). Moreover, the inequality shows that $f(z)/g(z)$ never takes non-positive real values on γ , so f/g never vanishes on γ and its image under f/g avoids the negative real axis.

1. Define $h(z) = \frac{f(z)}{g(z)}$. Then h is holomorphic on U minus the zeros of g , and never takes a value in $(-\infty, 0]$ on γ . In particular, $h(z) \neq 0$ on γ , so both f and g are nonvanishing on γ .
2. Consider the contour integral

$$\frac{1}{2\pi i} \int_{\gamma} \frac{h'(z)}{h(z)} dz.$$

(Definition A.3.2) By Theorem A.4.10, this equals $Z_h - P_h$, where Z_h is the number of zeros of h inside γ and P_h is the number of poles of h inside γ , both counted with multiplicity.

3. Observe that

$$\frac{h'(z)}{h(z)} = \frac{f'(z)}{f(z)} - \frac{g'(z)}{g(z)}.$$

Therefore

$$\begin{aligned} \frac{1}{2\pi i} \int_{\gamma} \frac{h'(z)}{h(z)} dz &= \frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz - \frac{1}{2\pi i} \int_{\gamma} \frac{g'(z)}{g(z)} dz \\ &= (N_f - P_f) - (N_g - P_g), \end{aligned}$$

(Theorem A.4.10) where N_f, P_f are the zero and pole counts of f inside γ , and N_g, P_g are those of g .

4. Since f and g are holomorphic (not merely meromorphic) on U , we have $P_f = P_g = 0$. Moreover, the inequality $|f - g| < |g|$ on γ forces $h(z) = f(z)/g(z)$ to lie in the right half-plane $\operatorname{Re}(h(z)) > 0$ on γ , so the image curve $h \circ \gamma$ does not wind around the origin. Hence

$$\frac{1}{2\pi i} \int_{\gamma} \frac{h'(z)}{h(z)} dz = 0$$

by Theorem A.4.10 applied to h .

5. Combining the above, we obtain

$$N_f - N_g = 0,$$

so $N_f = N_g$. Thus f and g have the same number of zeros inside γ , counted with multiplicity.

This completes the proof. □

A.5. Global Theorems of Complex Analysis.

Theorem A.5.1. Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be an entire function (Definition A.1.1). (♠ TODO: bounded function) (♠ TODO: constant function) If f is bounded, meaning there exists $M > 0$ such that $|f(z)| \leq M$ for all $z \in \mathbb{C}$, then f is constant.

Theorem A.5.2. Let $U \subset \mathbb{C}$ be a connected open subset (a domain) and let $f, g : U \rightarrow \mathbb{C}$ be two holomorphic functions (Definition A.1.1) on U . If the set $\{z \in U : f(z) = g(z)\}$ has an accumulation point — that is, there exists $z_0 \in U$ such that every neighborhood of z_0 contains infinitely many points $z \neq z_0$ with $f(z) = g(z)$ — then $f(z) = g(z)$ for all $z \in U$. In other words, $f \equiv g$ on U .

Theorem A.5.3. Let $U \subset \mathbb{C}$ be an open set and let $(f_n)_{n=1}^\infty$ be a sequence of holomorphic functions (Definition A.1.1) $f_n : U \rightarrow \mathbb{C}$. Suppose that (f_n) converges to a function $f : U \rightarrow \mathbb{C}$ uniformly on every compact subset of U . Then:

1. The limit function f is holomorphic on U .
2. For each $k = 0, 1, 2, \dots$, the sequence of derivatives (Definition A.1.1) $(f_n^{(k)})$ converges to $f^{(k)}$ uniformly on every compact subset of U .

Proof. (♠ TODO: AI generated, verify) We first establish the continuity of f , then prove that f is holomorphic, and finally prove the uniform convergence of derivatives on compact subsets.

We begin with the continuity of the limit function. Let $z_0 \in U$ be arbitrary. Since U is open, there exists a compact neighborhood K of z_0 contained in U (♠ TODO: state and reference a general fact that there is a compact subneighborhood of an open neighborhood — in particular, is this only true in Euclidean spaces or is it more true of other kinds of general topological spaces that euclidean spaces just happen to be an example of?). By the hypothesis of uniform convergence on compact subsets, (f_n) converges to f uniformly on K . Each f_n is holomorphic and hence continuous by Proposition A.1.2. By ??, the uniform limit f on K is continuous at z_0 . Since z_0 was arbitrary, f is continuous on U .

Next we prove that f is holomorphic. Fix $z_0 \in U$ and choose $r > 0$ such that the closed disk (Definition A.2.5) $\overline{D}(z_0, r)$ is contained in U . Let γ be the positively oriented circle of radius r centered at z_0 , parametrized by $\gamma(t) = z_0 + re^{it}$ for $t \in [0, 2\pi]$.

For each n , since f_n is holomorphic on U and the open disk $D(z_0, r)$ is simply connected, Theorem A.3.6 applied to f_n on $D(z_0, r)$ and the curve γ gives, for every $z \in D(z_0, r)$,

$$f_n(z) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f_n(w)}{w - z} dw.$$

The image $\gamma([0, 2\pi])$ is compact, so by the hypothesis of uniform convergence on compact subsets, (f_n) converges uniformly to f on $\gamma([0, 2\pi])$. For each fixed $z \in D(z_0, r)$, the function

$w \mapsto \frac{1}{w-z}$ is bounded on $\gamma([0, 2\pi])$. Hence,

$$\frac{f_n(\gamma(t))}{\gamma(t) - z} \gamma'(t) \longrightarrow \frac{f(\gamma(t))}{\gamma(t) - z} \gamma'(t)$$

uniformly on $[0, 2\pi]$ as $n \rightarrow \infty$.

By ?? applied separately to the real and imaginary parts of the integrand, the limit commutes with the integral. Therefore, for every $z \in D(z_0, r)$,

$$f(z) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{w - z} dw.$$

We now show that this integral representation implies that f is holomorphic on $D(z_0, r)$. For $z, z + h \in D(z_0, r)$ with $h \neq 0$, we compute

$$\begin{aligned} \frac{f(z + h) - f(z)}{h} &= \frac{1}{2\pi i} \oint_{\gamma} f(w) \cdot \frac{1}{h} \left(\frac{1}{w - z - h} - \frac{1}{w - z} \right) dw \\ &= \frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w - z - h)(w - z)} dw. \end{aligned}$$

We compare this with the candidate derivative $\frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w - z)^2} dw$. The difference is

$$\frac{f(z + h) - f(z)}{h} - \frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w - z)^2} dw = \frac{1}{2\pi i} \oint_{\gamma} f(w) \cdot \frac{h}{(w - z - h)(w - z)^2} dw.$$

For $|h|$ sufficiently small, specifically $|h| < \frac{r - |z - z_0|}{2}$, we have for all $w \in \gamma$ the estimates

$$|w - z| \geq r - |z - z_0|, \quad |w - z - h| \geq |w - z| - |h| \geq \frac{r - |z - z_0|}{2}.$$

Hence,

$$\begin{aligned} \left| \frac{f(z + h) - f(z)}{h} - \frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w - z)^2} dw \right| &\leq \frac{|h|}{2\pi} \oint_{\gamma} \frac{|f(w)|}{|w - z - h||w - z|^2} |dw| \\ &\leq \frac{|h|}{2\pi} \cdot \sup_{w \in \gamma} |f(w)| \cdot \frac{1}{(r - |z - z_0|)^3} \cdot (2\pi r). \end{aligned}$$

The right-hand side tends to 0 as $h \rightarrow 0$. Consequently, the limit of the difference quotient exists and equals $\frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w - z)^2} dw$, i.e.,

$$f'(z) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(w)}{(w - z)^2} dw$$

for every $z \in D(z_0, r)$. Thus f is complex differentiable on $D(z_0, r)$, hence holomorphic on $D(z_0, r)$ by Definition A.1.1. Since z_0 was arbitrary, f is holomorphic on all of U .

Finally we prove the uniform convergence of derivatives on compact subsets. Let $K \subset U$ be an arbitrary compact set. For each $z \in K$, choose $r_z > 0$ such that $\overline{D}(z, 2r_z) \subset U$. The

collection $\{D(z, r_z) : z \in K\}$ is an open cover of K , from which we extract a finite subcover $\{D(z_j, r_j)\}_{j=1}^m$.

For each j , let γ_j be the positively oriented circle of radius $2r_j$ centered at z_j , so that $\overline{D}(z_j, r_j) \subset D(z_j, 2r_j) \subset U$ and γ_j lies in U . By Theorem A.3.6, for each n and each $z \in D(z_j, r_j)$ we have

$$f_n^{(k)}(z) = \frac{k!}{2\pi i} \oint_{\gamma_j} \frac{f_n(w)}{(w-z)^{k+1}} dw.$$

Since we have already shown that f is holomorphic on U , the same theorem applied to f yields

$$f^{(k)}(z) = \frac{k!}{2\pi i} \oint_{\gamma_j} \frac{f(w)}{(w-z)^{k+1}} dw.$$

For $z \in \overline{D}(z_j, r_j)$ and $w \in \gamma_j$, we have $|w-z| \geq |w-z_j| - |z-z_j| \geq 2r_j - r_j = r_j$. Therefore,

$$\begin{aligned} |f_n^{(k)}(z) - f^{(k)}(z)| &\leq \frac{k!}{2\pi} \oint_{\gamma_j} \frac{|f_n(w) - f(w)|}{|w-z|^{k+1}} |dw| \\ &\leq \frac{k!}{2\pi} \cdot \frac{\sup_{w \in \gamma_j} |f_n(w) - f(w)|}{r_j^{k+1}} \cdot (4\pi r_j) \\ &= \frac{2k!}{r_j^k} \sup_{w \in \gamma_j} |f_n(w) - f(w)|. \end{aligned}$$

Since $f_n \rightarrow f$ uniformly on the compact set γ_j , the right-hand side tends to 0 as $n \rightarrow \infty$, uniformly for $z \in \overline{D}(z_j, r_j)$ and hence for $z \in D(z_j, r_j)$. Taking the maximum over the finite index set $\{1, \dots, m\}$, we obtain

$$\sup_{z \in K} |f_n^{(k)}(z) - f^{(k)}(z)| \rightarrow 0$$

as $n \rightarrow \infty$. Thus $(f_n^{(k)})$ converges to $f^{(k)}$ uniformly on K , and since K was an arbitrary compact subset of U , the convergence is uniform on every compact subset of U . $\square$

Theorem A.5.4. Let $U \subset \mathbb{C}$ be a connected open subset (a domain) and let $f : U \rightarrow \mathbb{C}$ be a non-constant holomorphic function (Definition A.1.1) on U . Then f is an open map: for every open subset $V \subset U$, the image $f(V)$ is open in $\mathbb{C}$.

Theorem A.5.5. Let $U \subset \mathbb{C}$ be a connected open subset (a domain) and let $f : U \rightarrow \mathbb{C}$ be a holomorphic function (Definition A.1.1) on U . If there exists a point $z_0 \in U$ such that $|f(z_0)| \geq |f(z)|$ for all $z \in U$, then f is constant on U . Equivalently, a non-constant holomorphic function on a connected open subset of $\mathbb{C}$ cannot attain its maximum modulus at an interior point.

Definition A.5.6 (Analytic continuation). Let U and V be non-empty open subsets of $\mathbb{C}$, and let $f : U \rightarrow \mathbb{C}$ and $g : V \rightarrow \mathbb{C}$ be holomorphic functions (Definition A.1.1). Suppose $U \cap V \neq \emptyset$ and there exists a non-empty open subset $W \subseteq U \cap V$ such that $f|_W = g|_W$. Then g is called an *analytic continuation of f to V* , and f is called an *analytic continuation of g to U* .

Definition A.5.7 (Meromorphic continuation). Let U and V be non-empty open subsets of $\mathbb{C}$, and let $f : U \rightarrow \mathbb{C}$ be a holomorphic function (Definition A.1.1). Let $g : V \rightarrow \hat{\mathbb{C}}$ be a meromorphic function (Definition A.4.8). Suppose $U \cap V \neq \emptyset$ and there exists a non-empty open subset $W \subseteq U \cap V$ such that $f|_W = g|_W$. Then g is called a *meromorphic continuation of f to V* .

Theorem A.5.8. (♠ TODO: AI generated, verify) Let $U \subset \mathbb{C}$ be a simply connected open subset with $U \neq \mathbb{C}$. Then there exists a biholomorphic map (Definition A.1.5) $f : U \xrightarrow{\sim} \mathbb{D}$ from U onto the open unit disk $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$. Moreover, given $z_0 \in U$ and $\theta \in \mathbb{R}$, there exists a unique such biholomorphic map satisfying $f(z_0) = 0$ and $\arg(f'(z_0)) = \theta$.

Proof. (♠ TODO: AI generated, verify) Let $U \subsetneq \mathbb{C}$ be simply connected and proper, and fix $z_0 \in U$. We construct a biholomorphism $\phi : U \rightarrow D = \{w \in \mathbb{C} : |w| < 1\}$ with $\phi(z_0) = 0$ and $\phi'(z_0) > 0$.

Define the extremal family

$$\mathcal{F} = \{f : U \rightarrow D \text{ injective holomorphic, } f(z_0) = 0, f'(z_0) > 0\}.$$

Since U is proper and simply connected, $\mathcal{F}$ is nonempty. Choose any $w_0 \notin U$. Because U is simply connected, the function $z - w_0$ admits a holomorphic square root on U , say $g(z) = \sqrt{z - w_0}$. The map g is injective and its image avoids one of the two rays from the origin. Compose with a Möbius transformation to obtain an element of $\mathcal{F}$.

Let $M = \sup\{f'(z_0) : f \in \mathcal{F}\} > 0$. Choose a sequence $f_n \in \mathcal{F}$ with $f'_n(z_0) \rightarrow M$. Since each $|f_n(z)| < 1$, the family is uniformly bounded on compact subsets of U . By Montel's theorem, there exists a subsequence converging locally uniformly to some holomorphic $\phi : U \rightarrow \overline{D}$.

We verify that ϕ is injective using Hurwitz's theorem: for any distinct $a, b \in U$, the functions $f_n(z) - f_n(a)$ have no zeros near b (by injectivity), so their limit $\phi(z) - \phi(a)$ has either a zero of infinite order or none at all. This implies $\phi(b) \neq \phi(a)$, hence ϕ is injective.

We show surjectivity onto D . If not, let $w_1 \in D \setminus \phi(U)$. The function $(\phi(z) - w_1)^{-1/2}$ composed with a suitable Möbius transformation gives an element of $\mathcal{F}$ with strictly larger derivative at z_0 , contradicting the maximality of M . Hence ϕ maps onto D .

Finally, uniqueness follows from the Schwarz lemma: if $\psi : D \rightarrow D$ is biholomorphic with $\psi(0) = 0$, then $\psi(w) = e^{i\alpha}w$ for some $\alpha \in \mathbb{R}$. The condition on the argument of $f'(z_0)$ fixes this rotation uniquely. $\square$

APPENDIX A. ANALYSIS

Theorem A.0.1 (Weierstrass M-test for uniform convergence of function series). Let $T = \widetilde{R}_{\geq 0}$ be the positive cone of a totally ordered ring $\widetilde{R}$ that is dense at zero. In particular, T is an ordered semiring with an adjoined infinity T^∞ . Let R be a commutative ring equipped with an absolute value valued in T . Let $(V, \|\cdot\|)$ be a complete extended normed module over R valued in T^∞ . Let S be a set and let $(f_n)_{n=1}^\infty$ be a sequence of functions $f_n : S \rightarrow V$.

Suppose there exists a sequence $(M_n)_{n=1}^\infty$ in $T_{\geq 0}$ such that:

1. For each n and each $x \in S$, $\|f_n(x)\| \leq M_n$;
2. The sequence (M_n) satisfies the Cauchy tails condition.

Then the series $\sum_{n=1}^\infty f_n(x)$ converges uniformly on S to a function $f : S \rightarrow V$.

In particular, since T is a subset of $\tilde{R}$, if the sequence of partial sums $(\sum_{n=1}^N M_n)_{N=1}^\infty$ converges in $\tilde{R}$, then (M_n) satisfies the Cauchy tails condition.

Proof. We prove pointwise convergence first, then uniform convergence.

Fix $x \in S$. The partial sums of the series $\sum f_n(x)$ form a sequence in V . For any $K > N$, by the triangle inequality for the norm, we have

$$\left\| \sum_{n=N+1}^K f_n(x) \right\| \leq \sum_{n=N+1}^K \|f_n(x)\| \leq \sum_{n=N+1}^K M_n.$$

By the Cauchy tails condition on (M_n) , for every $\varepsilon \in T_{>0}$ there exists N_0 such that for all $K > N \geq N_0$:

$$\sum_{n=N+1}^K M_n < \varepsilon.$$

Therefore, the partial sums satisfy, uniformly in x , the Cauchy condition:

$$\left\| \sum_{n=N+1}^K f_n(x) \right\| < \varepsilon \quad \text{for all } x \in S.$$

In other words, for any $K > N \geq N_0$, we have

$$\left\| \sum_{n=1}^K f_n(x) - \sum_{n=1}^N f_n(x) \right\| = \left\| \sum_{n=N+1}^K f_n(x) \right\| < \varepsilon,$$

i.e. the partial sums $(\sum_{n=1}^N f_n(x))_{N=1}^\infty$ form a Cauchy sequence in V ; recall that the extended metric on V is given by $d_V(v_1, v_2) = \|v_1 - v_2\|$. Since V is complete under this extended metric, for each fixed x the series $\sum f_n(x)$ thus converges to some limit, which we denote by $f(x)$. In fact, note that the convergence is uniformly Cauchy. By ??, since $(V, \|\cdot\|)$ is complete (with respect to the extended metric induced by $\|\cdot\|$), the convergence is thus uniform. $\square$

Theorem A.0.2. Let R be a commutative ring equipped with an absolute value valued in $\mathbb{R}$. Let $(V, \|\cdot\|)$ be a complete extended normed module over R valued in $\mathbb{R} \cup \{\infty\}$. Let S be a set and let $(f_n)_{n=1}^\infty$ be a sequence of functions $f_n : S \rightarrow V$.

Suppose there exists a sequence $(M_n)_{n=1}^\infty$ in $\mathbb{R}_{\geq 0}$ such that:

1. For each n and each $x \in S$, $\|f_n(x)\| \leq M_n$;

2. The series $\sum_{n=1}^{\infty} M_n$ converges.

Then the series $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on S to a function $f : S \rightarrow V$.

Proof. This is a consequence of Theorem A.0.1 by letting the ordered ring be $\mathbb{R}$ — note that $\mathbb{R}$ is totally ordered and dense at zero. Moreover, since the M_n are elements of $\mathbb{R}_{\geq 0}$, the convergence of $\sum_{n=1}^{\infty} M_n$ is equivalent to the Cauchy-tails condition of (M_n) . $\square$

Theorem A.0.3 (Comparison test for series of nonnegative elements in a totally ordered abelian group with the least upper bound property). (♠ TODO: AI generated, verify. Generalized from $\mathbb{R}$ to any totally ordered abelian group with the LUB property.) Let G be a totally ordered abelian group with the least upper bound property. Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be sequences in G such that $0 \leq a_n \leq b_n$ for all n . For a series of nonnegative terms, we say that it converges if its sequence of partial sums is bounded above; in this case the supremum of the partial sums exists by the least upper bound property, and the series is said to converge to that supremum.

1. If the series $\sum_{n=1}^{\infty} b_n$ converges, then the series $\sum_{n=1}^{\infty} a_n$ also converges.
2. If the series $\sum_{n=1}^{\infty} a_n$ diverges (i.e., its partial sums are not bounded above), then the series $\sum_{n=1}^{\infty} b_n$ also diverges.

Proof. (♠ TODO: AI generated, verify) Let $A_N = \sum_{n=1}^N a_n$ and $B_N = \sum_{n=1}^N b_n$ denote the partial sums. Since $0 \leq a_n \leq b_n$ for all n , the compatibility of the order with addition implies $0 \leq A_N \leq B_N$ for all N .

1. Suppose $\sum b_n$ converges, so the set $\{B_N : N \in \mathbb{N}\}$ is bounded above. By the least upper bound property of G , its supremum $\sup_N B_N$ exists. Since $A_N \leq B_N \leq \sup_N B_N$ for all N , the sequence (A_N) is also bounded above. Moreover, (A_N) is nondecreasing because each $a_n \geq 0$. Hence $\sum a_n$ converges.
2. This follows from part (1) by contraposition: if $\sum b_n$ converged, then $\sum a_n$ would converge; therefore, if $\sum a_n$ diverges, $\sum b_n$ must also diverge.

$\square$